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Multi-dimensional non-Markovian backward stochastic differential equations of interactively quadratic generators*

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Abstract

This paper is devoted to a general solvability of multi-dimensional non-Markovian backward stochastic differential equations (BSDEs) with interactively quadratic generators. General structures of the generator g are posed for both local and global existence and uniqueness results on a BSDE, which admit a general growth of the generator $g(\omega, t, y, z)$ in the state variable y , and a quadratic growth of the i th components $g^i(\omega, t, y, z)$ both in the j th row z^j of the state variable z for $j \neq i$ (which we call the “interactively quadratic” growth) and in the i th row z^i of z . We first establish an existence and uniqueness result on local bounded solutions and then several existence and uniqueness results on global bounded and unbounded solutions. They improve several existing works in the non-Markovian setting, and also incorporate some interesting examples, which include global bounded solution of backward stochastic Burger differential systems and a partial answer to the problem posed in Jackson [25] (see Example 2.16(ii) and Corollary 2.25, respectively). A comprehensive study on the bounded solution of scalar-valued quadratic BSDEs with unbounded stochastic parameters is carried out for deriving our main results.

Keywords: system of BSDEs; interactively quadratic generator; backward stochastic Burger differential systems; BMO martingale; bounded solution; unbounded solution.

MSC2020 subject classifications: 60H10.

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1 Introduction

Let $T \in (0, +\infty)$, $(B_t)_{t \in [0, T]}$ be a d -dimensional Brownian motion defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and $(\mathcal{F}_t)_{t \in [0, T]}$ be the augmented filtration generated by B . Consider the following backward stochastic differential equation (BSDE in short):

$$Y_t = \xi + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad t \in [0, T], \quad (1.1)$$

where the terminal value ξ is a d -dimensional $\mathcal{F}_T$ -measurable random vector, and the generator $g(\omega, t, y, z) : \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^n$ is $(\mathcal{F}_t)$ -progressively measurable for each pair (y, z) and is jointly continuous in the last two arguments y and z . By a solution $(Y_t, Z_t)_{t \in [0, T]}$ of the BSDE, we mean a pair of $(\mathcal{F}_t)$ -progressively measurable processes, taking values in $\mathbb{R}^n \times \mathbb{R}^{n \times d}$, such that it almost surely satisfies BSDE (1.1). The BSDE is said to be *quadratic* if the generator g has a quadratic growth in the last argument z .

Bismut [3] initially introduced in 1973 the above BSDE in the linear case, and further studied in 1976, in his alternative work [4], the backward stochastic Riccati differential system as a specially structured matrix-valued quadratic nonlinear BSDE. In 1990, Pardoux and Peng [38] established the existence and uniqueness result for a multidimensional uniformly Lipschitz continuous nonlinear BSDE. Since then, there has been an increasing interest in BSDEs and their applications in diverse fields such as mathematical finance, stochastic control, partial differential equations (PDEs), etc. The interested reader is referred to, among others, El Karoui et al. [13], Kobylanski [30], El Karoui and Hamadène [12], Hu et al. [23], Frei and Dos Reis [21], Kramkov and Pulido [34], Escariz et al. [14], Tian [42], and Weston [43].

A quadratic BSDE naturally arises from non-Markovian linear quadratic optimal stochastic control, risk-sensitive control, stochastic differential games, and stochastic economic equilibrium (see, in particular, Bismut [4] and El Karoui and Hamadène [12]). When $n = 1$ and the terminal value is bounded, the existence and uniqueness result was first given by Kobylanski [30], and subsequently by Kohlmann and Tang [31], Tevzadze [41], Briand and Elie [6], Fan [15], and Luo and Fan [36]. When $n = 1$ and the terminal value is allowed to be unbounded, existence and uniqueness of unbounded solutions of quadratic BSDEs was given in Briand and Hu [7, 8], Delbaen et al. [10, 11], Barrieu and El Karoui [1], and Fan et al. [16]. A system of quadratic BSDEs is widely known to be more difficult to be solved (see Peng [39]). In particular, the solution of general backward stochastic Riccati differential systems was already deeply studied by Bismut [4], as early as in 1976—almost from the birth of a BSDE. It is known as the so-called Bismut's open problem, and was partially solved by Kohlmann and Tang [32] and was completely solved by Tang [40]. A general existence and uniqueness, without any structural assumption on the quadratic growth of the generator, is recognized to be impossible, as shown by Frei and Dos Reis [21] with a counterexample. On the other hand, diverse systems of quadratic BSDEs have been derived from diverse problems such as financial price-impact modeling, financial market equilibrium among interacting agents, nonzero-sum risk-sensitive stochastic differential games, and stochastic equilibrium in an incomplete financial markets. Much progress on them has been made. When the terminal value is very small in the essential supremum norm, Tevzadze [41] established a general existence and uniqueness result, which has inspired numerous studies on the bounded or unbounded solution under some "smallness" of the terminal value, the terminal time, or the generator; see for example Frei [20], Jamneshan et al. [27], Kramkov and Pulido [34, 33], Harter and Richou [22] and Kardaras et al. [28]. In the particular Markovian setting, Cheridito and Nam [9] proved the global solvability of a special system of quadratic BSDEs associated to a forward SDE, and Xing and Žitković [45] formulated a large class of systems of quadratic BSDEs via the Bensoussan-Frehse (BF) condition

(inspired by Bensoussan and Frehse [2]) and the a priori boundedness (AB) condition on the generator (see (AB) in Section 2 for details); see Weston and Žitković [44] and Escariz et al. [14] for more developments. Recently, Jackson and Žitković [26] and Jackson [25] further extended the result of [45] to a non-Markovian system of quadratic BSDEs, but assuming an additional Malliavin-regularity on the generator, in addition to the (BF) and (AB) conditions.

Numerous works are also available on bounded solutions in the non-Markovian setting, without smallness nor Malliavin-regularity of the terminal time, the terminal value or the generator. Cheridito and Nam [9] considered multidimensional BSDEs with projectable quadratic generators and sub-quadratic generators. Hu and Tang [24] established the global solvability of multi-dimensional BSDEs, assuming that the i th component $g^i(t, y, z)$ of the generator $g(t, y, z)$ admits a quadratic growth only in the i th row z^i of the matrix variable z (i.e. in the so-called “*diagonally quadratic*” way) and is bounded in z^j for each $j \neq i$, and consequently answered the open problem arising from a nonzero-sum risk-sensitive stochastic differential game posed by El Karoui and Hamadène [12]. When $g^i(t, y, z)$ has a strictly quadratic growth in z^i for each $i = 1, \dots, n$, Fan et al. [17] further extended [24] to allow the generator $g(t, y, z)$ to have a sub-quadratic growth in z . Luo [37] solved a system of BSDEs, where $g^i(t, y, z)$ grows quadratically in z^j for any $1 \leq j \leq i$ (i.e. in the so-called “*triangularly quadratic*” way) and has a special form. Very recently, Fan et al. [19] generalized [17] to the system of quadratic backward stochastic Volterra integral equations, and Weston [43] applied the result of [17] to prove the global existence of a Radner equilibrium in a limited participation economy model.

The present paper focuses on both local and global solution of systems of non-Markovian BSDEs with *interactively quadratic* generators g , that is, the components $g^i(\omega, t, y, z)$ of the generators are allowed to have a quadratic growth both in z^i and in z^j with $j \neq i$. Our results unify and extend some existing results in the non-Markovian setting such as [24], [37] and [17], and show that under some circumstances, a system of quadratic BSDEs has a unique solution when the generator $g(t, y, z)$ has a quadratic growth in the product of $\sqrt{\theta}$ and some rows of the matrix variable z for a small constant θ , which is compared to some existing results mentioned in the second last paragraph. More specifically, we first establish an existence and uniqueness result on the local bounded solution of systems of quadratic BSDEs with bounded terminal values (see Theorem 2.3 in Section 2) under general assumptions, see Assumptions (B1) and (B2) in Section 2 for details. Assumption (B1) states that g^i grows in z^i in any of the following three different ways: *strictly quadratic*, *generally quadratic* and *linear*. g^i also has a general growth in y , and has a quadratic growth in $\sqrt{\theta}z^j$ for both $j \neq i$ and a small constant θ , and an interactively quadratic growth like the inner products $\langle z^i, z^j \rangle$ for $j < i$ and $\langle z^i, \theta z^j \rangle$ for both $j > i$ and a small constant θ . We note that generally speaking, those processes defined in Assumptions (B1) and (B2) (namely, $\alpha, \bar{\alpha}, \tilde{\alpha}$ and v) are all unbounded and stochastic, and satisfy various integrability conditions which seem to be necessary to the desired conclusions. At the same time, these conditions admit the unbounded solutions of systems of quadratic BSDEs. Then, under, respectively, an a priori boundedness assumption and two stronger assumptions than (B1) (see (AB), (C1a) and (C1b) in Section 2), we further derive from Theorem 2.3 three existence and uniqueness results on global bounded solutions of systems of quadratic BSDEs: Theorems 2.6, 2.10 and 2.14 in Section 2. These results could cover all those of Hu and Tang [24], Luo [37] and Fan et al. [17]. As a natural application of Theorems 2.10 and 2.14, with the help of the invertible linear transformation, we solve three kinds of specially structured systems of BSDEs in Theorems 2.20, 2.22 and 2.23 in Section 2, which are respectively compared with Jackson [25, Theorems 6.9 and 6.19] and Xing

and Žitković [45, Theorem 3.1]. By the way, using Proposition 2.24 and Corollary 2.25 in Section 2, we answer partially the open problem 6.25 of Jackson [25] on solution of a special system of BSDEs. Finally, from Theorems 2.6, 2.10 and 2.14, we give three existence and uniqueness results on unbounded solution of multi-dimensional quadratic BSDEs with unbounded BMO terminal values, where the generator g needs to be bounded in y , see Assumptions (D1) and (D2) as well as Theorems 2.26, 2.27 and 2.31 in Section 2 for details. These results also improve some existing results in the non-Markovian setting.

In comparison especially to Hu and Tang [24] and Fan et al. [17], our methods manifest the following several features. First, bounded solution of scalar-valued quadratic BSDEs with unbounded diversely integrable stochastic parameters, is comprehensively investigated with the BMO martingale (i.e., *bounded mean oscillation* martingale) theory, Girsanov's transform and some helpful inequalities. New results include existence, uniqueness and a priori estimates; see Propositions A.1 to A.3 in the Appendix for details. Second, in the proof of Theorem 2.3, we construct a novel contraction mapping on some closed convex subset in the space of bounded solutions of systems of quadratic BSDEs, via bounded solution of scalar-valued BSDEs in contrast to those of [24, 17]. Third, we use a delicate induction combined with Propositions A.1 to A.3 and Theorem 2.3 in the proof of Theorems 2.10 and 2.14. Finally, in the proof of Theorems 2.26, 2.27 and 2.31, we transfer the BSDE with an unbounded BMO terminal value into one with a bounded terminal value and a generator satisfying the assumptions of Theorems 2.6, 2.10 and 2.14. We note that for the unbounded global solution of a system of quadratic BSDEs, in [17, Theorem 2.8], the terminal value ξ can have only exponential moments of arbitrary order, while the generator g is forced to satisfy some stronger growth and continuity than Assumptions (D1) and (D2).

The rest of the paper is organized as follows. In Section 2, we introduce notations, state the main results, and present interesting examples, some of them arising from practical applications, along with remarks to illustrate the novelty of our results. The proof is given if it is not lengthy, and only outlined otherwise. In Section 3, we prove the local bounded solutions (Theorem 2.3), and in Section 4, we prove the global bounded solutions (Theorems 2.10 and 2.14). The global unbounded solution (Theorem 2.27) is proved finally in Section 5. In the Appendix, we give as auxiliary results bounded solutions of scalar-valued quadratic BSDEs with unbounded stochastic parameters.

2 Notations and statement of main results

2.1 Notations

First, let us fix a terminal time $T \in (0, \infty)$ and two positive integers n and d . Let $a \wedge b := \min\{a, b\}$, $a \vee b := \max\{a, b\}$, $a^+ := a \vee 0$, $a^- := -(a \wedge 0)$ and

$$\sum_{j=1}^0 b^j = \sum_{j=n+1}^n b^j = 0$$

for any real b^j . And, denote by $|\cdot|$ the Euclidean norm, $z^\top$ the transpose of vector (or matrix) z , $\mathbf{1}_A$ the indicator of set A , and $\text{sgn}(x) := \mathbf{1}_{x>0} - \mathbf{1}_{x\leq 0}$.

Throughout the paper, all processes are $(\mathcal{F}_t)_{t \in [0, T]}$ -progressively measurable, and all equalities and inequalities between random variables and processes are understood in the senses of $\mathbb{P}$ -a.s. and $d\mathbb{P} \times dt$ -a.e., respectively. We introduce the following spaces of random variables and processes.

- $L^\infty(\mathbb{R}^n)$: all $\mathbb{R}^n$ -valued and $\mathcal{F}_T$ -measurable random variables ξ satisfying

$$\|\xi\|_\infty := \text{ess sup}_{\omega \in \Omega} |\xi(\omega)| < +\infty.$$

- $\mathcal{S}^p(\mathbb{R}^n)$ for $p \geq 1$: all $\mathbb{R}^n$ -valued continuous adapted processes $(Y_t)_{t \in [0, T]}$ such that

$$\|Y\|_{\mathcal{S}^p} := \mathbb{E} \left[\sup_{t \in [0, T]} |Y_t|^p \right]^{\frac{1}{p}} < +\infty.$$

- $\mathcal{S}^\infty(\mathbb{R}^n)$: all $Y \in \bigcap_{p \geq 1} \mathcal{S}^p(\mathbb{R}^n)$ such that

$$\|Y\|_{\mathcal{S}^\infty} := \operatorname{ess\,sup}_{(\omega, t) \in \Omega \times [0, T]} |Y_t(\omega)| = \left\| \sup_{t \in [0, T]} |Y_t| \right\|_\infty < +\infty.$$

- $\mathcal{M}^\infty$: all real-valued non-negative progressively measurable process $(Y_t)_{t \in [0, T]}$ satisfying

$$\|Y\|_{\mathcal{M}^\infty} := \sup_{\tau \in \mathcal{T}_{[0, T]}} \left\| \mathbb{E}_\tau \left[\int_\tau^T Y_s ds \right] \right\|_\infty < +\infty,$$

here and hereafter $\mathcal{T}_{[a, b]}$ denotes the set of all $(\mathcal{F}_t)$ -stopping times τ valued in $[a, b] \subset [0, T]$, and $\mathbb{E}_\tau$ stands for the conditional expectation with respect to $\mathcal{F}_\tau$.

- $\mathcal{L}^\infty$: all $Y \in \mathcal{M}^\infty$ satisfying

$$\|Y\|_{\mathcal{L}^\infty} := \left\| \int_0^T Y_s ds \right\|_\infty < +\infty.$$

- $\mathcal{H}^p(\mathbb{R}^{n \times d})$ for $p \geq 1$: all $\mathbb{R}^{n \times d}$ -valued progressively measurable processes $(Z_t)_{t \in [0, T]}$ such that

$$\|Z\|_{\mathcal{H}^p} := \mathbb{E} \left[\left(\int_0^T |Z_s|^2 ds \right)^{\frac{p}{2}} \right]^{\frac{1}{p}} < +\infty.$$

- $\text{BMO}(\mathbb{R}^{n \times d})$: all $Z \in \mathcal{H}^2(\mathbb{R}^{n \times d})$ such that

$$\|Z\|_{\text{BMO}} := \sup_{\tau \in \mathcal{T}_{[0, T]}} \left\| \mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \right\|_\infty^{\frac{1}{2}} < +\infty.$$

We write $L^\infty(\mathbb{R}) := L^\infty(\mathbb{R}^1)$, $\mathcal{S}^\infty(\mathbb{R}) := \mathcal{S}^\infty(\mathbb{R}^1)$, $\mathcal{S}^p(\mathbb{R}) := \mathcal{S}^p(\mathbb{R}^1)$ for $p \geq 1$, $\mathcal{H}^p(\mathbb{R}) := \mathcal{H}^p(\mathbb{R}^{1 \times 1})$ for $p \geq 1$, and $\text{BMO}(\mathbb{R}) := \text{BMO}(\mathbb{R}^{1 \times 1})$. By $\mathcal{H}_{[a, b]}$, we denote the restriction to the subinterval $[a, b]$ of the space $\mathcal{H}$ of processes on $[0, T]$, i.e., the space of all processes which are restrictions to the interval $[a, b]$ of processes in $\mathcal{H}$. It is noted that the process $(\int_0^t Z_s dB_s)_{t \in [0, T]}$ is an n -dimensional BMO martingale for each $Z \in \text{BMO}(\mathbb{R}^{n \times d})$. We refer to Kazamaki [29] for more details on the BMO theory. For the reader's convenience, here we would like to recall the well-known John-Nirenberg inequality (see for example Lemma A.1 in Hu and Tang [24]): Let the process Z belong to the space of $\text{BMO}(\mathbb{R})$. If $\|Z\|_{\text{BMO}}^2 < 1$, then for any $\tau \in \mathcal{T}_{[0, T]}$,

$$\mathbb{E}_\tau \left[\exp \left(\int_\tau^T |Z_s|^2 ds \right) \right] \leq \frac{1}{1 - \|Z\|_{\text{BMO}}^2}.$$

Throughout the paper, we also always fix four real-valued non-negative progressively measurable processes

$$(v_t)_{t \in [0, T]} \in \text{BMO}(\mathbb{R}), \quad (\tilde{\alpha}_t)_{t \in [0, T]} \in \mathcal{L}^\infty, \quad (\bar{\alpha}_t)_{t \in [0, T]} \in \mathcal{M}^\infty$$

and $(\alpha_t)_{t \in [0, T]}$ having a certain integrability to be specified as well as a deterministic nondecreasing continuous function $\phi(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ and several real constants $\beta, \lambda, \bar{\lambda}, \theta, c, \bar{c} \geq 0$, $0 < \bar{\gamma} \leq \gamma$ and $\delta \in [0, 1)$.

Finally, for $i = 1, \dots, n$, denote by (z^i, y^i, g^i) the i th row of $(z, y, g) \in \mathbb{R}^{n \times d} \times \mathbb{R}^n \times \mathbb{R}^n$. In addition, for some real $r > 0$ and some time interval $[a, b] \subset [0, T]$, we let $\mathcal{E}_{[a, b]}^\infty(r)$ represent the set of all real-valued non-negative progressively measurable process $(\hat{\alpha}_t)_{t \in [0, T]}$ such that

$$\|\hat{\alpha}\|_{\mathcal{E}_{[a, b]}^\infty(r)} := \frac{1}{r} \ln \left(\sup_{\tau \in \mathcal{T}_{[a, b]}} \left\| \mathbb{E}_\tau \left[\exp \left(r \int_\tau^b \hat{\alpha}_s ds \right) \right] \right\|_\infty \right) < +\infty.$$

We also denote $\mathcal{E}_{[0, T]}^\infty(r)$ by $\mathcal{E}^\infty(r)$ simply.

Remark 2.1. Let $[a, b] \subset [0, T]$. By virtue of Jensen's inequality, the John-Nirenberg inequality and Hölder's inequality, we have the following assertions.

(i) The assertion $\hat{\alpha} \in \text{BMO}(\mathbb{R})$ holds if and only if $\hat{\alpha}^2 \in \mathcal{M}^\infty$, and $\|\hat{\alpha}^2\|_{\mathcal{M}^\infty} = \|\hat{\alpha}\|_{\text{BMO}}^2$.

If $\hat{\alpha} \in \mathcal{L}_{[a, b]}^\infty$, then $\hat{\alpha} \in \mathcal{E}_{[a, b]}^\infty(r)$ for each $r > 0$, and $\|\hat{\alpha}\|_{\mathcal{E}_{[a, b]}^\infty(r)} \leq \|\hat{\alpha}\|_{\mathcal{L}_{[a, b]}^\infty}$.

If $\hat{\alpha} \in \mathcal{E}_{[a, b]}^\infty(r)$ for some $r > 0$, then $\hat{\alpha} \in \mathcal{M}_{[a, b]}^\infty$, and $\|\hat{\alpha}\|_{\mathcal{M}_{[a, b]}^\infty} \leq \|\hat{\alpha}\|_{\mathcal{E}_{[a, b]}^\infty(r)}$.

If $\hat{\alpha} \in \mathcal{M}_{[a, b]}^\infty$, then $\hat{\alpha} \in \mathcal{E}_{[a, b]}^\infty(r)$ for each $0 < r < r_0 := \|\hat{\alpha}\|_{\mathcal{M}_{[a, b]}^\infty}$.

In addition, for any $0 < r < \bar{r}$, we have $\mathcal{E}^\infty(r) \subset \mathcal{E}_{[a, b]}^\infty(r)$ and $\mathcal{E}_{[a, b]}^\infty(\bar{r}) \subset \mathcal{E}_{[a, b]}^\infty(r)$.

(ii) If $\hat{\alpha}^1 \in \mathcal{E}_{[a, b]}^\infty(r)$ for some $r > 0$ and $\hat{\alpha}^2 \in \mathcal{L}_{[a, b]}^\infty$, then $\hat{\alpha} := \hat{\alpha}^1 + \hat{\alpha}^2 \in \mathcal{E}_{[a, b]}^\infty(r)$. Moreover,

$$\|\hat{\alpha}\|_{\mathcal{E}_{[a, b]}^\infty(r)} \leq \|\hat{\alpha}^1\|_{\mathcal{E}_{[a, b]}^\infty(r)} + \|\hat{\alpha}^2\|_{\mathcal{L}_{[a, b]}^\infty}.$$

(iii) If $\hat{\alpha}^1 \in \mathcal{E}_{[a, b]}^\infty(pr)$ and $\hat{\alpha}^2 \in \mathcal{E}_{[a, b]}^\infty(qr)$ for some $r > 0$ and $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$, then $\hat{\alpha} := \hat{\alpha}^1 + \hat{\alpha}^2 \in \mathcal{E}_{[a, b]}^\infty(r)$. Moreover, we have

$$\|\hat{\alpha}\|_{\mathcal{E}_{[a, b]}^\infty(r)} \leq \|\hat{\alpha}^1\|_{\mathcal{E}_{[a, b]}^\infty(pr)} + \|\hat{\alpha}^2\|_{\mathcal{E}_{[a, b]}^\infty(qr)}.$$

Let the $\mathbb{R}^n$ -valued function $g(\cdot, \cdot, y, z) : \Omega \times [0, T] \rightarrow \mathbb{R}^n$ be $(\mathcal{F}_t)$ -progressively measurable for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, and consider the following multidimensional BSDE:

$$Y_t = \xi + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad t \in [0, T], \quad (2.1)$$

or, equivalently,

$$Y_t^i = \xi^i + \int_t^T g^i(s, Y_s, Z_s) ds - \int_t^T Z_s^i dB_s, \quad t \in [0, T], \quad i = 1, \dots, n, \quad (2.2)$$

where ξ is an $\mathcal{F}_T$ -measurable $\mathbb{R}^n$ -valued random vector, and the solution (Y, Z) is a pair of $(\mathcal{F}_t)$ -progressively measurable processes, taking values in $\mathbb{R}^n \times \mathbb{R}^{n \times d}$.

2.2 Local bounded solution of multi-dimensional quadratic BSDEs

We first introduce the following two assumptions on the quadratic generator g .

(B1) For each fixed $i = 1, \dots, n$, any one of the following three conditions holds:

(i) $f := g^i$ or $f := -g^i$ satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{aligned} & \frac{\bar{\gamma}}{2} |z^i|^2 - \bar{\alpha}_t(\omega) - \phi(|y|) - \sum_{j=i+1}^n (\bar{\lambda} |z^j|^{1+\delta} + \theta |z^j|^2) - \bar{c} \sum_{j=1}^{i-1} |z^j|^2 \leq f(\omega, t, y, z) \\ & \leq \alpha_t(\omega) + \phi(|y|) + \sum_{j \neq i} (\lambda |z^j|^{1+\delta} + \theta |z^j|^2) + \frac{\gamma}{2} |z^i|^2; \end{aligned}$$

(ii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$|g^i(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \phi(|y|) + \frac{\gamma}{2}|z^i|^2 + \left\{ v_t(\omega) + \phi(|y|) + c \sum_{j=1}^{i-1} |z^j| + \sum_{j=i+1}^n (\lambda |z^j|^\delta + \theta |z^j|) \right\} |z^i|;$$

(iii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$|g^i(\omega, t, y, z)| \leq \bar{\alpha}_t(\omega) + \phi(|y|) + \bar{\lambda}|z^i| + \sum_{j \neq i} (\bar{\lambda}|z^j|^\delta \mathbf{1}_{i+1 \leq j \leq n+1} + \theta |z^j|^2).$$

(B2) For $i = 1, \dots, n$, g^i satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, \bar{y}, z, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \times \mathbb{R}^{n \times d}$,

$$|g^i(\omega, t, y, z) - g^i(\omega, t, \bar{y}, \bar{z})| \leq \phi(|y| \vee |\bar{y}|) \left\{ ([v_t(\omega)]^{1+\delta} + |z|^{1+\delta} + |\bar{z}|^{1+\delta}) |y - \bar{y}| + (v_t(\omega) + |z| + |\bar{z}|) \sum_{j=1}^i |z^j - \bar{z}^j| + [([v_t(\omega)]^\delta + |z|^\delta + |\bar{z}|^\delta) + \theta(v_t(\omega) + |z| + |\bar{z}|)] \sum_{j=i+1}^n |z^j - \bar{z}^j| \right\}.$$

Remark 2.2. Concerning Assumptions (B1) and (B2), we have the following comments.

- (i) Both assumptions (B1) and (B2) imply that g^j admits a strictly quadratic, generally quadratic or linear growth in z^j . g^j can also have a general growth in y , a quadratic growth in $\sqrt{\theta}z^k$ for both $k \neq j$ and a small constant θ , and an interactively quadratic growth like the inner products $\langle z^j, z^k \rangle$ for $k < j$ and $\langle z^j, \theta z^k \rangle$ for both $k > j$ and a small constant θ .
- (ii) Assumptions (B1)(i) and (B2) are more general than those of Luo [37, Theorem 2.2], where for each $j = 1, \dots, n$, g^j has a strictly quadratic growth in z^j and a stronger growth and continuity in both unknown variables y and z .
- (iii) The following assumptions (B1') and (B2') are made in [17, Theorem 2.1]:

(B1') For each $i = 1, \dots, n$, g^i satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$|g^i(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \phi(|y|) + \frac{\gamma}{2}|z^i|^2 + \lambda \sum_{j \neq i} |z^j|^{1+\delta};$$

(B2') For $i = 1, \dots, n$, g^i satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, \bar{y}, z, \bar{z}) \in \mathbb{R}^{2n} \times \mathbb{R}^{2(n \times d)}$,

$$|g^i(\omega, t, y, z) - g^i(\omega, t, \bar{y}, \bar{z})| \leq \phi(|y| \vee |\bar{y}|) \left[(1 + |z| + |\bar{z}|) (|y - \bar{y}| + |z^i - \bar{z}^i|) + (1 + |z|^\delta + |\bar{z}|^\delta) \sum_{j \neq i} |z^j - \bar{z}^j| \right].$$

It is easy to check that Assumption (B2) is strictly weaker than Assumption (B2'), while both assumptions (B1) and (B1') do not cover each other.

- (iv) In Assumption (B1), the set of integers $\{1, \dots, n\}$ is the sum of two disjoint sets I_1 and I_2 , either of which might be $\emptyset$, g^j satisfies any one of the three assertions (i)-(iii) for $j \in I_1$, and $-g^j$ satisfies assertion (i) for $j \in I_2$. Now, we define, for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\bar{g}^j(\omega, t, y, z) := \begin{cases} g^j(\omega, t, \bar{y}, \bar{z}), & j \in I_1; \\ -g^j(\omega, t, \bar{y}, \bar{z}), & j \in I_2 \end{cases}$$

with

$$\bar{y}^j := \begin{cases} y^j, & j \in I_1; \\ -y^j, & j \in I_2 \end{cases} \quad \text{and} \quad \bar{z}^j := \begin{cases} z^j, & j \in I_1; \\ -z^j, & j \in I_2. \end{cases}$$

It is not hard to verify that $\bar{g}$ satisfies (B1). Therefore, in Assumption (B1) we can assume, without loss of generality, that g^j satisfies any one of the three assertions (i)-(iii) for all $j = 1, \dots, n$. Furthermore, it is clear that $\bar{g}$ satisfies Assumption (B2) as g does.

We have the following general existence and uniqueness result on the local bounded solution of systems of BSDEs with interactively quadratic generators upon which more global arguments will rely.

Theorem 2.3. Let $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some $p > 1$ and the generator g satisfy Assumptions (B1) and (B2). Then, there exist two constants $\varepsilon_0 > 0$ and $\theta_0 > 0$ depending only on $(\|\xi\|_\infty, \|\alpha\|_{\mathcal{E}^\infty(p\gamma)}, \|\bar{\alpha}\|_{\mathcal{M}^\infty}, \|\bar{\alpha}\|_{\mathcal{L}^\infty}, \|v\|_{\text{BMO}}, n, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p)$ and $\phi(\cdot)$ together with a bounded subset $\mathcal{B}_{\varepsilon_0} \subset \mathcal{S}_{[T-\varepsilon_0, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-\varepsilon_0, T]}(\mathbb{R}^{n \times d})$ such that when $\theta \in [0, \theta_0]$, BSDE (2.1) has a unique local solution (Y, Z) on the time interval $[T - \varepsilon_0, T]$ with $(Y, Z) \in \mathcal{B}_{\varepsilon_0}$. Moreover, the above conclusion holds still for $p = 1$ when $\lambda = 0$ and $\theta_0 = 0$.

Note that if some parameters stated in Theorem 2.3 are not involved in assumptions of the generator g , then they can be arbitrarily designed within their predetermined bounds. We will always adopt this convention in the whole paper.

Sketch of the proof of Theorem 2.3. Let $K > 0$ be a carefully designed constant depending only on those parameters stated in Theorem 2.3, and $\varepsilon, \theta > 0$ be two sufficiently small constants relevant to K . Consider a pair of processes

$$(U, V) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{n \times d})$$

such that

$$\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K.$$

We will first prove that under the assumptions of Theorem 2.3, for each $i = 1, \dots, n$, the following scalar-valued BSDE

$$Y_t^i = \xi^i + \int_t^T f^i(s, Z_s^i) ds - \int_t^T Z_s^i dB_s, \quad t \in [T - \varepsilon, T] \quad (2.3)$$

admits, successively, a unique solution (Y^i, Z^i) in $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, where for each $(\omega, t, z) \in \Omega \times [T - \varepsilon, T] \times \mathbb{R}^{1 \times d}$,

$$\begin{cases} f^1(t, z) := g^1 \left(t, U_t, \left(z^\top, (V_t^2)^\top, \dots, (V_t^n)^\top \right)^\top \right), \\ f^i(t, z) := g^i \left(t, U_t, \left((Z_t^1)^\top, \dots, (Z_t^{i-1})^\top, z^\top, (V_t^{i+1})^\top, \dots, (V_t^n)^\top \right)^\top \right), \\ \hspace{20em} i = 2, \dots, n-1, \\ f^n(t, z) := g^n \left(t, U_t, \left((Z_t^1)^\top, \dots, (Z_t^{n-1})^\top, z^\top \right)^\top \right). \end{cases}$$

Moreover, we have

$$\|Y\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|Z\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K.$$

These assertions can be verified by an induction combined with some new existence and uniqueness results on the bounded solution of scalar-valued BSDEs along with some new a priori estimates under some certain integrability and growth conditions. Note that the construction of the preceding generator f^i is inspired by Luo [37].

Now, define the following complete metric space

$$\mathcal{B}_\varepsilon := \left\{ (U, V) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{n \times d}) : \|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K \right\},$$

which is a closed convex subset in the Banach space $\mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$, equipped with the norm

$$\|(U, V)\|_{\mathcal{B}_\varepsilon} := \left(\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right)^{1/2}, \quad (U, V) \in \mathcal{B}_\varepsilon.$$

Based on the preceding assertions, we define the following transformation

$$\Gamma : (U, V) \in \mathcal{B}_\varepsilon \mapsto \Gamma(U, V) := (Y, Z) \in \mathcal{B}_\varepsilon,$$

where for each $i = 1, \dots, n$, (Y^i, Z^i) , as the i th row of (Y, Z) , is the unique solution of BSDE (2.3) in the space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$.

Finally, using Assumption (B2), Hölder's inequality, Girsanov's transform and the BMO martingale theory along with a delicate induction, we see that for sufficiently small constants $\varepsilon, \theta > 0$, the transformation Γ is a contraction in $\mathcal{B}_\varepsilon$. The detailed proof is postponed to Section 3. $\square$

Remark 2.4. In view of (ii) and (iii) of Remark 2.2, we see that Theorem 2.3 improves Luo [37, Theorem 2.2], and does not cover Fan et al. [17, Theorem 2.1] nor Hu and Tang [24, Theorem 2.2]. In addition, in view of Fan et al. [17, Remark 2.2], we see that Fan et al. [17, Theorem 2.1] extends Hu and Tang [24, Theorem 2.2]; both of them do not cover Theorem 2.3. Indeed, in the proof of [24, Theorem 2.2] and [17, Theorem 2.1], the processes $Z_t^1, \dots, Z_t^{i-1}$ in the generator f^i of BSDE (2.3) need be replaced respectively with $V_t^1, \dots, V_t^{i-1}$, which explains why they can not be covered by Theorem 2.3. Fortunately, those results on the global solution of [24, 17] are covered by Theorem 2.14, see Remark 2.15 for more details.

2.3 Global bounded solution of multi-dimensional quadratic BSDEs

In this subsection, we will present three existence and uniqueness results on the global bounded solution of multi-dimensional BSDE (2.1). For the first one, let us introduce the following a priori boundedness assumption (AB) on the generator g , which was used in Jackson [25] and Jackson and Žitković [26]. The interested reader is referred to Xing and Žitković [45] and Escariz et al. [14] for other versions of this assumption.

(AB) There exists a finite collection $\{a_k\} = (a_1, \dots, a_K)$ of vectors in $\mathbb{R}^n$ such that

- (i) the collection of $(a_1, \dots, a_K)$ positively span $\mathbb{R}^n$, i.e., for any $a \in \mathbb{R}^n$ there exist nonnegative constants $\lambda_1, \dots, \lambda_K$ such that $\lambda_1 a_1 + \dots + \lambda_K a_K = a$;
- (ii) Almost everywhere in $\Omega \times [0, T]$, for any $k = 1, \dots, K$ and any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have $a_k^\top g(\omega, t, y, z) \leq \tilde{\alpha}_t(\omega) + \gamma |a_k^\top z|^2$.

Following the proof of Jackson [25, Lemma 6.6] and Jackson and Žitković [26, Proposition 3.8], we have

Proposition 2.5. Let $\xi \in L^\infty(\mathbb{R}^n)$ and the generator g satisfy Assumption (AB). Assume that for some $h \in (0, T]$, BSDE (2.1) has a solution $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}^n) \times$

$\text{BMO}_{[T-h, T]}(\mathbb{R}^{n \times d})$ on the time interval $[T - h, T]$. Then, there exists a positive constant $\tilde{K} > 0$ depending only on $(\|\xi\|_\infty, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, \{a_k\}, \gamma)$ and being independent of h such that

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty} + \|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq \tilde{K}.$$

With the help of Theorem 2.3, we closely follow the proof of Cheridito and Nam [9, Theorem 4.1] to give the next Theorem 2.6, as our first result on the global bounded solution of multi-dimensional BSDE (2.1). All details are omitted here.

Theorem 2.6. Let $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some $p > 1$ and the generator g satisfy Assumptions (B1), (B2) and (AB). Then, there exists a positive constant $\theta_0 > 0$ depending only on $(\|\xi\|_\infty, \|\alpha\|_{\mathcal{E}^\infty(p\gamma)}, \|\tilde{\alpha}\|_{\mathcal{M}^\infty}, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, \|v\|_{\text{BMO}}, n, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p)$ and $\phi(\cdot)$ such that when $\theta \in [0, \theta_0]$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$. Moreover, the above assertion is still true for $p = 1$ when $\lambda = 0$ and $\theta_0 = 0$.

In view of (i) of Remark 2.1, we immediately have

Corollary 2.7. Let $\alpha, \bar{\alpha} \in \mathcal{L}^\infty$ and the generator g satisfy Assumptions (B1), (B2) and (AB) with $\theta = 0$. Then for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

Remark 2.8. We have the following comments.

- (i) Compared with Jackson and Žitković [26, Theorem 3.5], the Malliavin-regularity condition on the generator g is not required in Theorem 2.6 via a new proof.
- (ii) By Fan et al. [17, Theorem 2.1] and Proposition 2.5, following the proof of Cheridito and Nam [9, Theorem 4.1], we still have the assertion of Corollary 2.7 when (B1) and (B2) are replaced with (B1') and (B2') in (iii) of Remark 2.2.
- (iii) The (AB) condition of the generator g seems difficult to be verified in the multi-dimensional case. In particular, it seems to imply that the generator is bounded in the state variable y , which is a too strong restriction. Consequently, in the sequel we shall search for some better conditions on the generator g for a unique global solution of systems of BSDEs (2.1).

To obtain the second result on the global bounded solution of multi-dimensional BSDE (2.1), the assumption (B1) needs to be strengthened to the following assumption (C1a). Before that, we introduce the following notations. For each $M \in \mathbb{R}^n$, $x \in \mathbb{R}$ and $i = 1, \dots, n$, denote by $M(x; i)$ the vector in $\mathbb{R}^n$ whose i th component is x and whose j th component is M^j for $j \neq i$. And, for each $H \in \mathbb{R}^{n \times d}$, $w \in \mathbb{R}^{1 \times d}$ and $i = 1, \dots, n$, denote by $H(w; i)$ the matrix in $\mathbb{R}^{n \times d}$ whose i th row is w and whose j th row is H^j for $j \neq i$.

(C1a) For each fixed $i = 1, \dots, n$, any one of the following three conditions holds:

- (i) The random field $f := g^i$ or $f(\omega, t, y, z) := -g^i(\omega, t, y(-y^i; i), z(-z^i; i))$ satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{aligned} & \frac{\bar{\gamma}}{2}|z^i|^2 - \bar{\alpha}_t(\omega) - \beta|y| - \sum_{j=i+1}^n (\bar{\lambda}|z^j|^{1+\delta} + \theta|z^j|^2) - \bar{c} \sum_{j=1}^{i-1} |z^j|^2 \leq f(\omega, t, y, z) \\ & \leq \alpha_t(\omega) + [\beta|y| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + \sum_{j \neq i} (\lambda|z^j|^\delta \mathbf{1}_{y^i < 0} + \theta|z^j|^2) + \frac{\gamma}{2}|z^i|^2; \end{aligned}$$

- (ii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\begin{aligned} & -[\beta|y^i| \mathbf{1}_{y^i < 0} + \phi(|y|) \mathbf{1}_{y^i > 0}] - l_i(\omega, t, z) \leq g^i(\omega, t, y, z) \\ & \leq [\beta|y^i| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + l_i(\omega, t, z) \end{aligned}$$

with

$$l_i(\omega, t, z) := \tilde{\alpha}_t(\omega) + \frac{\gamma}{2}|z^i|^2 + \left\{ v_t(\omega) + c \sum_{j=1}^{i-1} |z^j| + \sum_{j=i+1}^n (\lambda |z^j|^\delta + \theta |z^j|) \right\} |z^i|;$$

(iii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\begin{aligned} -[\beta |y^i| \mathbf{1}_{y^i < 0} + \phi(|y|) \mathbf{1}_{y^i > 0}] - \bar{l}_i(\omega, t, z) &\leq g^i(\omega, t, y, z) \\ &\leq [\beta |y^i| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + \bar{l}_i(\omega, t, z) \end{aligned}$$

with

$$\bar{l}_i(\omega, t, z) := \bar{\alpha}_t(\omega) + \bar{\lambda}|z| + \theta \sum_{j \neq i} |z^j|^2.$$

Remark 2.9. Assumptions (C1a) and (B2) allow the generator $g(t, y, z)$ to have a general growth in y and a general quadratic growth in z . If any one of the scalar-valued generators g^j for $j = 1, \dots, n$ satisfies either (C1a)(i) or (C1a)(ii), then the generator g is diagonally quadratic in z when $\theta = 0$ and $c = \bar{c} = 0$, and is triangularly quadratic in z when $\theta = 0$ and $c > 0$ or $\bar{c} > 0$. In addition, similar to (iv) of Remark 2.2, in Assumption (C1a) we can assume without loss of generality that g^j satisfies any one of the three assertions (i)-(iii) for all $j = 1, \dots, n$.

Theorem 2.10. Let $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma \exp(\beta T))$ for some $p > 1$ and the generator g satisfy Assumptions (C1a) and (B2). Then, there exists a positive constant $\theta_0 > 0$ depending only on $(\|\xi\|_\infty, \|\alpha\|_{\mathcal{E}^\infty(p\gamma \exp(\beta T))}, \|\bar{\alpha}\|_{\mathcal{M}^\infty}, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, \|v\|_{\text{BMO}}, n, \beta, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p)$ such that when $\theta \in [0, \theta_0]$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$. Moreover, the above assertion is still true for $p = 1$ when $\lambda = 0$ and $\theta_0 = 0$.

Sketch of the proof. First, in the spirit of Proposition 2.5, we need to verify a uniform a priori estimate on global solution of BSDE (2.1) when the generator g satisfy Assumption (C1a). Roughly speaking, we can prove that if for some $h \in (0, T]$, BSDE (2.1) has a solution $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{n \times d})$ on the time interval $[T-h, T]$, then there exists two constants $\tilde{K}, \theta_0 > 0$ being independent of h such that for each $\theta \in [0, \theta_0]$, we have

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty} + \|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq \tilde{K}.$$

The proof involves some refined estimates on the bounded solutions of scalar-valued BSDEs with generators satisfying certain growth conditions along with some induction arguments. Then, with the help of Theorem 2.3 and the previous uniform a priori estimate, we can closely follow the proof of Theorem 4.1 in Cheridito and Nam [9] to give the desired assertion. The details are deferred to Section 4. $\square$

Note that Theorem 2.10 can be compared with Jamneshan et al. [27, Theorem 2.3 and Remarks 2.4-2.5] to observe the role of θ in Assumptions (C1a) and (B2) for existence and uniqueness of the bounded solution of multi-dimensional interacting quadratic BSDEs. Roughly speaking, the latter considers the following generator: for each $i = 1, \dots, n$, g^i is independent of y such that $d\mathbb{P} \times dt - a.e.$, for any $(z, \bar{z}) \in \mathbb{R}^{n \times d} \times \mathbb{R}^{n \times d}$, we have

$$|g^i(\omega, t, z) - g^i(\omega, t, \bar{z})| \leq \theta(1 + |z| + |\bar{z}|)|z - \bar{z}| + \gamma(1 + |z^i| + |\bar{z}^i|)|z^i - \bar{z}^i|$$

and

$$|g^i(\omega, t, z)| \leq c + \theta|z|^2 + \gamma|z^i|^2.$$

BSDE (2.1) is then shown to have a unique bounded global solution when $\xi \in L^\infty(\mathbb{R}^n)$ and the parameter θ is small enough.

By virtue of Theorem 2.10 and (i) of Remark 2.1, we immediately have

Corollary 2.11. Let $\alpha, \bar{\alpha} \in \mathcal{L}^\infty$ and the generator g satisfy Assumptions (C1a) and (B2) with $\theta = 0$. Then, for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

Remark 2.12. If g satisfies those assumptions in Luo [37, Theorem 2.3], then Assumption (C1a)(i) with $\lambda = \theta = 0$ is true for each g^i with $i = 1, \dots, n$, and Assumption (B2) with $\theta = 0$ is true. Consequently, Corollary 2.11 improves Luo [37, Theorem 2.3]. In particular, the interacting term $z^i(z^j)^\top$ is excluded to appear in g^i for $1 \leq j < i \leq n$ in Luo [37, Theorem 2.3], but included in Corollary 2.11, see the next three subsections for more details. We especially mention that multidimensional BSDEs with this type of interacting term arise from many problems, such as price impact models (see Kramkov and Pulido [34]), incomplete stochastic equilibrium (see Xing and Žitković [45], Kardaras et al. [28], Escariz et al. [14], and Weston [43]), and risk-sensitive nonzero-sum stochastic games (see Xing and Žitković [45], Jackson and Žitković [26], and Jackson [25]).

Finally, let us further demonstrate the third existence and uniqueness result on the global bounded solution. In stating it, the following assumption (C1b) on the generator will be used, which is strictly stronger than (B1) with $\theta = 0$. It should be noted that Assumptions (C1b) and (C1a) with $\theta = 0$ do not cover each other.

(C1b) The set of integers $\{1, \dots, n\}$ consists of three mutually disjoint subsets J_1, J_2 and J_3 , any one of which can vanish, such that $J_1 + J_2 + J_3 = \{1, \dots, n\}$. For each fixed $i \in \{1, \dots, n\}$, any one of the following three conditions holds:

(i) If $i \in J_1$, then $f := g^i$ or $f(\omega, t, y, z) := -g^i(\omega, t, y(-y^i; i), z(-z^i; i))$ satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{aligned} \frac{\gamma}{2}|z^i|^2 - \tilde{\alpha}_t(\omega) - \beta|y| - \lambda \sum_{j \in J_1} |z^j|^{1+\delta} &\leq f(\omega, t, y, z) \\ &\leq \tilde{\alpha}_t(\omega) + [\beta|y| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + \lambda \sum_{j \in J_1} |z^j|^{1+\delta} + \frac{\gamma}{2}|z^i|^2; \end{aligned}$$

(ii) If $i \in J_2$, then $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\begin{aligned} -[\beta|y| \mathbf{1}_{y^i < 0} + \phi(|y|) \mathbf{1}_{y^i > 0}] - \tilde{l}_i(\omega, t, y, z) &\leq g^i(\omega, t, y, z) \\ &\leq [\beta|y| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + \tilde{l}_i(\omega, t, y, z) \end{aligned}$$

with

$$\tilde{l}_i(\omega, t, y, z) := \tilde{\alpha}_t(\omega) + \left[v_t(\omega) + \phi(|y|) + c \sum_{j=1}^{i-1} |z^j| + \lambda \sum_{j=i+1}^n |z^j|^\delta \right] |z^i| + \frac{\gamma}{2}|z^i|^2;$$

(iii) If $i \in J_3$, then $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\begin{aligned} -[\beta|y| \mathbf{1}_{y^i < 0} + \phi(|y|) \mathbf{1}_{y^i > 0}] - \hat{l}_i(\omega, t, z) &\leq g^i(\omega, t, y, z) \\ &\leq [\beta|y| \mathbf{1}_{y^i > 0} + \phi(|y|) \mathbf{1}_{y^i < 0}] + \hat{l}_i(\omega, t, z) \end{aligned}$$

with

$$\hat{l}_i(\omega, t, z) := \tilde{\alpha}_t(\omega) + \lambda \sum_{j \in J_3} |z^j|.$$

Remark 2.13. Similar to (iv) of Remark 2.2, in Assumption (C1b) we can assume without loss of generality that g^i satisfies any one of the three assertions (C1b)(i), (C1b)(ii) and (C1b)(iii) for all $i = 1, \dots, n$. In addition, in Assumption (C1b)(i) it gives no essential difference to replace the sum $\sum_{j \in J_1} |z^j|^{1+\delta}$ with the sum $\sum_{j \in J_1, j \neq i} |z^j|^{1+\delta}$.

Theorem 2.14. Let the generator g satisfy Assumptions (C1b) and (B2) with $\theta = 0$. Then, for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ on the time interval $[0, T]$.

Outline of the proof. First of all, in the spirit of Proposition 2.5, we need to verify a uniform a priori estimate on global solution of BSDE (2.1) when the generator g satisfy Assumption (C1b). Roughly speaking, we can prove that if for some $h \in (0, T]$, BSDE (2.1) has a solution $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{n \times d})$ on the time interval $[T-h, T]$, then there exists a constant $\tilde{K} > 0$ being independent of h such that

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty} \leq \tilde{K}.$$

The proof involves some delicate estimates on the bounded solutions of scalar-valued BSDEs with generators satisfying certain growth conditions along with perfect applications of Hölder's inequality, Jensen's inequality and Gronwall's inequality. Then, with the help of Theorem 2.3 and the previous uniform a priori estimate, we can follow the proof of Theorem 4.1 in Cheridito and Nam [9] to prove the desired assertion. The detailed proof is deferred to Section 4. $\square$

Remark 2.15. We have the following comments.

- (i) Assumptions (C1b)(ii)-(iii) generalize Fan et al. [19, Assumption (H3.2)]. Consequently, Theorem 2.14 improves Fan et al. [19, Theorem 3.6], and then Fan et al. [17, Theorems 2.4] and Hu and Tang [24, Theorem 2.3].
- (ii) In addition to (B1') and (B2') in (ii) of Remark 2.2, Theorem 2.5 of Fan et al. [17] requires the following two assumptions:

(B3') For each $i = 1, \dots, n$, g^i satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g^i(\omega, t, y, z) \operatorname{sgn}(y^i) \leq \tilde{\alpha}_t(\omega) + \beta|y| + \lambda \sum_{j \neq i} |z^j|^{1+\delta} + \frac{\gamma}{2} |z^i|^2;$$

(B4') For $i = 1, \dots, n$, $f := g^i$ or $f := -g^i$ satisfies that $d\mathbb{P} \times dt - a.e.$,

$$f(\omega, t, y, z) \geq \frac{\gamma}{2} |z^i|^2 - \tilde{\alpha}_t(\omega) - \beta|y| - \lambda \sum_{j \neq i} |z^j|^{1+\delta}, \quad \forall (y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}.$$

Assumptions (B1')-(B4') coincides with (C1b) where both J_2 and J_3 vanish. As a consequence, Theorem 2.14 improves Fan et al. [17, Theorem 2.5].

- (iii) Let the assumptions of Theorem 2.14 be satisfied except that $\theta = 0$. It can also be proved that the assertion of Theorem 2.14 is still true when θ is smaller than a constant $\theta_0 > 0$ depending only on $\|\xi\|_\infty$ and those parameters in Assumption (C1b).
- (iv) Let $l \in (1, 2]$ and $\{(a_t, b_t)\}_{t \in [0, T]} \in \mathcal{S}^\infty(\mathbb{R}^n \times \mathbb{R}^n)$. For the following generator

$$g(\omega, t, y, z) := a_t(\omega)|y| + b_t(\omega) \sin(|z|^l), \quad \forall (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d},$$

we see from the last comment (iii) that for each $\xi \in L^\infty(\mathbb{R}^n)$, there is $\theta_0 > 0$ such that when $\|b\|_{\mathcal{S}^\infty} \mathbf{1}_{l=2} \leq \theta_0$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$, which seems to be new.

2.4 Illustrating examples

First of all, Jamneshan et al. [27] address the following system of two BSDEs:

$$\begin{cases} Y_t^1 = \xi^1 + \int_t^T (\theta_1 |Z_s^1|^2 + \vartheta_1 |Z_s^2|^2) ds - \int_t^T Z_s^1 dB_s, \\ Y_t^2 = \xi^2 + \int_t^T (\vartheta_2 |Z_s^1|^2 + \theta_2 |Z_s^2|^2) ds - \int_t^T Z_s^2 dB_s, \quad t \in [0, T], \end{cases} \quad (2.4)$$

where $\xi^i \in L^\infty(\mathbb{R})$ and $(\theta_i, \vartheta_i)^T \in \mathbb{R}^2$ for $i = 1, 2$. It is not hard to verify that the generator $g^1(z) := \theta_1 |z^1|^2 + \vartheta_1 |z^2|^2$ satisfies Assumption (C1a)(i) with $\gamma = \bar{\gamma} = 2\theta_1$ and $\theta = \vartheta_1$ when $\theta_1 \neq 0$ and Assumption (C1a)(iii) with $\theta = \vartheta_1$ when $\theta_1 = 0$, and that the generator $g^2(z) := \vartheta_2 |z^1|^2 + \theta_2 |z^2|^2$ satisfies Assumption (C1a)(i) with $\gamma = \bar{\gamma} = 2\theta_2$ and $\bar{c} = \theta = \vartheta_2$ when $\theta_2 \neq 0$ and Assumption (C1a)(iii) with $\theta = \vartheta_2$ when $\theta_2 = 0$. Then, according to Theorem 2.10, we see that for each $\xi \in L^\infty(\mathbb{R}^2)$, there exists a constant $\vartheta_0 > 0$ depending only on $(\theta_1, \theta_2, \|\xi\|_\infty)$ such that for each pair $(\vartheta_1, \vartheta_2)$ satisfying $|\vartheta_1|, |\vartheta_2| \leq \vartheta_0$, the system (2.4) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$ on $[0, T]$. We note that this conclusion has been obtained in [27, Theorem 2.3].

On the other hand, in the case of $\theta_1 = \vartheta_1 = 0$, $\vartheta_2 = 1$ and $\theta_2 = 1/2$, Frei and Dos Reis [21, Theorem 2.1] show that for some $\xi \in L^\infty(\mathbb{R}^2)$, the system (2.4) fails to have a global bounded solution on the time interval $[0, T]$. However, when $\vartheta_1 = 0$ and $\vartheta_2 \theta_2 < 0$ (for example, $\vartheta_2 = 1$ and $\theta_2 = -1/2$), Corollary 2.11 asserts that for each $\xi \in L^\infty(\mathbb{R}^2)$, the system (2.4) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$ on $[0, T]$.

Furthermore, we consider the following variant of the system of BSDEs (2.4):

$$\begin{cases} Y_t^1 = \xi^1 + \int_t^T (\theta_1 |Z_s^1|^2 + \vartheta_1 |Z_s^2|^2) ds - \int_t^T Z_s^1 dB_s, \\ Y_t^2 = \xi^2 + \int_t^T (\vartheta_2 |Z_s^1|^2 + \theta_2 |Z_s^2|^2 + l Z_s^1 (Z_s^2)^\top) ds - \int_t^T Z_s^2 dB_s, \quad t \in [0, T], \end{cases} \quad (2.5)$$

where $l \in \mathbb{R}$ and $\xi^i \in L^\infty(\mathbb{R})$ and $(\theta_i, \vartheta_i)^T \in \mathbb{R}^2$ for $i = 1, 2$ such that $\theta_2 \vartheta_2 < 0$. Without loss of generality, we assume that $\vartheta_2 < 0$ and $\theta_2 > 0$. Observe that

$$-\frac{l^2}{2\theta_2} |Z_s^1|^2 - \frac{\theta_2}{2} |Z_s^2|^2 \leq l Z_s^1 (Z_s^2)^\top \leq -\frac{\vartheta_2}{2} |Z_s^1|^2 - \frac{l^2}{2\vartheta_2} |Z_s^2|^2$$

and then

$$\frac{\theta_2}{2} |Z_s^2|^2 - \left(\frac{l^2}{2\theta_2} - \vartheta_2 \right) |Z_s^1|^2 \leq \vartheta_2 |Z_s^1|^2 + \theta_2 |Z_s^2|^2 + l Z_s^1 (Z_s^2)^\top \leq \left(\theta_2 - \frac{l^2}{2\vartheta_2} \right) |Z_s^2|^2. \quad (2.6)$$

We get that the generator $f := g^1$ satisfies (C1a)(i) for $\theta_1 \neq 0$ and (C1a)(iii) for $\theta_1 = 0$, and $f := g^2$ satisfies (C1a)(i). It follows from Corollary 2.11 that for each $\xi \in L^\infty(\mathbb{R}^2)$, the system of BSDEs (2.5) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$. In addition, it is clear that the above assertion is still true as $\vartheta_2 = 0$.

We proceed to consider the following three-dimensional BSDE, which is another variant of the system of BSDEs (2.4):

$$\begin{cases} Y_t^1 = \xi^1 + \int_t^T (\vartheta_1 |Z_s^1|^2 + \theta_1 |Z_s^3|^2) ds - \int_t^T Z_s^1 dB_s, \\ Y_t^2 = \xi^2 + \int_t^T (\vartheta_2 |Z_s^1|^2 + \theta_2 |Z_s^2|^2 + l_{21} Z_s^2 (Z_s^1)^\top + k_2 |Z_s^3|^2) ds - \int_t^T Z_s^2 dB_s, \\ Y_t^3 = \xi^3 + \int_t^T (\vartheta_3 |Z_s^1|^2 + \theta_3 |Z_s^2|^2 + \kappa_3 |Z_s^3|^2) ds - \int_t^T Z_s^3 dB_s \\ \quad + \int_t^T (l_{31} Z_s^3 (Z_s^1)^\top + l_{32} Z_s^3 (Z_s^2)^\top + l_{33} Z_s^1 (Z_s^2)^\top + k_3 |Z_s^2|^2) ds; \quad t \in [0, T], \end{cases} \quad (2.7)$$

where $\xi^i \in L^\infty(\mathbb{R})$, $(\theta_i, \vartheta_i)^\top \in \mathbb{R}^2$ for $i = 1, 2, 3$, and $(\kappa_3, l_{21}, l_{31}, l_{32}, l_{33}, k_2, k_3)^\top \in \mathbb{R}^7$ with

$$\theta_2 \vartheta_2 < 0, \quad \kappa_3 \theta_3 < 0, \quad \kappa_3 \vartheta_3 < 0 \quad \text{and} \quad l_{33}^2 < 4\theta_3 \vartheta_3.$$

Without loss of generality, we assume that $\theta_2 > 0$, $\vartheta_2 < 0$, $\kappa_3 > 0$, $\theta_3 < 0$ and $\vartheta_3 < 0$. Observe that there exists a unique real number $\varepsilon \in (0, 1]$ such that

$$l_{33}^2 = 4(1 - \varepsilon)^2 \theta_3 \vartheta_3$$

and

$$\vartheta_3(1 - \varepsilon)|Z_s^1|^2 + \theta_3(1 - \varepsilon)|Z_s^2|^2 + l_{33}Z_s^1(Z_s^2)^\top = -(1 - \varepsilon) \left| \sqrt{|\vartheta_3|}Z_s^1 - \text{sgn}(l_{33})\sqrt{|\theta_3|}Z_s^2 \right|^2.$$

Using a similar argument to (2.6), we directly verify that

$$\Delta_s \leq \left(\kappa_3 - \frac{l_{31}^2}{4\varepsilon\vartheta_3} - \frac{l_{32}^2}{4\varepsilon\theta_3} \right) |Z_s^3|^2$$

and

$$\Delta_s \geq \frac{\kappa_3}{2}|Z_s^3|^2 - \left(\frac{1}{2} + \frac{l_{31}^2}{\kappa_3} - \vartheta_3 \right) |Z_s^1|^2 - \left(\frac{l_{33}^2}{2} + \frac{l_{32}^2}{\kappa_3} - \theta_3 \right) |Z_s^2|^2,$$

where

$$\Delta_s := \vartheta_3|Z_s^1|^2 + \theta_3|Z_s^2|^2 + \kappa_3|Z_s^3|^2 + l_{31}Z_s^3(Z_s^1)^\top + l_{32}Z_s^3(Z_s^2)^\top + l_{33}Z_s^1(Z_s^2)^\top.$$

Consequently, the generator g of the system of BSDEs (2.7) satisfies Assumption (C1a). It follows from Corollary 2.11 that for each $\xi \in L^\infty(\mathbb{R}^2)$, the system (2.7) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$. In addition, it can also be proved that when $\vartheta_2 = k_2 = 0$, $\kappa_3 \theta_3 < 0$, $\kappa_3 \vartheta_3 < 0$, $l_{31} = l_{32} = 0$ and $l_{33}^2 = 4\theta_3 \vartheta_3$ holds or $\vartheta_2 = k_2 = 0$ and $\theta_3 = \vartheta_3 = l_{33} = k_3 = 0$ holds, the above assertion is still true.

Finally, let us present several specific examples of multi-dimensional solvable BSDEs with interactively quadratic generators, to which one of Theorem 2.10, Corollary 2.11 and Theorem 2.14 applies, but no existing results could do, to our best knowledge.

Example 2.16. We have the following assertions.

- (i) Define the following generator $g := (g^1, \dots, g^n)^\top$: for each $i = 1, \dots, n$, and $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g^i(\omega, t, y, z) := \tilde{\alpha}_t(\omega) + (-1)^i \left(\sum_{j=1}^{i-1} a_{ij} |z^j|^2 - a_{ii} |z^i|^2 + \sum_{j=i+1}^n a_{ij} |z^j|^{1+\delta} \right) + h^i(y, z),$$

where any element of matrix $A := (a_{ij})_{n \times n}$ is positive, and for each $i = 1, \dots, n$, $h^i(y, z) : \mathbb{R}^n \times \mathbb{R}^{n \times d} \rightarrow \mathbb{R}$ is any Lipschitz continuous function. It is not hard to verify that this generator g satisfies Assumption (B2) with $\theta = 0$, and for each $i = 1, \dots, n$, $f := g^i$ or $f(\omega, t, y, z) := -g^i(\omega, t, y(-y^i; i), z(-z^i; i))$ satisfies Assumption (C1a)(i) with $\theta = 0$. Then, from Corollary 2.11 we see that for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$. Note that in Assumption (C1a)(i), both terms $|z^i|^2$ and $|z^j|^2$ for $j < i$ share opposite signs.

- (ii) Let $n = d$ and the generator $g(y, z) := zy$ for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times n}$. Then g satisfies Assumption (B2) with $\theta = 0$, and for each $i = 1, \dots, n$, $g^i(y, z) = z^i y$ satisfies Assumption (C1b)(ii). From Theorem 2.14, we see that BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ for each $\xi \in L^\infty(\mathbb{R}^n)$. Note that when the terminal value is Markovian, this BSDE is associated to a Burger system. Therefore, it is reasonably called the *backward stochastic Burger system* or *equation*.

- (iii) Define the following generator $g := (g^1, \dots, g^n)^\top$: for each $i = 1, \dots, n$, and $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g^i(\omega, t, y, z) := \tilde{\alpha}_t(\omega) + \left(v_t(\omega) + e^{|y|}\right) |z^i| + e^{-y_i} |z^i|^{\frac{3}{2}} + z^i \sum_{j=1}^i c_{ij} (z^j)^\top,$$

where $C = (c_{ij})_{n \times n}$ is any real matrix. It is easy to verify that this generator g satisfies Assumption (B2) with $\theta = 0$, and for each $i = 1, \dots, n$, $f := g^i$ satisfies Assumption (C1b)(ii). Then, from Theorem 2.14, we see that BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ for each $\xi \in L^\infty(\mathbb{R}^n)$.

- (iv) Let $n = 5$ and $d = 2$. Define the following generator g : for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g(\omega, t, y, z) := \begin{pmatrix} g^1 \\ g^2 \\ g^3 \\ g^4 \\ g^5 \end{pmatrix} (\omega, t, y, z) := \begin{pmatrix} e^{-y^1} - |y| + |z^1|^2 - |z^2|^{\frac{4}{3}} + \sin |z^3| \\ |y| \cos |y| - |z^2|^2 + |z^1|^{\frac{5}{4}} - \cos |z^4| \\ |y| + z^3(2z^1 - 3z^2)^\top + z^3 A (z^3)^\top - \arcsin |z^5| \\ 2|y| \sin |y| + |z^4| - |z^5| + \arccos |z^1| \\ y^1 + 3y^3 - y^4 + y^5 - |z^4| + 2|z^5| - \arctan |z^2| \end{pmatrix}$$

with the matrix

$$A := \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

It is easy to check that this generator g satisfies (B2) with $\theta = 0$, and that $f := g^i$ satisfies (C1b)(i) for $i = 1, 2$, (C1b)(ii) for $i = 3$, and (C1b)(iii) for $i = 4, 5$. Then, from Theorem 2.14, we see that BSDE (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ for each $\xi \in L^\infty(\mathbb{R}^n)$.

2.5 Connections to existing results

With an invertible linear transformation as in Xing and Žitković [45] and Weston [43], we immediately have the following result after an application of Itô formula.

Proposition 2.17. Let $A \in \mathbb{R}^{n \times n}$ be a real invertible matrix, and A^{-1} its inverse. Then, $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ is a solution of BSDE (2.1) if and only if $(\bar{Y}, \bar{Z}) := (AY, AZ) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ is a solution of the following BSDE

$$\bar{Y}_t = \bar{\xi} + \int_t^T \bar{g}(s, \bar{Y}_s, \bar{Z}_s) ds - \int_t^T \bar{Z}_s dB_s, \quad t \in [0, T], \quad (2.8)$$

where $\bar{\xi} := A\xi$ and

$$\bar{g}(\omega, t, \bar{y}, \bar{z}) := Ag(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z}), \quad \forall (\omega, t, \bar{y}, \bar{z}) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}.$$

Since the condition (AB) is invariant under an invertible linear transformation of $\mathbb{R}^n$, as shown in Remark 2.12 of Xing and Žitković [45], we immediately have from Corollary 2.7 and Proposition 2.17 the following assertion.

Theorem 2.18. Let $\alpha, \bar{\alpha}, \tilde{\alpha} \in \mathcal{L}^\infty$ and the generator g satisfy (AB). If there is a real invertible matrix $A \in \mathbb{R}^{n \times n}$ such that the generator $\bar{g}$ defined in Proposition 2.17 satisfies (B1) and (B2) with $\theta = 0$, then for each $\xi \in L^\infty(\mathbb{R}^n)$, the system of BSDEs (2.1) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

From Proposition 2.17, we easily see the following slight extension of Theorem 2.1 of Frei and Dos Reis [21].

Theorem 2.19. Let $\theta_1 = \vartheta_1 = 0$ and $\theta_2 \vartheta_2 > 0$. Then, there is a terminal value $\xi \in L^\infty(\mathbb{R}^2)$ such that the system of BSDEs (2.4) has no global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$.

Proof. We use a contradiction argument. Suppose that for any $\xi \in L^\infty(\mathbb{R}^2)$, the system of BSDEs (2.4) admits a global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$. Define

$$(\bar{Y}, \bar{Z}) := \begin{pmatrix} \sqrt{2\theta_2 \vartheta_2} & 0 \\ 0 & 2\vartheta_2 \end{pmatrix} (Y, Z).$$

Then, we have from Proposition 2.17 that $(\bar{Y}, \bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$ solves the following system of BSDEs

$$\begin{cases} \bar{Y}_t^1 = \sqrt{2\theta_2 \vartheta_2} \xi^1 - \int_t^T \bar{Z}_s^1 dB_s, \\ \bar{Y}_t^2 = 2\vartheta_2 \xi^2 + \int_t^T \left(|\bar{Z}_s^1|^2 + \frac{1}{2} |\bar{Z}_s^2|^2 \right) ds - \int_t^T \bar{Z}_s^2 dB_s, \quad t \in [0, T], \end{cases}$$

which is a contradiction to Theorem 2.1 of Frei and Dos Reis [21] as ξ is arbitrary. The proof is then complete. $\square$

The following theorem is a nonlinear extension of Theorem 6.9 of Jackson [25] with a different proof.

Theorem 2.20. Define the generator g as follows: $\forall (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g(\omega, t, y, z) := f(\omega, t, y, z) + zh(b^\top z),$$

where $b := (b_1, \dots, b_n)^\top \in \mathbb{R}^n$ with $b_1 \neq 0$, the vector function $h : \mathbb{R}^{1 \times d} \rightarrow \mathbb{R}^d$ is Lipschitz continuous, and $f : \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^n$ has the following stronger continuity than Assumption (B2): $d\mathbb{P} \times dt - a.e.$, for each $(y, \bar{y}, z, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \times \mathbb{R}^{n \times d}$,

$$|f(\omega, t, y, z) - f(\omega, t, \bar{y}, \bar{z})| \leq \phi(|y| \vee |\bar{y}|) (v_t(\omega) + |z|^\delta + |\bar{z}|^\delta) (|y - \bar{y}| + |z - \bar{z}|), \quad (2.9)$$

and the following growth: $d\mathbb{P} \times dt - a.e.$,

$$|b^\top f(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \beta|y| + \frac{\gamma}{2} |b^\top z|^2, \quad \forall (y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d} \quad (2.10)$$

and

$$|f^i(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \beta|y| + \frac{\gamma}{2} |z^i|^2, \quad i = 2, \dots, n. \quad (2.11)$$

Then, for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) with the above generator g admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

Proof. The following matrix

$$A := \begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \quad (2.12)$$

is invertible. For each $(\bar{y}, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, let $(y, z) := A^{-1}(\bar{y}, \bar{z})$, and then $(\bar{y}, \bar{z}) = A(y, z)$. Clearly, $\bar{y}^1 = b^\top y$, $\bar{z}^1 = b^\top z$, $\bar{y}^i = y^i$ and $\bar{z}^i = z^i$ for each $i = 2, \dots, n$. By (2.9)-(2.11) together with the Lipschitz continuity of h , the following generator

$$\bar{g}(\omega, t, \bar{y}, \bar{z}) := Ag(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z}) = Af(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z}) + \begin{pmatrix} \bar{z}^1 h(\bar{z}^1) \\ \bar{z}^2 h(\bar{z}^1) \\ \vdots \\ \bar{z}^n h(\bar{z}^1) \end{pmatrix}$$

satisfies Assumption (B2) with $\theta = 0$, and Assumption (C1b)(ii). From Theorem 2.14 we see that for each $\xi \in L^\infty(\mathbb{R}^n)$, the system of BSDEs (2.8) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$. Finally, by Proposition 2.17 we know that $(A^{-1}\bar{Y}, A^{-1}\bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ is just the desired unique solution of (2.1). The proof is complete. $\square$

Remark 2.21. If the generator f is Lipschitz continuous in the last two variables (y, z) and bounded in the last variable z , then (2.9)-(2.11) hold naturally. On the other hand, if $b_1 = 0$ but $b_{i_0} \neq 0$ for some $i_0 > 1$, Theorem 2.20 is still true when (2.11) is satisfied for all integer $i \neq i_0$ instead of for all $i \geq 2$, just using an obvious invertible transformation in the proof.

Similar to the proof of Theorem 2.20, we have the following Theorem 2.22.

Theorem 2.22. Consider the following generator g : $\forall (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$g(\omega, t, y, z) := f(\omega, t, y, z) + zh(b^\top z) + a\bar{h}(b^\top z),$$

where $a := (a_1, \dots, a_n)^\top \in \mathbb{R}^n$ and $b := (b_1, \dots, b_n)^\top \in \mathbb{R}^n$ with $a_1 \neq 0$ and $b^\top a \neq 0$, the vector function $h : \mathbb{R}^{1 \times d} \rightarrow \mathbb{R}^d$ is Lipschitz continuous, the function $\bar{h} : \mathbb{R}^{1 \times d} \rightarrow \mathbb{R}$ satisfies that $\bar{h}(0) = 0$ and

$$|\bar{h}(w_1) - \bar{h}(w_2)| \leq \gamma(1 + |w_1| + |w_2|)|w_1 - w_2|, \quad \forall w_1, w_2 \in \mathbb{R}^{1 \times d}, \quad (2.13)$$

and $f : \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^n$ has the continuity (2.9) and the following growth: $dP \times dt - a.e.$, for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$|f(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \beta|y|. \quad (2.14)$$

Then, for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) with the above generator g admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

It is quite related to Theorem 6.19 of Jackson [25], which requires the a priori boundedness condition (AB) and the Malliavin regular condition on the generator g . However, both do not cover each other.

Proof. The following matrix

$$A := \begin{pmatrix} b_1 & b_2 & b_3 & \cdots & b_n \\ -a_2 & a_1 & & & \\ -a_3 & & a_1 & & \\ \vdots & & & \ddots & \\ -a_n & & & & a_1 \end{pmatrix}$$

has the determinant $\det(A) = a_1^{n-2}b^\top a \neq 0$, and is thus invertible. For each $(\bar{y}, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, let $(y, z) := A^{-1}(\bar{y}, \bar{z})$, and then $(\bar{y}, \bar{z}) = A(y, z)$. Clearly, $\bar{y}^1 = b^\top y$, $\bar{z}^1 = b^\top z$, $\bar{y}^i = -a_i y^1 + a_1 y^i$ and $\bar{z}^i = -a_i z^1 + a_1 z^i$ for each $i = 2, \dots, n$. From (2.9) and (2.13)-(2.14) together with the Lipschitz continuity of h , we see that the following generator

$$\bar{g}(\omega, t, \bar{y}, \bar{z}) := Af(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z}) + \begin{pmatrix} \bar{z}^1 h(\bar{z}^1) + b^\top a \bar{h}(\bar{z}^1) \\ \bar{z}^2 h(\bar{z}^1) \\ \vdots \\ \bar{z}^n h(\bar{z}^1) \end{pmatrix}$$

satisfies Assumption (B2) with $\theta = 0$, and Assumption (C1b)(ii). Proceeding identically as in Theorem 2.20, we have the desired assertion. $\square$

Furthermore, we have the following theorem, which and Theorem 3.1 of Xing and Žitković [45] do not cover each other.

Theorem 2.23. Define the generator $g := (g^1, \dots, g^n)^\top$ as follows: for each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{cases} g^1(\omega, t, y, z) := f^1(\omega, t, y, z) + z^1 h(b^\top z) - \bar{h}_1(b^\top z) - \frac{1}{b_1} \sum_{j=2}^n a_j b_j |z^j|^2; \\ g^i(\omega, t, y, z) := f^i(\omega, t, y, z) + z^i h(b^\top z) - \bar{h}_i(b^\top z) + a_i |z^i|^2, \quad i = 2, \dots, n. \end{cases} \quad (2.15)$$

Here, $a := (0, a_2, \dots, a_n)^\top \in \mathbb{R}^n$, $b := (b_1, \dots, b_n)^\top \in \mathbb{R}^n$ with $b_1 \neq 0$, the vector function $h : \mathbb{R}^{1 \times d} \rightarrow \mathbb{R}^d$ is Lipschitz continuous with Lipschitz constant $L > 0$ with $|h(0)| \leq L$, the vector function $\bar{h} = (\bar{h}_1, \dots, \bar{h}_n)^\top : \mathbb{R}^{1 \times d} \rightarrow \mathbb{R}^n$ satisfies that $\bar{h}(0) = 0$ and

$$|\bar{h}(w_1) - \bar{h}(w_2)| \leq L(1 + |w_1| + |w_2|)|w_1 - w_2|, \quad \forall w_1, w_2 \in \mathbb{R}^{1 \times d}, \quad (2.16)$$

and $f = (f^1, \dots, f^n)^\top : \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^n$ satisfies that $dP \times dt - a.e.$, $|f(\omega, t, 0, 0)| \leq \tilde{\alpha}_t(\omega)$ and for each $(y_1, y_2, z_1, z_2) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \times \mathbb{R}^{n \times d}$,

$$|f(\omega, t, y_1, z_1) - f(\omega, t, y_2, z_2)| \leq \beta|y_1 - y_2| + \gamma|z_1 - z_2|. \quad (2.17)$$

Assume further that either

$$\begin{aligned} \inf_{w \in \mathbb{R}^{1 \times d}} \left[wh(w) - b^\top \bar{h}(w) - \frac{\bar{\gamma}}{2}|w|^2 \right] &\geq -c \quad \text{or} \\ \sup_{w \in \mathbb{R}^{1 \times d}} \left[wh(w) - b^\top \bar{h}(w) + \frac{\bar{\gamma}}{2}|w|^2 \right] &\leq c, \end{aligned} \quad (2.18)$$

and for each $i = 2, \dots, n$, either

$$\begin{aligned} a_i > 0 \quad \text{and} \quad \inf_{w \in \mathbb{R}^{1 \times d}} \left[\bar{h}_i(w) - \frac{\bar{\gamma}}{2}|w|^2 \right] &\geq -c \quad \text{or} \\ a_i < 0 \quad \text{and} \quad \sup_{w \in \mathbb{R}^{1 \times d}} \left[\bar{h}_i(w) + \frac{\bar{\gamma}}{2}|w|^2 \right] &\leq c. \end{aligned} \quad (2.19)$$

Then, for each $\xi \in L^\infty(\mathbb{R}^n)$, BSDE (2.1) with the above generator g admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

Proof. We only give the proof when the first conditions of (2.18) and (2.19) are satisfied. The other cases are proved identically.

Let the invertible matrix A be defined in (2.12). For each $(\bar{y}, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, let $(y, z) := A^{-1}(\bar{y}, \bar{z})$, and then $(\bar{y}, \bar{z}) = A(y, z)$. i.e., $\bar{y}^1 = b^\top y$, $\bar{z}^1 = b^\top z$, $\bar{y}^i = y^i$ and $\bar{z}^i = z^i$ for each $i = 2, \dots, n$. From (2.16), (2.17) and the Lipschitz continuity of h , the following generator

$$\bar{g}(\omega, t, \bar{y}, \bar{z}) := Af(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z}) + \begin{pmatrix} \bar{z}^1 h(\bar{z}^1) - b^\top \bar{h}(\bar{z}^1) \\ \bar{z}^2 h(\bar{z}^1) - \bar{h}_i(\bar{z}^1) + a_i |\bar{z}^2|^2 \\ \vdots \\ \bar{z}^n h(\bar{z}^1) - \bar{h}_n(\bar{z}^1) + a_n |\bar{z}^n|^2 \end{pmatrix} \quad (2.20)$$

satisfies Assumption (B2) with $\theta = 0$. Furthermore, we have from (2.17) that

$$|Af(\omega, t, A^{-1}\bar{y}, A^{-1}\bar{z})| \leq |A|\alpha_t(\omega) + \beta|\bar{y}| + \gamma|\bar{z}|. \quad (2.21)$$

From (2.16), (2.18), (2.20) and (2.21) together with the Lipschitz continuity of h , we see that $\bar{g}^1$ satisfies Assumption (C1a)(i) with $\theta = 0$ since

$$\bar{g}^1(\omega, t, \bar{y}, \bar{z}) \geq \frac{\bar{\gamma}}{2}|\bar{z}^1|^2 - c - |A|\alpha_t(\omega) - \beta|\bar{y}| - \gamma|\bar{z}|$$

and

$$\bar{g}^1(\omega, t, \bar{y}, \bar{z}) \leq |A|\alpha_t(\omega) + \beta|\bar{y}| + \gamma|\bar{z}| + L|\bar{z}^1|(1 + |\bar{z}^1|) + L|b|(1 + |\bar{z}^1|)|\bar{z}^1|.$$

From (2.16), (2.19), (2.20) and (2.21) together with the Lipschitz continuity of h and Young's inequality, we see that for each $i = 2, \dots, n$, $\bar{g}^i$ satisfies (C1a)(i) with $\theta = 0$ since

$$\begin{aligned} \bar{g}^i(\omega, t, \bar{y}, \bar{z}) &\geq a_i|\bar{z}^i|^2 - L|\bar{z}^i|(1 + |\bar{z}^1|) - L(1 + |\bar{z}^1|)|\bar{z}^1| - |A|\alpha_t(\omega) - \beta|\bar{y}| - \gamma|\bar{z}| \\ &\geq \frac{a_i}{2}|\bar{z}^i|^2 - \left(\frac{L^2}{2a_i} + L\right)|\bar{z}^1|^2 - L|\bar{z}^i| - L|\bar{z}^1| - |A|\alpha_t(\omega) - \beta|\bar{y}| - \gamma|\bar{z}| \end{aligned}$$

and

$$\begin{aligned} \bar{g}^i(\omega, t, \bar{y}, \bar{z}) &\leq a_i|\bar{z}^i|^2 + L|\bar{z}^i|(1 + |\bar{z}^1|) - \frac{\bar{\gamma}}{2}|\bar{z}^1|^2 + c + |A|\alpha_t(\omega) + \beta|\bar{y}| + \gamma|\bar{z}| \\ &\leq \left(a_i + \frac{L^2}{2\bar{\gamma}}\right)|\bar{z}^i|^2 + L|\bar{z}^i| + c + |A|\alpha_t(\omega) + \beta|\bar{y}| + \gamma|\bar{z}|. \end{aligned}$$

Consequently, we have from Corollary 2.11 that for each $\xi \in L^\infty$, the system of BSDEs (2.8) with the generator $\bar{g}$ defined in (2.20) admits a unique global solution $(\bar{Y}, \bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$. Finally, by Proposition 2.17 we know that $(A^{-1}\bar{Y}, A^{-1}\bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ is just the desired unique solution of the system of BSDEs (2.1). The proof is complete. $\square$

The following proposition and its corollary partially answer the problem 6.25 of Jackson [25].

Proposition 2.24. Assume that $n = 2$ and $d = 1$. Consider the following generator g

$$g(z) := \begin{pmatrix} g^1(z) \\ g^2(z) \end{pmatrix} = \begin{pmatrix} z^\top A_1 z + z^\top k_1 + l_1 \\ z^\top A_2 z + z^\top k_2 + l_2 \end{pmatrix}, \quad z \in \mathbb{R}^2,$$

where $A_i \in \mathbb{R}^{2 \times 2}$, $k_i \in \mathbb{R}^2$ and $l_i \in \mathbb{R}$ for $i = 1, 2$. Assume further that there exist three constants $a, b, \iota \in \mathbb{R}$ with $a \neq 0$ such that

$$aA_1 + bA_2 = \iota \begin{pmatrix} a \\ b \end{pmatrix} (a, b) = \begin{pmatrix} a^2\iota & ab\iota \\ ab\iota & b^2\iota \end{pmatrix} \quad (2.22)$$

and either $\alpha_{11} = 0$ and $\alpha_{22} \neq 0$ or $\alpha_{11}\alpha_{22} < 0$ with

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{12} & \alpha_{22} \end{pmatrix} := \frac{1}{a^2} \begin{pmatrix} 1 & 0 \\ -b & a \end{pmatrix} A_2 \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix}$$

Then for each $\xi \in L^\infty(\mathbb{R}^2)$, BSDE (2.1) with the above generator g admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times 1})$.

Proof. For simplicity, we only prove the case of $k_1 = k_2 = 0$ and $l_1 = l_2 = 0$. Let

$$A := \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

For each $\bar{z} \in \mathbb{R}^2$, let

$$z := A^{-1}\bar{z} = \frac{1}{a} \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix} \bar{z}.$$

Then, in view of (2.22), we have

$$\bar{g}^1(\bar{z}) := ag^1(z) + bg^2(z) = \frac{1}{a^2} \bar{z}^\top \begin{pmatrix} 1 & 0 \\ -b & a \end{pmatrix} (aA_1 + bA_2) \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix} \bar{z} = \iota (\bar{z}^1)^2$$

and

$$\bar{g}^2(\bar{z}) = g^2(z) = z^\top A_2 z = \frac{1}{a^2} \bar{z}^\top \begin{pmatrix} 1 & 0 \\ -b & a \end{pmatrix} A_2 \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix} \bar{z}.$$

According to the conditions of Proposition 2.24 combined with an argument similar to (2.6), we know that $\bar{g}$ satisfies Assumptions (B2) and (C1b) with $\theta = 0$. Proceeding identically as in Theorem 2.20, we have the desired assertion. $\square$

The following result is a direct consequence of Proposition 2.24.

Corollary 2.25. Assume that $n = 2$ and $d = 1$. Consider the following generator g

$$g(z) := \begin{pmatrix} g^1(z) \\ g^2(z) \end{pmatrix} = \begin{pmatrix} z^1(z^1 + z^2) - \frac{\alpha}{2}(z^1)^2 \\ z^2(z^1 + z^2) - \frac{\beta}{2}(z^2)^2 \end{pmatrix}, \quad z \in \mathbb{R}^2,$$

where $\alpha, \beta \in \mathbb{R}^*$ are two constants. If $1/\alpha + 1/\beta = 1$, then for each $\xi \in L^\infty(\mathbb{R}^2)$, BSDE (2.1) with this generator g admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times 1})$.

Proof. In Proposition 2.24, let $k_1 = k_2 = 0$, $l_1 = l_2 = 0$,

$$A_1 := \begin{pmatrix} 1 - \frac{\alpha}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \quad \text{and} \quad A_2 := \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 1 - \frac{\beta}{2} \end{pmatrix}.$$

There are two cases: $\alpha = 2$ and $\alpha \neq 2$. In the case of $\alpha = 2$, we have $A_1 = A_2$. When $a = 1$, $b = -1$ and $\iota = 0$, we have (2.22), $\alpha_{11} = 0$ and $\alpha_{22} = 1$, and thus all assumptions of Proposition 2.24 are satisfied. In the other case of $\alpha \neq 2$, we take $a = 1 - \alpha/2 \neq 0$, $b = 1 - \beta/2$ and $\iota = 1$. Since $\alpha + \beta = \alpha\beta$, we have

$$\frac{a+b}{2} = 1 - \frac{\alpha+\beta}{4} = 1 - \frac{\alpha+\beta}{2} + \frac{\alpha\beta}{4} = \left(1 - \frac{\alpha}{2}\right) \left(1 - \frac{\beta}{2}\right) = ab.$$

It can also be directly verified that all of (2.22), $\alpha_{11} = 0$ and $\alpha_{22} \neq 0$ hold, and then all assumptions of Proposition 2.24 are satisfied. Thus, the desired assertion follows immediately from Proposition 2.24. $\square$

2.6 Examples arising from practical problems

In this subsection, we will present several practical examples to which our theoretical results can be applied.

First, the authors in [45, 28] connected the existence of incomplete stochastic equilibrium in financial markets, to the system of quadratic BSDEs (1.1) with $d = 2$ and the following Markovian generator $g := (g^1, \dots, g^n)^\top$:

$$\begin{cases} g^i(z) := -\frac{1}{2}\nu^i(\nu^i + 2\nu^n) - \mu^i \left(\sum_{j=1}^{n-1} a_j \mu^j + \mu^n \right), & i = 1, \dots, n-1; \\ g^n(z) := -\frac{1}{2}[(\mu^n)^2 + (\nu^n)^2] + \left(\sum_{j=1}^{n-1} a_j \mu^j \right)^2, & z := ((\mu^1, \nu^1)^\top, \dots, (\mu^n, \nu^n)^\top)^\top \in \mathbb{R}^{n \times 2}, \end{cases} \quad (2.23)$$

where the constant $a_j \in (0, 1)$ for each $j = 1, \dots, n-1$. Now, we study the non-Markovian case. For each $\bar{z} \in \mathbb{R}^{n \times 2}$, define $\bar{g}(\bar{z}) := Ag(A^{-1}\bar{z})$ with the following invertible matrix

$$A := \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 \end{pmatrix}.$$

Then, for each $\bar{z} := ((\bar{\mu}^1, \bar{\nu}^1)^\top, \dots, (\bar{\mu}^n, \bar{\nu}^n)^\top)^\top \in \mathbb{R}^{n \times 2}$, we have

$$\begin{cases} \bar{g}^1(\bar{z}) = -\frac{1}{2}[(\bar{\mu}^1)^2 + (\bar{\nu}^1)^2] + \left(\sum_{j=2}^n a_{j-1} \bar{\mu}^j\right)^2; \\ \bar{g}^i(\bar{z}) = -\frac{1}{2}\bar{\nu}^i(\bar{\nu}^i + 2\bar{\nu}^1) - \bar{\mu}^i \left(\sum_{j=2}^n a_{j-1} \bar{\mu}^j + \bar{\mu}^1\right), \quad i = 2, \dots, n. \end{cases}$$

We verify that $\bar{g}$ satisfies Assumption (B2), $\bar{g}^1$ satisfies Assumption (C1a)(i) and $\bar{g}^i$ satisfies Assumption (C1a)(ii) for each $i = 2, \dots, n$. From Theorem 2.10 and Proposition 2.17, we see that for each $\xi \in L^\infty(\mathbb{R}^n)$, there exists a constant $\theta_0 > 0$ depending only on $\|\xi\|_\infty$ such that when $\sum_{j=1}^{n-1} a_j \leq \theta_0$, the system of quadratic BSDEs (1.1) with the generator g defined in (2.23) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times 2})$.

Second, Xing and Žitković [45] connected the Nash equilibrium in a stochastic game to the system of quadratic BSDEs (1.1) with $n = 2$ and the following generator $g := (g^1, g^2)^\top$ in a Markovian setting:

$$\begin{cases} g^1(z) := a_\vartheta |z^1|^2 + b_\vartheta (z^1)^\top z^2 + c_\vartheta |z^2|^2, \\ g^2(z) := c_\vartheta |z^1|^2 + b_\vartheta (z^1)^\top z^2 + a_\vartheta |z^2|^2; \quad z := ((z^1)^\top, (z^2)^\top)^\top \in \mathbb{R}^{2 \times d}, \end{cases} \quad (2.24)$$

where $\vartheta \neq \pm 1$ is a constant called the cooperation penalty and

$$a_\vartheta := \frac{-2\vartheta^3 + 2\vartheta - 1}{2(1 - \vartheta^2)^2}, \quad b_\vartheta := \frac{2\vartheta^3 - \vartheta^2 - \vartheta + 1}{(1 - \vartheta^2)^2} \quad \text{and} \quad c_\vartheta := \frac{-\vartheta^2}{2(1 - \vartheta^2)^2}.$$

We now turn to the non-Markovian case and the case that $\vartheta = 1/2$. In this case, we have $a_\vartheta = c_\vartheta = -2/9$ and $b_\vartheta = 8/9$. For each $\bar{z} \in \mathbb{R}^{2 \times d}$, define $\bar{g}(\bar{z}) := Ag(A^{-1}\bar{z})$ with the following invertible matrix

$$A := \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Then, for each $\bar{z} := ((\bar{z}^1)^\top, (\bar{z}^2)^\top)^\top \in \mathbb{R}^{2 \times d}$, we have

$$\bar{g}^1(\bar{z}) \equiv 0 \quad \text{and} \quad \bar{g}^2(\bar{z}) = -\frac{2}{9}|\bar{z}^1|^2 + \frac{4}{9}(\bar{z}^1)^\top \bar{z}^2 + \frac{4}{9}|\bar{z}^2|^2.$$

An argument similar to (2.6) shows that $\bar{g}$ satisfies Assumptions (B2) with $\theta = 0$ and (C1b). Then from Theorem 2.14 and Proposition 2.17, we see that for each $\xi \in L^\infty(\mathbb{R}^2)$, the system of quadratic BSDEs (1.1) with the generator g defined in (2.24) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^2) \times \text{BMO}(\mathbb{R}^{2 \times d})$.

Third, in a financial price-impact model, Kramkov and Pulido [34] characterized the prices of risky assets in terms of a system of quadratic BSDEs (1.1) with the following generator $g := (g^1, \dots, g^n)^\top$: for each $z := ((z^1)^\top, (\bar{z})^\top)^\top$ with $z^1 \in \mathbb{R}^{1 \times d}$ and $\bar{z} := ((z^2)^\top, \dots, (z^n)^\top)^\top \in \mathbb{R}^{(n-1) \times d}$,

$$\begin{cases} g^1(t, z) := \frac{1}{2}(|z^1|^2 - |\bar{z}^\top \kappa_t|^2); \\ g^i(t, z) := z^i \left((z^1)^\top + \bar{z}^\top \kappa_t \right), \quad i = 2, \dots, n, \end{cases} \quad (2.25)$$

where $(\kappa_t)_{t \in [0, T]}$ is a bounded $\mathbb{R}^{n-1}$ -valued progressively measurable process. Note that for each $z \in \mathbb{R}^{n \times d}$, we have

$$|z^\top|^2 = |z|^2 = \sum_{j=1}^n |z^j|^2 \quad \text{and} \quad |z^\top| = |z| \leq \sum_{j=1}^n |z^j|.$$

It is easy to verify that $\bar{g}$ satisfies Assumption (B2), $\bar{g}^0$ satisfies Assumption (C1a)(i) and $\bar{g}^i$ satisfies Assumption (C1a)(ii) for each $i = 1, \dots, n$. It then follows from Theorem 2.10 that for each $\xi \in L^\infty(\mathbb{R}^n)$, there exists a constant $\theta_0 > 0$ depending only on $\|\xi\|_\infty$ such that when $|\kappa| \leq \theta_0$, the system of quadratic BSDEs (1.1) with the generator g defined in (2.25) admits a unique global solution $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$.

Finally, we would like to mention that in light of (i) of Remark 2.15 along with Proposition 2.24, some non-Markovian systems of quadratic BSDEs, arising naturally from a risk-sensitive nonzero-sum stochastic game and an equilibrium with limited participation, can also be addressed in terms of our Theorem 2.14 and Theorem 2.3. The interested reader is referred to [24, 43, 25] for more details.

2.7 Global unbounded solution of multi-dimensional quadratic BSDEs

In this subsection, we will present three existence and uniqueness results on the global unbounded solution of multi-dimensional BSDE (2.1). For this, let us first introduce the following assumption (D2), which is strictly stronger than the previous assumption (B2). In particular, the generator g satisfying (B2) can have a general growth in the state variable y , while the generator g satisfying (D2) can only have a linear growth in y .

(D2) For $i = 1, \dots, n$, g^i satisfies that $d\mathbb{P} \times dt - a.e.$, for each $(y, \bar{y}, z, \bar{z}) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times d} \times \mathbb{R}^{n \times d}$,

$$|g^i(\omega, t, y, z) - g^i(\omega, t, \bar{y}, \bar{z})| \leq \gamma(v_t(\omega) + |z| + |\bar{z}|) \left(|y - \bar{y}| + \sum_{j=1}^i |z^j - \bar{z}^j| \right) \\ + [\gamma([v_t(\omega)]^\delta + |z|^\delta + |\bar{z}|^\delta) + \theta(v_t(\omega) + |z| + |\bar{z}|)] \sum_{j=i+1}^n |z^j - \bar{z}^j|.$$

The following is the first existence and uniqueness result of this subsection.

Theorem 2.26. Let the generator g satisfy Assumptions (B1), (D2) and (AB) with $\theta = 0$ and $\phi(\cdot) \equiv 0$, and let

$$\xi = \bar{\xi} + \int_0^T H_s dB_s$$

with $\bar{\xi} \in L^\infty(\mathbb{R}^n)$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$. When g^i satisfies (ii) in Assumption (B1) and violates (B1) (i) and (iii), and $-g^i$ violates (B1) (i), we further assume that $H^i \equiv 0$. If $\alpha, \bar{\alpha}, |H|^2 \in \mathcal{L}^\infty$, then BSDE (2.1) admits a unique global solution (Y, Z) such that

$$\left(Y - \int_0^\cdot H_s dB_s, Z \right) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d}).$$

Proof. Define

$$\bar{g}(t, y, z) := g\left(t, y + \int_0^t H_s dB_s, z + H_t\right), \quad (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R}^n \times \mathbb{R}^{n \times d}. \quad (2.26)$$

Consider the following multi-dimensional BSDE

$$\bar{Y}_t = \bar{\xi} + \int_t^T \bar{g}(s, \bar{Y}_s, \bar{Z}_s) ds - \int_t^T \bar{Z}_s dB_s, \quad t \in [0, T], \quad (2.27)$$

or, equivalently,

$$\bar{Y}_t^i = \bar{\xi}^i + \int_t^T \bar{g}^i(s, \bar{Y}_s, \bar{Z}_s) ds - \int_t^T \bar{Z}_s^i dB_s, \quad t \in [0, T], \quad i = 1, \dots, n. \quad (2.28)$$

It is not difficult to check that BSDE (2.27) admits a unique global solution $(\bar{Y}, \bar{Z}) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ if and only if BSDE (2.1) admits a unique global solution $(Y, Z) := (Y + \int_0^\cdot H_s dB_s, \bar{Z} + H)$ on the time interval $[0, T]$ such that

$$\left(Y - \int_0^\cdot H_s dB_s, Z\right) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d}).$$

Consequently, in view of Corollary 2.7, for completing the proof of Theorem 2.26 it suffices to prove that the generator $\bar{g}$ defined in (2.26) also satisfies Assumptions (B1), (D2) and (AB) with $\theta = 0$. In view of assumptions of the generator g and parameters $(\alpha, \bar{\alpha}, \tilde{\alpha}, H)$, it is straightforward to verify the above statement. The proof is then complete. $\square$

The second existence and uniqueness result requires the following assumption (D1), which is strictly stronger than the previous assumption (C1a), for g admits a general growth in the state variable y in (C1a), instead of only a linear growth in y in (D1).

(D1) For each fixed $i = 1, \dots, n$, any one of the following three conditions holds:

(i) $f := g^i$ or $f := -g^i$ satisfies that $d\mathbb{P} \times dt - a.e.$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{aligned} \frac{\bar{\gamma}}{2}|z^i|^2 - \bar{\alpha}_t(\omega) - \beta|y| - \sum_{j=i+1}^n (\bar{\lambda}|z^j|^{1+\delta} + \theta|z^j|^2) - \bar{c} \sum_{j=1}^{i-1} |z^j|^2 &\leq f(\omega, t, y, z) \\ &\leq \alpha_t(\omega) + \beta|y| + \sum_{j \neq i} (\lambda|z^j| + \theta|z^j|^2) + \frac{\gamma}{2}|z^i|^2; \end{aligned}$$

(ii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\begin{aligned} |g^i(\omega, t, y, z)| &\leq \tilde{\alpha}_t(\omega) + \beta|y^i| + \frac{\gamma}{2}|z^i|^2 \\ &\quad + \left\{ v_t(\omega) + c \sum_{j=1}^{i-1} |z^j| + \sum_{j=i+1}^n (\lambda|z^j|^\delta + \theta|z^j|) \right\} |z^i|; \end{aligned}$$

(iii) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$|g^i(\omega, t, y, z)| \leq \bar{\alpha}_t(\omega) + \beta|y^i| + \bar{\lambda}|z| + \theta \sum_{j \neq i} |z^j|^2.$$

The following Theorem 2.27 and Jamneshan et al. [27, Theorem 2.6] do not cover each other, and so are Theorem 2.27 and Frei [20, Proposition 2.1].

Theorem 2.27. Let the generator g satisfy Assumptions (D1) and (D2) with $\beta = 0$, and let

$$\xi = \bar{\xi} + \int_0^T H_s dB_s$$

with $\bar{\xi} \in L^\infty(\mathbb{R}^n)$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$. When g^i satisfies (ii) in Assumption (D1) and violates (D1) (i) and (iii), and $-g^i$ violates (D1) (i), we further assume that $H^i \equiv 0$. If $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some $p > 1$, and $|H|^2 \in \mathcal{E}^\infty(2\bar{p}(q\gamma)^2)$ for some $\bar{p} > 1$ with $q = p/(p-1)$ such that $1/p + 1/q = 1$, then there exists a positive constant $\theta_0 > 0$ depending only on $(\|\bar{\xi}\|_\infty, \|\alpha\|_{\mathcal{E}^\infty(p\gamma)}, \|\bar{\alpha}\|_{\mathcal{M}^\infty}, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, \|v\|_{\text{BMO}}, \|H\|_{\mathcal{E}^\infty(2\bar{p}(q\gamma)^2)}, n, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p, \bar{p})$ such that when $\theta \in [0, \theta_0]$, BSDE (2.1) admits a unique global solution (Y, Z) on $[0, T]$ such that

$$\left(Y - \int_0^\cdot H_s dB_s, Z\right) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d}).$$

Remark 2.28. By the John-Nirenberg inequality, if $\|H\|_{\text{BMO}} < 1/(\sqrt{2\bar{p}q\gamma})$, then $|H|^2 \in \mathcal{E}^\infty(2\bar{p}(q\gamma)^2)$. Then, we have a unique solution for the multi-dimensional BSDE, as stated in Theorem 3.1 of Kramkov and Pulido [34].

Outline of the proof of Theorem 2.27. In light of Theorem 2.10 and an identical proof of Theorem 2.26, it suffices to verify that the generator $\bar{g}$ defined in (2.26) satisfies Assumptions (D1) and (D2) with $\beta = 0$ and some other appropriate parameters when the constant θ is smaller than a given constant $\bar{\theta}_0$. This assertion can be achieved by the integrability condition of $|H|^2$ combined with some subtle estimates. The detailed proof is postponed to Section 5. $\square$

In view of Assertion (i) of Remark 2.1, we have immediately from Theorem 2.27 the following corollary.

Corollary 2.29. Let the generator g satisfy (D1) and (D2) with $\beta = 0$ and $\theta = 0$, and let

$$\xi = \bar{\xi} + \int_0^T H_s dB_s$$

with $\bar{\xi} \in L^\infty(\mathbb{R}^n)$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$. When g^i satisfies (ii) in Assumption (D1) and violates (D1) (i) and (iii), and $-g^i$ violates (D1) (i), we further assume that $H^i \equiv 0$. If $\alpha, |H|^2 \in \mathcal{L}^\infty$, then BSDE (2.1) admits a unique global solution (Y, Z) on the time interval $[0, T]$ such that

$$\left(Y - \int_0^\cdot H_s dB_s, Z\right) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d}).$$

Remark 2.30. If the constant c appearing in (ii) of Assumption (D1) vanishes, then the condition that $H^i \equiv 0$ appearing in Theorem 2.27 and Corollary 2.29 can be weakened to that both $|H^i|$ and $|H^i|v$ belong to $\mathcal{L}^\infty$.

We end this subsection with the following theorem.

Theorem 2.31. Let the generator g satisfy Assumptions (C1b) and (D2) with $\beta = c = 0$ and $\phi(\cdot) \equiv 0$, and

$$\xi = \bar{\xi} + \int_0^T H_s dB_s$$

with $\bar{\xi} \in L^\infty(\mathbb{R}^n)$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$. When g^i satisfies (ii) in Assumption (B1) and violates (B1) (i) and (iii), and $-g^i$ violates (B1) (i), we further assume that $|H^i|^2, |H^i|v \in \mathcal{L}^\infty$. If $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some $p > 1$, and $|H|^2 \in \mathcal{E}^\infty(2\bar{p}(q\gamma)^2)$ for some $\bar{p} > 1$ with $q = p/(p-1)$ such that $1/p + 1/q = 1$, then BSDE (2.1) admits a unique global solution (Y, Z) such that

$$\left(Y - \int_0^\cdot H_s dB_s, Z\right) \in \mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d}).$$

The proof is identical to that of Theorem 2.27 with the help of Theorem 2.14, and thus is omitted here.

3 Local unbounded solution: Proof of Theorem 2.3

Let us begin with the following technical lemma for subsequent arguments.

Lemma 3.1. Let $p > 1, r > 0, \bar{\delta} \in [0, 1), i \in \{1, \dots, n\}, \bar{t} \in [0, T]$ and $V, \bar{V} \in \text{BMO}_{[\bar{t}, T]}(\mathbb{R}^{n \times d})$ such that, with $q := \frac{p}{p-1} > 1$ satisfying that $\frac{1}{p} + \frac{1}{q} = 1$,

$$2n(qr \vee 1)\theta \left(\sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[\bar{t}, T]}}^2 + \|V\|_{\text{BMO}_{[\bar{t}, T]}}^2 \right) \leq 1.$$

(i) Define

$$\check{\alpha}_s := \hat{\alpha}_s + \sum_{j=1}^{i-1} \left(\lambda |\bar{V}_s^j|^{1+\bar{\delta}} + \theta |\bar{V}_s^j|^2 \right) + \sum_{j=i+1}^n \left(\lambda |V_s^j|^{1+\bar{\delta}} + \theta |V_s^j|^2 \right), \quad s \in [\bar{t}, T]. \quad (3.1)$$

If $\hat{\alpha} \in \mathcal{E}_{[\bar{t}, T]}^\infty(pr)$, then $\check{\alpha} \in \mathcal{E}_{[\bar{t}, T]}^\infty(r)$. And, for each $t \in [\bar{t}, T]$, we have

$$\begin{aligned} \|\check{\alpha}\|_{\mathcal{E}_{[\bar{t}, T]}^\infty(r)} &\leq \|\hat{\alpha}\|_{\mathcal{E}_{[\bar{t}, T]}^\infty(pr)} + \frac{\ln 2}{r} + C_{p,n,r,\lambda,\bar{\delta}} \sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[\bar{t}, T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} (T-t) \\ &\quad + C_{p,n,r,\lambda,\bar{\delta}} \sum_{j=i+1}^n \|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} (T-t), \end{aligned} \quad (3.2)$$

where

$$C_{p,n,r,\lambda,\bar{\delta}} := \lambda \frac{1-\bar{\delta}}{2} \left(\frac{pnr\lambda(1+\bar{\delta})}{p-1} \right)^{\frac{1+\bar{\delta}}{1-\bar{\delta}}}. \quad (3.3)$$

Moreover, it is clear that the previous conclusion holds also for $p = 1$ and $q = 0$ when $\lambda = 0$ and $\theta = 0$ if we let $C_{p,n,r,\lambda,\bar{\delta}} := 0$ when $\lambda = 0$ and $p = 1$.

(ii) Define

$$\check{\alpha}_s := \hat{\alpha}_s + \bar{c} \sum_{j=1}^{i-1} |\bar{V}_s^j|^2 + \sum_{j=i+1}^n \left(\bar{\lambda} |V_s^j|^{1+\bar{\delta}} + \theta |V_s^j|^2 \right), \quad s \in [\bar{t}, T]. \quad (3.4)$$

If $\hat{\alpha} \in \mathcal{M}_{[\bar{t}, T]}^\infty$, then $\check{\alpha} \in \mathcal{M}_{[\bar{t}, T]}^\infty$. And, for each $t \in [\bar{t}, T]$,

$$\|\check{\alpha}\|_{\mathcal{M}_{[\bar{t}, T]}^\infty} \leq 1 + \|\hat{\alpha}\|_{\mathcal{M}_{[\bar{t}, T]}^\infty} + \bar{c} \sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[\bar{t}, T]}}^2 + n\bar{\lambda} \|V\|_{\text{BMO}_{[\bar{t}, T]}}^{1+\bar{\delta}} (T-t)^{\frac{1-\bar{\delta}}{2}}. \quad (3.5)$$

(iii) Define

$$\check{\alpha}_s := \hat{\alpha}_s + \sum_{j=1}^{i-1} (\bar{\lambda} |\bar{V}_s^j| + \theta |\bar{V}_s^j|^2) + \sum_{j=i+1}^n \left(\bar{\lambda} |V_s^j|^{1+\bar{\delta}} + \theta |V_s^j|^2 \right), \quad s \in [\bar{t}, T]. \quad (3.6)$$

If $\hat{\alpha} \in \mathcal{M}_{[\bar{t}, T]}^\infty$, then $\check{\alpha} \in \mathcal{M}_{[\bar{t}, T]}^\infty$. And, for each $t \in [\bar{t}, T]$, we have

$$\begin{aligned} \|\check{\alpha}\|_{\mathcal{M}_{[\bar{t}, T]}^\infty} &\leq 1 + \|\hat{\alpha}\|_{\mathcal{M}_{[\bar{t}, T]}^\infty} + \bar{\lambda} \sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[\bar{t}, T]}} (T-t)^{\frac{1}{2}} \\ &\quad + \bar{\lambda} \sum_{j=i+1}^n \|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^{1+\bar{\delta}} (T-t)^{\frac{1-\bar{\delta}}{2}}. \end{aligned} \quad (3.7)$$

Proof. (i) Since $V, \bar{V} \in \text{BMO}(\mathbb{R}^{n \times d})$, it follows from Young's inequality that for $p > 1$, $\lambda \geq 0$, $0 \leq \bar{t} \leq t \leq s \leq T$ and $j = 1, \dots, n$, we have

$$\begin{aligned} \lambda |V_s^j|^{1+\bar{\delta}} &= \lambda \left(\left(\frac{pnr\lambda(1+\bar{\delta})}{p-1} \right)^{\frac{1+\bar{\delta}}{1-\bar{\delta}}} \|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} \right)^{\frac{1-\bar{\delta}}{2}} \\ &\quad \times \left(\frac{(p-1)|V_s^j|^2}{pnr\lambda(1+\bar{\delta})\|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^2} \right)^{\frac{1+\bar{\delta}}{2}} \\ &\leq \frac{p-1}{2pnr\|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^2} |V_s^j|^2 + C_{p,n,r,\lambda,\bar{\delta}} \|V^j\|_{\text{BMO}_{[\bar{t}, T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} \end{aligned} \quad (3.8)$$

and

$$\lambda |\bar{V}_s^j|^{1+\bar{\delta}} \leq \frac{p-1}{2pnr \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^2} |\bar{V}_s^j|^2 + C_{p,n,r,\lambda,\bar{\delta}} \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}}, \quad (3.9)$$

where the constant $C_{p,n,r,\lambda,\bar{\delta}}$ is defined in (3.3). Note that $q := \frac{p}{p-1} > 1$ satisfying that $\frac{1}{p} + \frac{1}{q} = 1$. By (3.8) and (3.9) we know that for each $t \in [\bar{t}, T]$, $\check{\alpha}_s \leq \hat{\alpha}_s + \check{\alpha}_s^{1,t} + \check{\alpha}_s^{2,t}$, $s \in [t, T]$, where

$$\check{\alpha}_s^{1,t} := \sum_{j=1}^{i-1} \frac{1}{2qn r \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^2} |\bar{V}_s^j|^2 + \sum_{j=i+1}^n \frac{1}{2qn r \|V^j\|_{\text{BMO}_{[t,T]}}^2} |V_s^j|^2 + \theta \left(\sum_{j=1}^{i-1} |\bar{V}_s^j|^2 + |V_s|^2 \right)$$

and

$$\check{\alpha}_s^{2,t} := C_{p,n,r,\lambda,\bar{\delta}} \left(\sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} + \sum_{j=i+1}^n \|V^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} \right).$$

It follows from Hölder's inequality and the John-Nirenberg inequality that for each $t \in [\bar{t}, T]$ and $\tau \in \mathcal{T}_{[t,T]}$,

$$\begin{aligned} \mathbb{E}_\tau \left[\exp \left(qr \int_\tau^T \check{\alpha}_s^{1,t} ds \right) \right] &\leq \prod_{j=1}^{i-1} \left(\mathbb{E}_\tau \left[\exp \left(\frac{1}{2\|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^2} \int_\tau^T |\bar{V}_s^j|^2 ds \right) \right] \right)^{\frac{1}{n}} \\ &\quad \times \left(\mathbb{E}_\tau \left[\exp \left(nqr\theta \int_\tau^T \left(\sum_{j=1}^{i-1} |\bar{V}_s^j|^2 + |V_s|^2 \right) ds \right) \right] \right)^{\frac{1}{n}} \\ &\quad \times \prod_{j=i+1}^n \left(\mathbb{E}_\tau \left[\exp \left(\frac{1}{2\|V^j\|_{\text{BMO}_{[t,T]}}^2} \int_\tau^T |V_s^j|^2 ds \right) \right] \right)^{\frac{1}{n}} \\ &\leq 2, \end{aligned}$$

which means that $\check{\alpha}^{1,t} \in \mathcal{E}_{[t,T]}^\infty(qr)$ and $\|\check{\alpha}^{1,t}\|_{\mathcal{E}_{[t,T]}^\infty(qr)} \leq \frac{\ln 2}{qr}$. On the other hand, it is clear that

$$\|\check{\alpha}^{2,t}\|_{\mathcal{L}_{[t,T]}^\infty} = C_{p,n,r,\lambda,\bar{\delta}} \left(\sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} + \sum_{j=i+1}^n \|V^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} \right) (T-t), \quad t \in [\bar{t}, T].$$

It then follows from (ii) and (iii) of Remark 2.1 that for each $t \in [\bar{t}, T]$, $\check{\alpha} \in \mathcal{E}_{[t,T]}^\infty(r)$ and

$$\begin{aligned} \|\check{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(r)} &\leq \|\hat{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(pr)} + \frac{\ln 2}{qr} \\ &\quad + C_{p,n,r,\lambda,\bar{\delta}} \left(\sum_{j=1}^{i-1} \|\bar{V}^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} + \sum_{j=i+1}^n \|V^j\|_{\text{BMO}_{[t,T]}}^{2\frac{1+\bar{\delta}}{1-\bar{\delta}}} \right) (T-t), \end{aligned}$$

from which the desired conclusion (3.2) follows immediately.

(ii) It follows from Hölder's inequality that for each $t \in [\bar{t}, T]$, $j = 1, \dots, n$ and $\tau \in \mathcal{T}_{[t,T]}$,

$$\mathbb{E}_\tau \left[\int_\tau^T |V_s^j|^{1+\bar{\delta}} ds \right] \leq \left(\mathbb{E}_\tau \left[\int_\tau^T |V_s^j|^2 ds \right] \right)^{\frac{1+\bar{\delta}}{2}} (T-t)^{\frac{1-\bar{\delta}}{2}} \leq \|V^j\|_{\text{BMO}_{[t,T]}}^{1+\bar{\delta}} (T-t)^{\frac{1-\bar{\delta}}{2}}.$$

Then, the desired conclusion (3.5) follows from (3.4) and the last inequality.

(iii) It follows from Hölder's inequality that for each $t \in [\bar{t}, T]$, $j = 1, \dots, n$ and $\tau \in \mathcal{T}_{[t,T]}$,

$$\mathbb{E}_\tau \left[\int_\tau^T |\bar{V}_s^j| ds \right] \leq \|\bar{V}^j\|_{\text{BMO}_{[t,T]}} (T-t)^{\frac{1}{2}}.$$

Then, the desired conclusion (3.7) follows from (3.6) and the last two inequalities. $\square$

Now, we can give the proof of Theorem 2.3. Assume first that $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some real $p > 1$ and the generator g satisfies Assumptions (B1) and (B2). Define

$$\begin{aligned} C_1 := & \frac{2+\bar{\gamma}}{\bar{\gamma}} \left\{ 8 + \frac{\ln 2}{\gamma} + 3\|\xi\|_\infty + \|\alpha\|_{\mathcal{E}^\infty(p\gamma)} + 2\|\bar{\alpha}\|_{\mathcal{M}^\infty} + nC_{p,n,\gamma,\lambda,\delta}T + 2n\bar{c} \right\} \\ & + \frac{4(\gamma+1)}{\gamma^2} \exp\{4\gamma(\|\xi\|_\infty + \|\bar{\alpha}\|_{\mathcal{L}^\infty} + 1)\} (9 + \|\bar{\alpha}\|_{\mathcal{L}^\infty} + 5\|v\|_{\text{BMO}}^2 + 5n^2c^2) \\ & + 1 + c_0 \exp(2\bar{\lambda}^2T) (54 + \|\xi\|_\infty^2 + 6\|\bar{\alpha}\|_{\mathcal{M}^\infty}^2 + 6n^2\bar{\lambda}^2T) \end{aligned}$$

and

$$\begin{aligned} C_2 := & \frac{2+\bar{\gamma}}{\bar{\gamma}} (2C_{p,n,\gamma,\lambda,\delta}T + 4\bar{c}) + \frac{20nc^2(\gamma+1)}{\gamma^2} \exp\{4\gamma(\|\xi\|_\infty + \|\bar{\alpha}\|_{\mathcal{L}^\infty} + 1)\} \\ & + 12nc_0\bar{\lambda}^2T \exp(2\bar{\lambda}^2T), \end{aligned}$$

where the uniform constant $c_0 > 0$ is defined in (i) of Proposition A.3 in Appendix, and

$$C_{p,n,\gamma,\lambda,\delta} := \lambda \frac{1-\delta}{2} \left(\frac{pn\gamma\lambda(1+\delta)}{p-1} \right)^{\frac{1+\delta}{1-\delta}} + 1. \quad (3.10)$$

Furthermore, let $C_1^0 := 0$ and for $i = 1, \dots, n$, recursively define

$$C_1^i := C_1^{i-1} + C_1 + C_2 [C_1^{i-1}]^{\frac{1+\delta}{1-\delta}}. \quad (3.11)$$

Finally, define $K := C_1^n$ and let the constant ε satisfy

$$\frac{1}{\varepsilon} \geq \max \left\{ \frac{1}{T}, 1 + \phi(K), nC_{p,n,\gamma,\lambda,\delta}K^{\frac{1+\delta}{1-\delta}}, \left(1 + 2n\bar{\lambda}K^{\frac{1+\delta}{1-\delta}}\right)^{\frac{2}{1-\delta}}, (1 + 5n\lambda^2K^\delta)^{\frac{1}{1-\delta}} \right\}. \quad (3.12)$$

It is clear that all constants defined above (except ε) depend only on $\|\xi\|_\infty$, $\|\alpha\|_{\mathcal{E}^\infty(p\gamma)}$, $\|\bar{\alpha}\|_{\mathcal{M}^\infty}$, $\|\bar{\alpha}\|_{\mathcal{L}^\infty}$, $\|v\|_{\text{BMO}}$, $n, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T$ and p , and that ε also depends on $\phi(\cdot)$.

In the sequel, for each $H \in \mathbb{R}^{n \times d}$, $i \in \{1, \dots, n\}$ and $z^{(1)} \dots, z^{(i)} \in \mathbb{R}^{1 \times d}$, define

$$\mathcal{M}(z^{(1)}, \dots, z^{(i)}, H^{i+1}, \dots, H^n) := \left((z^{(1)})^\top, \dots, (z^{(i)})^\top, (H^{i+1})^\top, \dots, (H^n)^\top \right)^\top.$$

Let $q := \frac{p}{p-1}$. Given a pair of processes $(U, V) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{n \times d})$ and a real θ satisfying

$$\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K \quad \text{and} \quad 0 \leq \theta \leq \frac{1}{4n(q\gamma \vee 1)K}. \quad (3.13)$$

We will first prove that for each $i = 1, \dots, n$, the following scalar-valued BSDE

$$Y_t^i = \xi^i + \int_t^T f^i(s, Z_s^i) ds - \int_t^T Z_s^i dB_s, \quad t \in [T-\varepsilon, T] \quad (3.14)$$

admits, successively, a unique solution (Y^i, Z^i) in $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, where for each $(\omega, t, z) \in \Omega \times [T-\varepsilon, T] \times \mathbb{R}^{1 \times d}$,

$$\begin{cases} f^1(\omega, t, z) := g^1(\omega, t, U_t(\omega), \mathcal{M}(z, V_t^2(\omega), \dots, V_t^n(\omega))), \\ f^i(\omega, t, z) := g^i(\omega, t, U_t(\omega), \mathcal{M}(Z_t^1(\omega), \dots, Z_t^{i-1}(\omega), z, V_t^{i+1}(\omega), \dots, V_t^n(\omega))), \\ \quad i = 2, \dots, n-1, \\ f^n(\omega, t, z) := g^n(\omega, t, U_t(\omega), \mathcal{M}(Z_t^1(\omega), \dots, Z_t^{n-1}(\omega), z)). \end{cases}$$

Moreover, it holds that for each $i = 1, \dots, n$,

$$\|Y^i\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|Z^i\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq C_1 + C_2 \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^{2\frac{1+\delta}{1-\delta}}, \quad (3.15)$$

$$\sum_{j=1}^i \left(\|Y^j\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|Z^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) \leq C_1^i \leq K \quad (3.16)$$

and

$$2n(q\gamma \vee 1)\theta \left(\sum_{j=1}^i \|Z^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) \leq 1. \quad (3.17)$$

In particular, letting $i = n$ in (3.16) yields that

$$\|Y\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|Z\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K. \quad (3.18)$$

For this, let us use an induction to prove the following proposition.

Proposition 3.2. Assume that $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma)$ for $p > 1$ and the generator g satisfies Assumptions (B1) and (B2). Let (3.11)-(3.13) be in force. Then, we have the following two assertions.

- (a) For $i = 1$, BSDE (3.14) admits a unique solution (Y^1, Z^1) in the product space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, and inequalities (3.15)-(3.17) are satisfied.
- (b) For $i = 2, \dots, n$, assume that for each $l = 1, \dots, i-1$, the following BSDE

$$Y_t^l = \xi^l + \int_t^T f^l(s, Z_s^l) ds - \int_t^T Z_s^l dB_s, \quad t \in [T-\varepsilon, T] \quad (3.19)$$

admits, successively, a unique solution (Y^l, Z^l) in the product space of processes $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, and the following inequalities hold:

$$\sum_{j=1}^l \left(\|Y^j\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|Z^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) \leq C_1^l \leq K \quad (3.20)$$

and

$$2n(q\gamma \vee 1)\theta \left(\sum_{j=1}^l \|Z^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) \leq 1. \quad (3.21)$$

Then, BSDE (3.14) also admits a unique solution (Y^i, Z^i) in the space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, and the inequalities (3.15)-(3.17) also hold.

Proof. We first prove the assertion (b). Assume that for some $i \in \{2, \dots, n\}$ and each $l = 1, \dots, i-1$, BSDE (3.19) admits a unique solution $(Y^l, Z^l) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$, and the inequalities (3.20)-(3.21) are true. Since the generator g satisfies Assumption (B2), it follows from the definition of f^i that $d\mathbb{P} \times dt - a.e.$ on $\Omega \times [T-\varepsilon, T]$, for each $(z, \bar{z}) \in \mathbb{R}^{1 \times d} \times \mathbb{R}^{1 \times d}$,

$$|f^i(t, z) - f^i(t, \bar{z})| \leq \phi(\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}) \left(v_t + 2 \sum_{j=1}^{i-1} |Z_t^j| + 2|V_t| + |z| + |\bar{z}| \right) |z - \bar{z}|,$$

which means that on $\Omega \times [T-\varepsilon, T]$, f^i satisfies Assumption (A5) in Appendix with $\bar{\beta} = 0$,

$$k = \phi(\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}) \quad \text{and} \quad \bar{v} = v + 2 \sum_{j=1}^{i-1} |Z^j| + 2|V| \in \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}).$$

And, since g satisfies Assumption (B1), in view of (iv) of Remark 2.2, we need only to consider the following three cases:

(1) g^i satisfies (i) of Assumption (B1). For this case, it follows from the definition of f^i that $d\mathbb{P} \times dt - a.e.$ on $\Omega \times [T - \varepsilon, T]$, for each $z \in \mathbb{R}^{1 \times d}$, we have

$$\frac{\bar{\gamma}}{2}|z|^2 - \dot{\alpha}_t \leq f^i(t, z) \leq \check{\alpha}_t + \frac{\gamma}{2}|z|^2, \quad (3.22)$$

where

$$\check{\alpha}_t := \alpha_t + \phi(|U_t|) + \sum_{j=1}^{i-1} \left(\lambda |Z_t^j|^{1+\delta} + \theta |Z_t^j|^2 \right) + \sum_{j=i+1}^n \left(\lambda |V_t^j|^{1+\delta} + \theta |V_t^j|^2 \right)$$

and

$$\dot{\alpha}_t := \bar{\alpha}_t + \phi(|U_t|) + \bar{c} \sum_{j=1}^{i-1} |Z_t^j|^2 + \sum_{j=i+1}^n \left(\bar{\lambda} |V_t^j|^{1+\delta} + \theta |V_t^j|^2 \right).$$

Since $\alpha \in \mathcal{E}_{[T-\varepsilon, T]}^\infty(p\gamma)$, $\bar{\alpha} \in \mathcal{M}_{[T-\varepsilon, T]}^\infty$ and $\phi(|U|) \in \mathcal{L}_{[T-\varepsilon, T]}^\infty$, in view of (ii) of Remark 2.1, (3.13) and (3.21) with $l = i - 1$, using (i) and (ii) of Lemma 3.1 with $r = \gamma$, $\bar{\delta} = \delta$, $\bar{t} = T - \varepsilon$, $\bar{V} = Z$, $\check{\alpha} = \check{\alpha}$ and $\hat{\alpha} = \alpha + \phi(|U|)$ (resp. $\check{\alpha} = \dot{\alpha}$ and $\hat{\alpha} = \bar{\alpha} + \phi(|U|)$) we can deduce that

$$\begin{aligned} \|\check{\alpha}\|_{\mathcal{E}_{[t, T]}^\infty(\gamma)} &\leq \|\alpha\|_{\mathcal{E}_{[t, T]}^\infty(p\gamma)} + \phi(K)(T - t) + \frac{\ln 2}{\gamma} + nC_{p, n, \gamma, \lambda, \delta} K^{\frac{1+\delta}{1-\delta}}(T - t) \\ &\quad + C_{p, n, \gamma, \lambda, \delta} T \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}}^{2\frac{1+\delta}{1-\delta}} < +\infty, \quad t \in [T - \varepsilon, T] \end{aligned} \quad (3.23)$$

and

$$\begin{aligned} \|\dot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} &\leq 1 + \|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + \phi(K)(T - t) + n\bar{\lambda} K^{\frac{1+\delta}{2}}(T - t)^{\frac{1-\delta}{2}} \\ &\quad + \bar{c} \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}}^2 < +\infty, \quad t \in [T - \varepsilon, T], \end{aligned} \quad (3.24)$$

where the constant $C_{p, n, \gamma, \lambda, \delta}$ is defined in (3.10). The last three inequalities (3.22)-(3.24) imply that on $\Omega \times [T - \varepsilon, T]$, the generator f^i satisfies Assumption (A1) in Appendix with $\bar{\beta} = 0$ and $\varphi(\cdot) \equiv 0$. Since f^i also satisfies Assumption (A5), it follows from (iii) of Proposition A.1 that BSDE (3.14) admits a unique solution $(Y^i, Z^i) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$. Moreover, by (i) of Proposition A.1 we have

$$\begin{aligned} \|Y^i\|_{\mathcal{S}_{[t, T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t, T]}}^2 &\leq \frac{2 + \bar{\gamma}}{\gamma} \left\{ 2 + \frac{\ln 2}{\gamma} + 3\|\xi^i\|_\infty + \|\alpha\|_{\mathcal{E}_{[t, T]}^\infty(p\gamma)} + 2\|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} \right. \\ &\quad \left. + 3\phi(K)(T - t) + (C_{p, n, \gamma, \lambda, \delta} T + 2\bar{c}) \sum_{j=1}^{i-1} \left(\|Z^j\|_{\text{BMO}_{[t, T]}}^{2\frac{1+\delta}{1-\delta}} + \|Z^j\|_{\text{BMO}_{[t, T]}}^2 \right) \right. \\ &\quad \left. + nC_{p, n, \gamma, \lambda, \delta} K^{\frac{1+\delta}{1-\delta}}(T - t) + 2n\bar{\lambda} K^{\frac{1+\delta}{2}}(T - t)^{\frac{1-\delta}{2}} \right\}, \quad t \in [T - \varepsilon, T]. \end{aligned} \quad (3.25)$$

Then, in view of the inequality

$$a^2 \leq 1 + a^{2\frac{1+\delta}{1-\delta}}, \quad \forall a \geq 0, \quad (3.26)$$

it follows from (3.25) and (3.12) together with the definitions of C_1 and C_2 that the desired inequality (3.15) holds. Moreover, in view of the following inequality

$$\sum_{j=1}^m a_j^{\frac{1+\delta}{1-\delta}} \leq \left(\sum_{j=1}^m a_j \right)^{\frac{1+\delta}{1-\delta}}, \quad \forall m \geq 1 \text{ and } \forall a_j \geq 0, \quad j = 1, \dots, m,$$

combining (3.15) and (3.20) with $l = i - 1$ as well as (3.11) we can derive that inequality (3.16) also holds. Finally, the desired inequality (3.17) follows from (3.13) and (3.16). Consequently, the assertion (b) is proved in this case.

(2) g^i satisfies (ii) of Assumption (B1). For this case, it follows from the definition of f^i that $d\mathbb{P} \times dt - a.e.$ on $\Omega \times [T - \varepsilon, T]$, for each $z \in \mathbb{R}^{1 \times d}$, we have

$$|f^i(t, z)| \leq \ddot{\alpha}_t + \bar{u}_t |z| + \frac{\gamma}{2} |z|^2,$$

where $\ddot{\alpha} := \ddot{\alpha} + \phi(|U|) \in \mathcal{L}_{[T-\varepsilon, T]}^\infty$ and

$$\bar{u} := c \sum_{j=1}^{i-1} |Z^j| + \sum_{j=i+1}^n (\lambda |V^j|^\delta + \theta |V^j|) + \phi(|U|) + v \in \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}).$$

This means that on $\Omega \times [T - \varepsilon, T]$, the generator f^i satisfies Assumption (A3) in Appendix with $\bar{\beta} = 0$ and $\varphi(\cdot) \equiv 0$. Since f^i also satisfies Assumption (A5), it follows from (iii) of Proposition A.2 in Appendix that BSDE (3.14) has a unique solution $(Y^i, Z^i) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$. Moreover, in view of (i) of Proposition A.2 with $r = T$, (3.13) and (3.12) along with the fact that for each $t \in [T - \varepsilon, T]$, $\tau \in \mathcal{T}_{[t, T]}$ and $j = i + 1, \dots, n$, it holds that $5n\theta^2 \|V\|_{\text{BMO}_{[t, T]}}^2 \leq 5n\theta K \leq 2$ and

$$\mathbb{E}_\tau \left[\int_\tau^T |V_s^j|^{2\delta} ds \right] \leq \left(\mathbb{E}_\tau \left[\int_\tau^T |V_s^j|^2 ds \right] \right)^\delta (T - t)^{1-\delta} \leq K^\delta \varepsilon^{1-\delta} \leq \frac{K^\delta}{1 + 5n\lambda^2 K^\delta},$$

we have for each $t \in [T - \varepsilon, T]$,

$$\begin{aligned} \|Y^i\|_{\mathcal{S}_{[t, T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t, T]}}^2 &\leq \frac{4(\gamma + 1)}{\gamma^2} \exp \left\{ 4\gamma \left(\|\xi^i\|_\infty + \|\ddot{\alpha}\|_{\mathcal{L}_{[t, T]}^\infty} + 1 \right) \right\} \\ &\times \left(9 + \|\ddot{\alpha}\|_{\mathcal{L}_{[t, T]}^\infty} + 9\|v\|_{\text{BMO}_{[t, T]}}^2 + 5nc^2 \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}}^2 \right). \end{aligned} \quad (3.27)$$

Then, in view of (3.26), it follows from (3.27) together with the definitions of constants C_1 and C_2 that the desired inequality (3.15) holds. And, in the same way as in (1) we can deduce that (3.16) and (3.17) also hold. Thus, the assertion (b) is proved in this case.

(3) g^i satisfies (iii) of Assumption (B1). For this case, it follows from the definition of f^i that $d\mathbb{P} \times dt - a.e.$ on $\Omega \times [T - \varepsilon, T]$, for each $z \in \mathbb{R}^{1 \times d}$, we have

$$|f^i(t, z)| \leq \ddot{\alpha}_t + \bar{\lambda} |z|, \quad (3.28)$$

where

$$\ddot{\alpha}_t := \bar{\alpha}_t + \phi(|U_t|) + \sum_{j=1}^{i-1} \left(\bar{\lambda} |Z_t^j| + \theta |Z_t^j|^2 \right) + \sum_{j=i+1}^n \left(\bar{\lambda} |V_t^j|^{1+\delta} + \theta |V_t^j|^2 \right).$$

Since $\bar{\alpha} \in \mathcal{M}_{[T-\varepsilon, T]}^\infty$ and $\phi(|U|) \in \mathcal{L}_{[T-\varepsilon, T]}^\infty$, in view of (3.12), (3.13) and (3.21) with $l = i - 1$, using (iii) of Lemma 3.1 with $\bar{t} = T - \varepsilon$, $\bar{\delta} = \delta$, $\bar{V} = Z$ and $\hat{\alpha} = \bar{\alpha} + \phi(|U|)$ we can deduce that

$$\begin{aligned} \|\ddot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} &\leq 1 + \|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + \phi(K)(T - t) + n\bar{\lambda} K^{\frac{1+\delta}{2}} (T - t)^{\frac{1-\delta}{2}} + \bar{\lambda} \sqrt{T} \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}} \\ &\leq 3 + \|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + \bar{\lambda} \sqrt{T} \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}} < +\infty, \quad t \in [T - \varepsilon, T]. \end{aligned} \quad (3.29)$$

Combining (3.28) and (3.29) yields that on $\Omega \times [T - \varepsilon, T]$, the generator f^i satisfies Assumption (A4) in Appendix with $\dot{\alpha} = \check{\alpha}$, $\bar{\beta} = 0$ and $\varphi(\cdot) \equiv 0$. Since f^i also satisfies Assumption (A5), it follows from (iii) of Proposition A.3 in Appendix that BSDE (3.14) admits a unique solution (Y^i, Z^i) in the space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$. Moreover, in view of (3.26), by (i) of Proposition A.3 we have for each $t \in [T - \varepsilon, T]$,

$$\begin{aligned} \|Y^i\|_{\mathcal{S}_{[t, T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t, T]}}^2 &\leq 1 + c_0 \exp(2\bar{\lambda}^2 T) \left(54 + \|\xi^i\|_\infty^2 + 6\|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty}^2 \right) \\ &\quad + 12nc_0\bar{\lambda}^2 T \exp(2\bar{\lambda}^2 T) \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}}^2, \end{aligned} \quad (3.30)$$

where the uniform constant $c_0 > 0$ is defined in (i) of Proposition A.3. Then, it follows from (3.30) together with the definitions of C_1 and C_2 that the desired inequality (3.15) holds. And, in the same way as in (1) we can deduce that (3.16) and (3.17) hold. Thus, the assertion (b) is also proved in this case.

Next, we prove the assertion (a). Indeed, in view of (3.13), by applying the above argument to $i = 1$ we can deduce that for $i = 1$, BSDE (3.14) admits a unique solution (Y^1, Z^1) in the space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$ and (3.15) holds. In addition, in the case of $i = 1$, (3.16) is just (3.15), and (3.17) is trivially satisfied by (3.13). Thus, the assertion (a) is also true, and the proof of Proposition 3.2 is then complete. $\square$

Now, for each real $\varepsilon > 0$ satisfying (3.12), define the following complete metric space

$$\mathcal{B}_\varepsilon := \left\{ (U, V) \in \mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{n \times d}) : \|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty} + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq K \right\},$$

which is a closed convex subset in the Banach space $\mathcal{S}^\infty(\mathbb{R}^n) \times \text{BMO}(\mathbb{R}^{n \times d})$ with the norm

$$\|(U, V)\|_{\mathcal{B}_\varepsilon} := \sqrt{\|U\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2}, \quad \forall (U, V) \in \mathcal{B}_\varepsilon.$$

Based on Proposition 3.2, we know that those assertions from (3.14) to (3.18) are all true. Thus, in the case of $0 \leq \theta \leq 1/[4n(q\gamma \vee 1)K]$ we can define the following map from $\mathcal{B}_\varepsilon$ to itself:

$$\Gamma : (U, V) \in \mathcal{B}_\varepsilon \mapsto \Gamma(U, V) := (Y, Z) \in \mathcal{B}_\varepsilon,$$

where for each $i = 1, \dots, n$, (Y^i, Z^i) (the i th component of Y and the i th row of Z) is the unique solution of BSDE (3.14) in the space $\mathcal{S}_{[T-\varepsilon, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-\varepsilon, T]}(\mathbb{R}^{1 \times d})$.

It remains to show that there exists a real $\varepsilon_0 > 0$ satisfying (3.12) and a real $\theta_0 \in (0, 1/[4n(q\gamma \vee 1)K])$ (both depending only on $\|\xi\|_\infty$, $\|\alpha\|_{\mathcal{E}^\infty(p\gamma)}$, $\|\bar{\alpha}\|_{\mathcal{M}^\infty}$, $\|\tilde{\alpha}\|_{\mathcal{L}^\infty}$, $\|v\|_{\text{BMO}}$, $n, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p$ and $\phi(\cdot)$) such that for each $\theta \in [0, \theta_0]$, Γ is a contraction in $\mathcal{B}_{\varepsilon_0}$.

In the sequel, let $0 \leq \theta \leq 1/[4n(q\gamma \vee 1)K]$. For any fixed ε satisfying (3.12) as well as $(U, V) \in \mathcal{B}_\varepsilon$ and $(\tilde{U}, \tilde{V}) \in \mathcal{B}_\varepsilon$, we set

$$(Y, Z) := \Gamma(U, V), \quad (\tilde{Y}, \tilde{Z}) := \Gamma(\tilde{U}, \tilde{V}).$$

That is, for $i = 1, \dots, n$ and $t \in [T - \varepsilon, T]$, we have

$$Y_t^i = \xi^i + \int_t^T g^i(s, U_s, V_s(Z_s, Z_s^i; i)) ds - \int_t^T Z_s^i dB_s$$

and

$$\tilde{Y}_t^i = \xi^i + \int_t^T g^i(s, \tilde{U}_s, \tilde{V}_s(\tilde{Z}_s, \tilde{Z}_s^i; i)) ds - \int_t^T \tilde{Z}_s^i dB_s.$$

Here and hereafter, for each $i = 1, \dots, n$, $w \in \mathbb{R}^{1 \times d}$ and $H, \bar{H} \in \mathbb{R}^{n \times d}$, we denote by $H(\bar{H}, w; i)$ the matrix in $\mathbb{R}^{n \times d}$ whose i th row is w , and whose j th row is $\bar{H}^j$ for

$j = 1, \dots, i-1$ with $i \neq 1$ and H^j for $j = i+1, \dots, n$ with $i \neq n$. Then, for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$, we have

$$\begin{aligned} Y_\tau^i - \tilde{Y}_\tau^i + \int_\tau^T (Z_s^i - \tilde{Z}_s^i) dB_s \\ - \int_\tau^T \underbrace{\left(g^i(s, U_s, V_s(Z_s, Z_s^i; i)) - g^i(s, U_s, V_s(Z_s, \tilde{Z}_s^i; i)) \right)}_{:= \Delta_s^{1,i}} ds \\ = \int_\tau^T \underbrace{\left(g^i(s, U_s, V_s(Z_s, \tilde{Z}_s^i; i)) - g^i(s, \tilde{U}_s, \tilde{V}_s(\tilde{Z}_s, \tilde{Z}_s^i; i)) \right)}_{:= \Delta_s^{2,i}} ds. \end{aligned} \quad (3.31)$$

It follows from Assumption (B2) that for each $s \in [T-\varepsilon, T]$ and each $i = 1, \dots, n$, we have

$$|\Delta_s^{1,i}| \leq \phi(|U_s|) \left(v_s + 2|V_s| + 2|Z_s| + |\tilde{Z}_s| \right) |Z_s^i - \tilde{Z}_s^i| \quad (3.32)$$

and

$$|\Delta_s^{2,i}| \leq \phi(|U_s| \vee |\tilde{U}_s|) \left[\tilde{v}_s |U_s - \tilde{U}_s| + \hat{v}_s \sum_{j=1}^{i-1} |Z_s^j - \tilde{Z}_s^j| + \sqrt{n}(\tilde{v}_s + \theta \hat{v}_s) |V_s - \tilde{V}_s| \right] \quad (3.33)$$

with

$$\tilde{v}_s := v_s^{1+\delta} + 3|V_s|^{1+\delta} + 2|\tilde{V}_s|^{1+\delta} + 3|Z_s|^{1+\delta} + 5|\tilde{Z}_s|^{1+\delta}, \quad (3.34)$$

$$\hat{v}_s := v_s + |V_s| + |\tilde{V}_s| + |Z_s| + 2|\tilde{Z}_s| \quad (3.35)$$

and

$$\tilde{v}_s := v_s^\delta + |V_s|^\delta + |\tilde{V}_s|^\delta + |Z_s|^\delta + 2|\tilde{Z}_s|^\delta. \quad (3.36)$$

For $i = 1, \dots, n$, define $G_s(i) \equiv 0$, $s \in [0, T-\varepsilon]$ and

$$G_s(i) =: \frac{(Z_s^i - \tilde{Z}_s^i)^\top}{|Z_s^i - \tilde{Z}_s^i|^2} \mathbf{1}_{|Z_s^i - \tilde{Z}_s^i| \neq 0} \Delta_s^{1,i}, \quad s \in [T-\varepsilon, T].$$

By (3.32) we know that for each $i = 1, \dots, n$ and $s \in [T-\varepsilon, T]$,

$$\Delta_s^{1,i} = (Z_s^i - \tilde{Z}_s^i) G_s(i) \quad \text{and} \quad |G_s(i)| \leq \phi(|U_s|) \left(v_s + 2|V_s| + 2|Z_s| + |\tilde{Z}_s| \right). \quad (3.37)$$

Then for each $i = 1, \dots, n$, $\tilde{B}_t(i) := B_t - \int_0^t G_s(i) ds$ is a Brownian motion under the probability measure $\mathbb{P}^i$ defined by

$$\frac{d\mathbb{P}^i}{d\mathbb{P}} := \exp \left\{ \int_0^T [G_s(i)]^\top dB_s - \frac{1}{2} \int_0^T |G_s(i)|^2 ds \right\},$$

and from the definition of $\mathcal{B}_\varepsilon$, there exists a constant $\bar{K} > 0$ such that

$$\| [G(i)]^\top \|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \bar{K} := 4[\phi(K)]^2 (\|v\|_{\text{BMO}}^2 + 9K).$$

Furthermore, it follows from Kazamaki [29, Theorem 3.3] that there exist constants $0 < L_1 \leq 1$ and $L_2 \geq 1$ depending only on $\bar{K}$ such that for any $M \in \text{BMO}(\mathbb{R}^{n \times d})$ or $M \in \text{BMO}(\mathbb{R}^{1 \times d})$, we have, for each $i = 1, \dots, n$,

$$L_1 \|M\|_{\text{BMO}_{[T-\varepsilon, T]}} \leq \|M\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)} \leq L_2 \|M\|_{\text{BMO}_{[T-\varepsilon, T]}}, \quad (3.38)$$

where

$$\|M\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)} := \sup_{\tau \in \mathcal{T}_{[T-\varepsilon, T]}} \left\| \mathbb{E}_{\tau}^i \left[\int_{\tau}^T |M_s|^2 ds \right] \right\|_{\infty}^{\frac{1}{2}}$$

and $\mathbb{E}_{\tau}^i$ denotes the conditional expectation with respect to $\mathcal{F}_{\tau}$ under the measure $\mathbb{P}^i$.

It follows from (3.31) and (3.37) that

$$Y_{\tau}^i - \tilde{Y}_{\tau}^i + \int_{\tau}^T (Z_s^i - \tilde{Z}_s^i) d\tilde{B}_s(i) = \int_{\tau}^T \Delta_s^{2,i} ds, \quad i = 1, \dots, n, \quad \tau \in \mathcal{T}_{[T-\varepsilon, T]}.$$

Taking square and then the conditional mathematical expectation under $\mathbb{P}^i$ on both sides of the last equation, in view of (3.33) and the definition of $\mathcal{B}_{\varepsilon}$ together with Hölder's inequality we can deduce that for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$,

$$\begin{aligned} |Y_{\tau}^i - \tilde{Y}_{\tau}^i|^2 + \mathbb{E}_{\tau}^i \left[\int_{\tau}^T |Z_s^i - \tilde{Z}_s^i|^2 ds \right] &\leq 3[\phi(K)]^2 \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T \tilde{v}_s ds \right)^2 \right] \|U - \tilde{U}\|_{\mathcal{S}_{[T-\varepsilon, T]}^{\infty}}^2 \\ &+ 3n[\phi(K)]^2 \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |\hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \sum_{j=1}^{i-1} \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |Z_s^j - \tilde{Z}_s^j|^2 ds \right)^2 \right]^{\frac{1}{2}} \\ &+ 3n[\phi(K)]^2 \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |\check{v}_s + \theta \hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |V_s - \tilde{V}_s|^2 ds \right)^2 \right]^{\frac{1}{2}}. \end{aligned} \quad (3.39)$$

It follows from the energy inequality for BMO martingales (see for example Section 2.1 in Kazamaki [29]) together with (3.38) that for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$,

$$\begin{aligned} \sum_{j=1}^{i-1} \mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |Z_s^j - \tilde{Z}_s^j|^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \sum_{j=1}^{i-1} \|Z^j - \tilde{Z}^j\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\ &\leq \sqrt{2} L_2^2 \sum_{j=1}^{i-1} \|Z^j - \tilde{Z}^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \end{aligned}$$

and

$$\mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |V_s - \tilde{V}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \leq \sqrt{2} \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \leq \sqrt{2} L_2^2 \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2.$$

Furthermore, in view of (3.34)-(3.36), using Hölder's inequality and the energy inequality for BMO martingales together with (3.38) and the definition of $\mathcal{B}_{\varepsilon}$ we can derive that for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$,

$$\mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T \tilde{v}_s ds \right)^2 \right] \leq 10 \varepsilon^{1-\delta} L_4^2 \|v\|_{\text{BMO}}^4 + 1000 \varepsilon^{1-\delta} (1 + L_2^4 K^2), \quad (3.40)$$

$$\mathbb{E}_{\tau}^i \left[\left(\int_{\tau}^T |\hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \leq 400 L_2^2 (\|v\|_{\text{BMO}}^2 + K), \quad (3.41)$$

and

$$\begin{aligned} & \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\check{v}_s + \theta \hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \\ & \leq 800 \theta^2 L_2^2 (\|v\|_{\text{BMO}}^2 + K) + 800 L_2^2 (\|v\|_{\text{BMO}}^2 + K + 1) \varepsilon^{1-\delta}. \end{aligned} \quad (3.42)$$

Their proof will be postponed to the end of this section. Combining those inequalities from (3.39) to (3.42) yields that for each $i = 1, \dots, n$,

$$\begin{aligned} & \|Y^i - \tilde{Y}^i\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + L_1^2 \|Z^i - \tilde{Z}^i\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \\ & \leq C_3 \varepsilon^{1-\delta} \left(\|U - \tilde{U}\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) + 2\theta^2 C_4 \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \\ & \quad + C_4 \sum_{j=1}^{i-1} \|Z^j - \tilde{Z}^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2, \end{aligned}$$

and then,

$$\begin{aligned} & \|Y^i - \tilde{Y}^i\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|Z^i - \tilde{Z}^i\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \\ & \leq \frac{C_3 \varepsilon^{1-\delta} + 2\theta^2 C_4}{L_1^2} \left(\|U - \tilde{U}\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right) \\ & \quad + \frac{C_4}{L_1^2} \sum_{j=1}^{i-1} \|Z^j - \tilde{Z}^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2, \end{aligned} \quad (3.43)$$

where

$$C_3 := 3[\phi(K)]^2 \left\{ 10L_2^4 \|v\|_{\text{BMO}}^4 + 1000 (1 + L_2^4 K^2) + 800\sqrt{2}nL_2^4 (\|v\|_{\text{BMO}}^2 + K + 1) \right\}$$

and

$$C_4 := 1200\sqrt{2}nL_2^4 [\phi(K)]^2 (\|v\|_{\text{BMO}}^2 + K).$$

Next, in view of (3.43), by induction for i we can deduce that for each $i = 1, \dots, n$,

$$\begin{aligned} & \sum_{j=1}^i \|Y^j - \tilde{Y}^j\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \sum_{j=1}^i \|Z^j - \tilde{Z}^j\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \\ & \leq \frac{C_3 \varepsilon^{1-\delta} + 2\theta^2 C_4}{L_1^2} C_5^i \left(\|U - \tilde{U}\|_{\mathcal{S}_{[T-\varepsilon, T]}^\infty}^2 + \|V - \tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \right), \end{aligned}$$

where

$$C_5^1 := 1 \quad \text{and} \quad C_5^i := C_5^{i-1} + 1 + \frac{C_4}{L_1^2} C_5^{i-1} \quad \text{for } i = 2, \dots, n, \text{ recursively.}$$

In particular, letting $i = n$ in the last equation yields that

$$\left\| (Y - \tilde{Y}, Z - \tilde{Z}) \right\|_{\mathcal{B}_\varepsilon}^2 \leq \frac{C_3 \varepsilon^{1-\delta} + 2\theta^2 C_4}{L_1^2} C_5^n \left\| (U - \tilde{U}, V - \tilde{V}) \right\|_{\mathcal{B}_\varepsilon}^2.$$

Consequently, there exist two very small positive numbers ε_0 and θ_0 such that for each $\theta \in [0, \theta_0]$, the solution map Γ is a contraction on the previously given set $\mathcal{B}_{\varepsilon_0}$.

Finally, we remark that in view of (i) of Lemma 3.1, all above arguments remain valid for $p = 1$ and $q = 0$ when $\lambda = \theta = 0$. The proof of Theorem 2.3 is then completed.

In a similar way to Hu and Tang [24, pages 1078-1079], we now prove (3.40)-(3.42). First of all, from the energy inequality for BMO martingales, (3.38) and the definition of $\mathcal{B}_\varepsilon$, we have for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$,

$$\begin{aligned}
 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |V_s|^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \|V\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\
 &\leq \sqrt{2} L_2^2 \|V\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \sqrt{2} L_2^2 K, \\
 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \|\tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\
 &\leq \sqrt{2} L_2^2 \|\tilde{V}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \sqrt{2} L_2^2 K, \\
 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |Z_s|^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \|Z\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\
 &\leq \sqrt{2} L_2^2 \|Z\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \sqrt{2} L_2^2 K, \\
 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \|\tilde{Z}\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\
 &\leq \sqrt{2} L_2^2 \|\tilde{Z}\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \sqrt{2} L_2^2 K, \\
 \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^2 \right]^{\frac{1}{2}} &\leq \sqrt{2} \|v\|_{\text{BMO}_{[T-\varepsilon, T]}(\mathbb{P}^i)}^2 \\
 &\leq \sqrt{2} L_2^2 \|v\|_{\text{BMO}_{[T-\varepsilon, T]}}^2 \leq \sqrt{2} L_2^2 \|v\|_{\text{BMO}}^2.
 \end{aligned} \tag{3.44}$$

Furthermore, by Hölder's inequality we have that for each $i = 1, \dots, n$ and $\tau \in \mathcal{T}_{[T-\varepsilon, T]}$,

$$\begin{aligned}
 &\mathbb{E}_\tau^i \left[\left(\int_\tau^T \tilde{v}_s ds \right)^2 \right] \\
 &\leq 5\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^{1+\delta} \right] + 45\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[\left(\int_\tau^T |V_s|^2 ds \right)^{1+\delta} \right] \\
 &\quad + 5\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[4 \left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^{1+\delta} + 9 \left(\int_\tau^T |Z_s|^2 ds \right)^{1+\delta} + 25 \left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^{1+\delta} \right] \\
 &\leq 240\varepsilon^{1-\delta} + 5\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^2 \right] + 45\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[\left(\int_\tau^T |V_s|^2 ds \right)^2 \right] \\
 &\quad + 125\varepsilon^{1-\delta} \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^2 + \left(\int_\tau^T |Z_s|^2 ds \right)^2 + \left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^2 \right],
 \end{aligned} \tag{3.45}$$

$$\begin{aligned}
& \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\check{v}_s|^2 ds \right)^2 \right] \\
& \leq 25 \mathbb{E}_\tau^i \left[\left(\int_\tau^T \left(v_s^{2\delta} + |V_s|^{2\delta} + |\tilde{V}_s|^{2\delta} + |Z_s|^{2\delta} + 4|\tilde{Z}_s|^{2\delta} \right) ds \right)^2 \right] \\
& \leq 125 \varepsilon^{2(1-\delta)} \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^{2\delta} + \left(\int_\tau^T |V_s|^2 ds \right)^{2\delta} + \left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^{2\delta} \right] \\
& \quad + 125 \varepsilon^{2(1-\delta)} \mathbb{E}_\tau^i \left[\left(\int_\tau^T |Z_s|^2 ds \right)^{2\delta} + 16 \left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^{2\delta} \right] \\
& \leq 2500 \varepsilon^{2(1-\delta)} + 125 \varepsilon^{2(1-\delta)} \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^2 + \left(\int_\tau^T |V_s|^2 ds \right)^2 \right] \\
& \quad + 2000 \varepsilon^{2(1-\delta)} \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^2 + \left(\int_\tau^T |Z_s|^2 ds \right)^2 + \left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^2 \right]
\end{aligned} \tag{3.46}$$

and

$$\begin{aligned}
& \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\hat{v}_s|^2 ds \right)^2 \right] \\
& \leq 25 \mathbb{E}_\tau^i \left[\left(\int_\tau^T \left(v_s^2 + |V_s|^2 + |\tilde{V}_s|^2 + |Z_s|^2 + 4|\tilde{Z}_s|^2 \right) ds \right)^2 \right] \\
& \leq 125 \mathbb{E}_\tau^i \left[\left(\int_\tau^T v_s^2 ds \right)^2 + \left(\int_\tau^T |V_s|^2 ds \right)^2 + \left(\int_\tau^T |\tilde{V}_s|^2 ds \right)^2 + \left(\int_\tau^T |Z_s|^2 ds \right)^2 \right] \\
& \quad + 2000 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\tilde{Z}_s|^2 ds \right)^2 \right].
\end{aligned} \tag{3.47}$$

Finally, in view of

$$\begin{aligned}
& \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\check{v}_s + \theta \hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} \\
& \leq 2 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\check{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}} + 2\theta^2 \mathbb{E}_\tau^i \left[\left(\int_\tau^T |\hat{v}_s|^2 ds \right)^2 \right]^{\frac{1}{2}},
\end{aligned}$$

it is immediate to have the desired inequalities (3.40)-(3.42) from (3.44)-(3.47).

4 Global bounded solution: Proof of Theorems 2.10 and 2.14

To prove the existence of global bounded solution, we need some uniform estimates of the solution.

4.1 Proof of Theorem 2.10

We need the following proposition.

Proposition 4.1. Let $\xi \in L^\infty(\mathbb{R}^n)$, $\alpha \in \mathcal{E}^\infty(p\gamma \exp(\beta T))$ for $p > 1$ and the generator g satisfy Assumption (C1a). Assume that for some $h \in (0, T]$, BSDE (2.1) has a solution $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{n \times d})$ on the time interval $[T-h, T]$. Then, there exists a $\tilde{K} > 0$ depending only on $(\|\xi\|_\infty, \|\alpha\|_{\mathcal{E}^\infty(p\gamma \exp(\beta T))}, \|\tilde{\alpha}\|_{\mathcal{M}^\infty}, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, \|v\|_{\text{BMO}}, n, \beta, \gamma, \bar{\gamma}, \lambda, \bar{\lambda}, c, \bar{c}, \delta, T, p)$ and being independent of h such that for each $\theta \in [0, 1/[4n(q\gamma \vee 1)\tilde{K}]]$ with $q := \frac{p}{p-1}$, we have

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty} + \|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq \tilde{K}.$$

Moreover, the preceding assertion is still true for $p = 1$ and $q = 0$ when $\lambda = 0$ and $\theta = 0$.

Proof. We only prove the case of $p > 1$. The other case can be proved in the same way.

First of all, define an increasing non-negative real-valued function $\Phi : [0, +\infty) \rightarrow [0, +\infty)$ as follows:

$$\begin{aligned} \Phi(x) := & \frac{2(1 + \beta T) + \bar{\gamma}}{\bar{\gamma}} \exp(2\beta T) \\ & \times \left[2 + 2n\bar{\lambda}T^{\frac{1-\delta}{2}} + \frac{\ln 2}{\gamma \exp(\beta T)} + \|\alpha\|_{\mathcal{E}^\infty(p\gamma \exp(\beta T))} + 2\|\tilde{\alpha}\|_{\mathcal{M}^\infty} + 3x \right] \\ & + \frac{4(\gamma + 1)}{\gamma^2} \exp\{4\gamma \exp(\beta T) (\|\xi\|_\infty + \|\tilde{\alpha}\|_{\mathcal{L}^\infty})\} \\ & \times \left[\beta T \exp(\beta T) (\|\xi\|_\infty + \|\tilde{\alpha}\|_{\mathcal{L}^\infty}) + 3 + 4n\lambda^2 T^{1-\delta} + \|\tilde{\alpha}\|_{\mathcal{L}^\infty} + 4\|v\|_{\text{BMO}_{[t, T]}}^2 \right] \\ & + 1 + c_0 \exp(2\beta T + 2\bar{\lambda}^2 T) (6 + 6\|\tilde{\alpha}\|_{\mathcal{M}^\infty}^2 + x^2), \end{aligned}$$

and define the following four constants:

$$\begin{aligned} \bar{C}_1 &:= \Phi(\|\xi\|_\infty), \quad \bar{C}_2 := \frac{6\beta(1 + \beta T) + 3\beta\bar{\gamma}}{\bar{\gamma}} \exp(2\beta T), \\ \bar{C}_3 &:= \frac{2(1 + \beta T) + \bar{\gamma}}{\bar{\gamma}} \exp(2\beta T) (C_{p, n, \beta, \gamma, \lambda, T} + 2n\bar{\lambda}) \\ &+ \frac{16n\lambda^2(\gamma + 1)}{\gamma^2} \exp\{4\gamma \exp(\beta T) (\|\xi\|_\infty + \|\tilde{\alpha}\|_{\mathcal{L}^\infty})\} + 12nc_0\bar{\lambda}^2 \exp(2\beta T + 2\bar{\lambda}^2 T) \end{aligned}$$

and

$$\bar{C}_4 := \frac{4\bar{c}(1 + \beta T) + 2\bar{c}\bar{\gamma}}{\bar{\gamma}} \exp(2\beta T) + \frac{16nc^2(\gamma + 1)}{\gamma^2} \exp\{4\gamma \exp(\beta T) (\|\xi\|_\infty + \|\tilde{\alpha}\|_{\mathcal{L}^\infty})\},$$

where the uniform constant $c_0 > 0$ is defined in (i) of Proposition A.3 in Appendix, and

$$C_{p, n, \beta, \gamma, \lambda, T} := \frac{n\gamma p\lambda^2}{2(p-1)} \exp(\beta T). \quad (4.1)$$

Note that $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty \times \text{BMO}_{[T-h, T]}$ is a solution of BSDE (2.2) on $[T-h, T]$. Then for each $i = 1, \dots, n$, (Y, Z) also solves the following BSDE:

$$Y_t^i = \xi^i + \int_t^T f^i(s, Y_s^i, Z_s^i) ds - \int_t^T Z_s^i dB_s, \quad t \in [T-h, T], \quad (4.2)$$

where for each $(\omega, t, y, z) \in \Omega \times [T-h, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d}$,

$$f^i(t, y, z) := g^i(t, Y_t(y; i), Z_t(z; i)). \quad (4.3)$$

In the sequel, let $q := \frac{p}{p-1}$ and the constant $\theta \in [0, 1]$ always satisfy

$$4n(q\gamma \vee 1)\theta\|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq 1. \quad (4.4)$$

We will first prove that for each $i = 1, \dots, n$ and each $t \in [T-h, T]$, if $T-t \leq 1$, then

$$\begin{aligned} & \|Y^i\|_{\mathcal{S}_{[T-t, T]}^\infty}^2 + \|Z^i\|_{\text{BMO}_{[T-t, T]}}^2 \\ & \leq \bar{C}_1 + \bar{C}_2\|Y\|_{\mathcal{S}_{[T-t, T]}^\infty}(T-t) + \bar{C}_3\|Z\|_{\text{BMO}_{[T-t, T]}}^2(T-t)^{\frac{1-\delta}{2}} \\ & \quad + \bar{C}_4 \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[T-t, T]}}^2. \end{aligned} \quad (4.5)$$

Indeed, for any fixed $i = 1, \dots, n$, since g satisfies Assumption (C1a), in view of Remark 2.9, we need to consider the following three cases.

(1) g^i satisfies (i) of Assumption (C1a). In this case, it follows from (4.3) that $d\mathbb{P} \times dt - a.e.$ on $\Omega \times [T-h, T]$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$f^i(t, y, z) \mathbf{1}_{y>0} \leq \check{\alpha}_t + \beta|y| + \frac{\gamma}{2}|z|^2 \quad \text{and} \quad f^i(t, y, z) \geq \frac{\bar{\gamma}}{2}|z|^2 - \check{\alpha}_t - \beta|y|, \quad (4.6)$$

where

$$\check{\alpha}_t := \alpha_t + \beta|Y_t| + \sum_{j \neq i} \left(\lambda|Z_t^j| + \theta|Z_t^j|^2 \right)$$

and

$$\hat{\alpha}_t := \bar{\alpha}_t + \beta|Y_t| + \bar{c} \sum_{j=1}^{i-1} |Z_t^j|^2 + \sum_{j=i+1}^n \left(\bar{\lambda}|Z_t^j|^{1+\delta} + \theta|Z_t^j|^2 \right).$$

Since $\alpha \in \mathcal{E}_{[T-h, T]}^\infty(p\gamma \exp(\beta T))$, $\bar{\alpha} \in \mathcal{M}_{[T-h, T]}^\infty$ and $|Y| \in \mathcal{L}_{[T-h, T]}^\infty$, in view of (ii) and (i) of Remark 2.1 as well as (4.4), applying (i) of Lemma 3.1 with

$$r = \gamma \exp(\beta T), \quad \bar{\delta} = 0, \quad \bar{t} = T-h, \quad V = \bar{V} = Z, \quad \check{\alpha} = \check{\alpha} \quad \text{and} \quad \hat{\alpha} = \alpha + \beta|Y|$$

and (ii) of Lemma 3.1 with

$$\bar{\delta} = \delta, \quad \bar{t} = T-h, \quad V = \bar{V} = Z, \quad \check{\alpha} = \check{\alpha} \quad \text{and} \quad \hat{\alpha} = \bar{\alpha} + \beta|Y|$$

we can deduce that

$$\begin{aligned} \|\check{\alpha}\|_{\mathcal{E}_{[t, T]}^\infty(\gamma \exp(\beta T))} & \leq \|\alpha\|_{\mathcal{E}_{[t, T]}^\infty(p\gamma \exp(\beta T))} + \beta\|Y\|_{\mathcal{S}_{[t, T]}^\infty}(T-t) + \frac{\ln 2}{\gamma \exp(\beta T)} \\ & \quad + C_{p, n, \beta, \gamma, \lambda, T} \|Z\|_{\text{BMO}_{[t, T]}}^2(T-t) < +\infty, \quad t \in [T-h, T] \end{aligned} \quad (4.7)$$

and

$$\begin{aligned} \|\hat{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} & \leq 1 + \|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + \beta\|Y\|_{\mathcal{S}_{[t, T]}^\infty}(T-t) + n\bar{\lambda}\|Z\|_{\text{BMO}_{[t, T]}}^{1+\delta}(T-t)^{\frac{1-\delta}{2}} \\ & \quad + \bar{c} \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t, T]}}^2 < +\infty, \quad t \in [T-h, T], \end{aligned} \quad (4.8)$$

where the constant $C_{p, n, \beta, \gamma, \lambda, T}$ is defined in (4.1). Combining (4.6), (4.7) and (4.8) yields that on $\Omega \times [T-h, T]$, the generator f^i satisfies the second inequality for the case of

$y > 0$ and the first inequality in Assumption (A1) with $\bar{\beta} = \beta$ and $\varphi(\cdot) \equiv 0$. It then follows from (i) of Proposition A.1 that for each $t \in [T - h, T]$ such that $T - t \leq 1$, we have

$$\begin{aligned} \|Y^i\|_{S_{[t,T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t,T]}}^2 &\leq \frac{2(1 + \beta T) + \bar{\gamma}}{\bar{\gamma}} \exp(2\beta T) \left\{ 2 + 2n\bar{\lambda}T^{\frac{1-\delta}{2}} + \frac{\ln 2}{\bar{\gamma}\exp(\beta T)} \right. \\ &\quad + 3\|\xi^i\|_\infty + \|\alpha\|_{\mathcal{E}^\infty(p\gamma\exp(\beta T))} + 2\|\bar{\alpha}\|_{\mathcal{M}^\infty} + 3\beta\|Y\|_{S_{[t,T]}^\infty}(T-t) \\ &\quad \left. + (C_{p,n,\beta,\gamma,\lambda,T} + 2n\bar{\lambda})\|Z\|_{\text{BMO}_{[t,T]}}^2(T-t)^{\frac{1-\delta}{2}} + 2\bar{c}\sum_{j=1}^{i-1}\|Z^j\|_{\text{BMO}_{[t,T]}}^2 \right\}. \end{aligned} \quad (4.9)$$

Then, it follows from (4.9) together with the definitions of constants $\bar{C}_1$, $\bar{C}_2$, $\bar{C}_3$ and $\bar{C}_4$ that the desired inequality (4.5) holds in this case when $T - t \leq 1$.

(2) g^i satisfies (ii) of Assumption (C1a). For this case, it follows from (4.3) that $dP \times dt - a.e.$ on $\Omega \times [T - h, T]$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$f^i(t, y, z) \operatorname{sgn}(y) \leq \bar{\alpha}_t + \beta|y| + \bar{u}_t|z| + \frac{\gamma}{2}|z|^2,$$

where

$$\bar{u} := c \sum_{j=1}^{i-1} |Z^j| + \sum_{j=i+1}^n (\lambda|Z^j|^\delta + \theta|Z^j|) + v \in \text{BMO}_{[T-h,T]}(\mathbb{R}).$$

This means that on $\Omega \times [T - h, T]$, the generator f^i satisfies the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in (A3) with $\ddot{\alpha} = \bar{\alpha}$ and $\bar{\beta} = \beta$. Then, in view of (4.4) and the fact that for each $t \in [T - h, T]$, $\tau \in \mathcal{T}_{[t,T]}$ and $j = i + 1, \dots, n$, it holds that $5n\theta^2\|Z\|_{\text{BMO}_{[t,T]}}^2 \leq 2$ and

$$\mathbb{E}_\tau \left[\int_\tau^T |Z_s^j|^{2\delta} ds \right] \leq \left(\mathbb{E}_\tau \left[\int_\tau^T |Z_s^j|^2 ds \right] \right)^\delta (T-t)^{1-\delta} \leq (1 + \|Z\|_{\text{BMO}_{[t,T]}}^2)(T-t)^{1-\delta},$$

it follows from (i) of Proposition A.2 with $r = T$ that for each $t \in [T - h, T]$,

$$\begin{aligned} \|Y^i\|_{S_{[t,T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t,T]}}^2 &\leq \frac{4(\gamma + 1)}{\gamma^2} \exp\{4\gamma\exp(\beta T)(\|\xi^i\|_\infty + \|\bar{\alpha}\|_{\mathcal{L}^\infty})\} \left(3 + 4n\lambda^2 T^{1-\delta} \right. \\ &\quad + \|\bar{\alpha}\|_{\mathcal{L}^\infty} + \beta T \exp(\beta T)(\|\xi^i\|_\infty + \|\bar{\alpha}\|_{\mathcal{L}^\infty}) + 4\|v\|_{\text{BMO}_{[t,T]}}^2 \\ &\quad \left. + 4nc^2 \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[t,T]}}^2 + 4n\lambda^2 \|Z\|_{\text{BMO}_{[t,T]}}^2 (T-t)^{1-\delta} \right). \end{aligned} \quad (4.10)$$

Then, it follows from (4.10) together with the definitions of constants $\bar{C}_1$, $\bar{C}_2$, $\bar{C}_3$ and $\bar{C}_4$ that the desired inequality (4.5) holds for each $t \in [T - h, T]$ in this case.

(3) g^i satisfies (iii) of Assumption (C1a). For this case, it follows from (4.3) that $dP \times dt - a.e.$ on $\Omega \times [T - h, T]$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$f^i(t, y, z) \operatorname{sgn}(y) \leq \bar{\alpha}_t + \beta|y| + \bar{\lambda}|z|, \quad (4.11)$$

where

$$\bar{\alpha}_t := \bar{\alpha}_t + \sum_{j \neq i} \left(\bar{\lambda}|Z_t^j| + \theta|Z_t^j|^2 \right).$$

In view of (4.4), using (iii) of Lemma 3.1 with $\bar{\delta} = 0$, $\bar{t} = T - h$, $V = \bar{V} = Z$ and $\hat{\alpha} = \bar{\alpha}$ we can deduce that

$$\|\bar{\alpha}\|_{\mathcal{M}_{[t,T]}^\infty} \leq 1 + \|\bar{\alpha}\|_{\mathcal{M}_{[t,T]}^\infty} + \bar{\lambda}\sqrt{n}\|Z\|_{\text{BMO}_{[t,T]}}(T-t)^{\frac{1}{2}} < +\infty, \quad t \in [T - h, T]. \quad (4.12)$$

Combining (4.11) and (4.12) yields that on $\Omega \times [T - h, T]$, the generator f^i satisfies the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in Assumption (A4) with $\dot{\alpha} = \check{\alpha}$ and $\dot{\beta} = \beta$. It follows from (i) of Proposition A.3 that for each $t \in [T - h, T]$,

$$\begin{aligned} \|Y^i\|_{\mathcal{S}_{[t,T]}^\infty} + \|Z^i\|_{\text{BMO}_{[t,T]}}^2 &\leq 1 + c_0 \exp(2\beta T + 2\bar{\lambda}^2 T) (6 + \|\xi^i\|_\infty^2 + 6\|\bar{\alpha}\|_{\mathcal{M}^\infty}^2) \\ &\quad + 12nc_0\bar{\lambda}^2 \exp(2\beta T + 2\bar{\lambda}^2 T) \|Z\|_{\text{BMO}_{[t,T]}}^2 (T - t). \end{aligned} \quad (4.13)$$

Then, it follows from (4.13) together with the definitions of constants $\bar{C}_1$, $\bar{C}_2$, $\bar{C}_3$ and $\bar{C}_4$ that the desired inequality (4.5) holds in this case when $T - t \leq 1$.

Furthermore, in view of (4.5), by induction for i it is not difficult to derive that for each $i = 1, \dots, n$ and each $t \in [T - h, T]$, if $T - t \leq 1$, then

$$\begin{aligned} \sum_{j=1}^i \left(\|Y^j\|_{\mathcal{S}_{[t,T]}^\infty} + \|Z^j\|_{\text{BMO}_{[t,T]}}^2 \right) \\ \leq \left(\bar{C}_1 + \bar{C}_2 \|Y\|_{\mathcal{S}_{[T-t,T]}^\infty} (T - t) + \bar{C}_3 \|Z\|_{\text{BMO}_{[T-t,T]}}^2 (T - t)^{\frac{1-\delta}{2}} \right) \bar{C}_5^i, \end{aligned} \quad (4.14)$$

where

$$\bar{C}_5^1 := 1 \quad \text{and} \quad \bar{C}_5^i := \bar{C}_5^{i-1} + 1 + \bar{C}_4 \bar{C}_5^{i-1} \quad \text{for } i = 2, \dots, n, \text{ recursively.} \quad (4.15)$$

In particular, letting $i = n$ in (4.14) yields that for each $t \in [T - h, T]$, if $T - t \leq 1$, then

$$\begin{aligned} \|Y\|_{\mathcal{S}_{[t,T]}^\infty} + \|Z\|_{\text{BMO}_{[t,T]}}^2 &\leq \bar{C}_1 \bar{C}_5^n + \bar{C}_2 \bar{C}_5^n \|Y\|_{\mathcal{S}_{[T-t,T]}^\infty} (T - t) \\ &\quad + \bar{C}_3 \bar{C}_5^n \|Z\|_{\text{BMO}_{[T-t,T]}}^2 (T - t)^{\frac{1-\delta}{2}}. \end{aligned}$$

Then, we have

$$\|Y\|_{\mathcal{S}_{[T-\varepsilon,T]}^\infty} + \|Z\|_{\text{BMO}_{[T-\varepsilon,T]}}^2 \leq 2\bar{C}_1 \bar{C}_5^n, \quad (4.16)$$

where

$$\varepsilon := \min \left\{ h, 1, \frac{1}{2\bar{C}_2 \bar{C}_5^n}, \left(\frac{1}{2\bar{C}_3 \bar{C}_5^n} \right)^{\frac{2}{1-\delta}} \right\} > 0.$$

Finally, for $m \geq 1$, we define successively the following constants:

$$\bar{C}_6^1 := 2\bar{C}_1 \bar{C}_5^n \quad \text{and} \quad \bar{C}_6^{m+1} := \bar{C}_6^m + 2\Phi(\bar{C}_6^m) \bar{C}_5^n, \quad (4.17)$$

where $\bar{C}_5^n$ is defined in (4.15). And, we let m_0 be the unique positive integer satisfying $T - h \in [T - m_0\varepsilon, T - (m_0 - 1)\varepsilon)$, or equivalently,

$$\frac{h}{\varepsilon} \leq m_0 < \frac{h}{\varepsilon} + 1. \quad (4.18)$$

If $m_0 = 1$, it then follows from (4.16) and (4.17) that

$$\|Y\|_{\mathcal{S}_{[T-h,T]}^\infty} + \|Z\|_{\text{BMO}_{[T-h,T]}}^2 \leq \bar{C}_6^1 = \bar{C}_6^{m_0}.$$

If $m_0 = 2$, it then follows from (4.16) and (4.17) that

$$\|Y_{T-\varepsilon}\|_\infty \leq \|Y\|_{\mathcal{S}_{[T-\varepsilon,T]}^\infty} \leq \bar{C}_6^1. \quad (4.19)$$

Now, consider the following system of BSDEs

$$Y_t = Y_{T-\varepsilon} + \int_t^{T-\varepsilon} g(s, Y_s, Z_s) ds - \int_t^{T-\varepsilon} Z_s dB_s, \quad t \in [T - h, T - \varepsilon].$$

In view of (4.19) and the definition of the function $\Phi(\cdot)$, by virtue of Propositions A.1 to A.3 we can use a similar argument as that obtaining (4.5) to get that for each $i = 1, \dots, n$,

$$\begin{aligned} & \|Y^i\|_{\mathcal{S}_{[T-h, T-\varepsilon]}^\infty}^2 + \|Z^i\|_{\text{BMO}_{[T-h, T-\varepsilon]}}^2 \\ & \leq \Phi(\bar{C}_1^1) + \bar{C}_2 \|Y\|_{\mathcal{S}_{[T-h, T-\varepsilon]}^\infty}^2 \varepsilon + \bar{C}_3 \|Z\|_{\text{BMO}_{[T-h, T-\varepsilon]}}^2 \varepsilon^{\frac{1-\delta}{2}} + \bar{C}_4 \sum_{j=1}^{i-1} \|Z^j\|_{\text{BMO}_{[T-h, T-\varepsilon]}}^2. \end{aligned}$$

We would like to mention that for the proof of the last assertion, we need to use $T - \varepsilon$ instead of T and $Y_{T-\varepsilon}$ instead of η in Propositions A.1 and A.3, and use $T - \varepsilon$ instead of r in Proposition A.2. Moreover, in view of the definition of ε , by induction and a similar argument as that from (4.14) to (4.16) we can further get that

$$\|Y\|_{\mathcal{S}_{[T-h, T-\varepsilon]}^\infty}^2 + \|Z\|_{\text{BMO}_{[T-h, T-\varepsilon]}}^2 \leq 2\Phi(\bar{C}_6^1) \bar{C}_5^n.$$

Combining the last inequality and (4.16) yields that, in view of (4.17),

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty}^2 + \|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq \bar{C}_6^1 + 2\Phi(\bar{C}_6^1) \bar{C}_5^n = \bar{C}_6^2 = \bar{C}_6^{m_0},$$

Proceeding the above computation gives that if m_0 satisfies (4.18), then

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty}^2 + \|Z\|_{\text{BMO}_{[T-h, T]}}^2 \leq \bar{C}_6^{m_0} = \bar{C}_6^{\lceil \frac{h}{\varepsilon} \rceil} \leq \bar{C}_6^{1 + \lceil \frac{T}{\varepsilon_0} \rceil} =: \tilde{K}, \quad (4.20)$$

where $\bar{C}_6^{m_0}$ is defined in (4.17), $\lceil x \rceil$ stands for the minimum of integers which is equal to or bigger than $x \in \mathbb{R}$, and

$$\varepsilon_0 := \min \left\{ \frac{1}{2\bar{C}_2 \bar{C}_5^n}, \left(\frac{1}{2\bar{C}_3 \bar{C}_5^n} \right)^{\frac{2}{1-\delta}} \right\} \in (0, 1).$$

Then, the desired conclusion follows from (4.4) and (4.20). The proof is complete. $\square$

Proof of Theorem 2.10. With Theorem 2.3 and Proposition 4.1 in hands, we can closely follow the proof of Theorem 4.1 in Cheridito and Nam [9] to prove our Theorem 2.10. All the details are omitted here. $\square$

4.2 Proof of Theorem 2.14

To prove Theorem 2.14, we need the following proposition.

Proposition 4.2. Let $\xi \in L^\infty(\mathbb{R}^n)$ and the generator g satisfy Assumption (C1b). Assume that for some $h \in (0, T]$, BSDE (2.1) has a solution $(Y, Z) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}^n) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{n \times d})$ on the time interval $[T - h, T]$. Then, there exists a positive constant $\tilde{K} > 0$ depending only on $(\|\xi\|_\infty, \|\tilde{\alpha}\|_{\mathcal{L}^\infty}, n, \beta, \gamma, \bar{\gamma}, \lambda, \delta, T)$ and being independent of h such that

$$\|Y\|_{\mathcal{S}_{[T-h, T]}^\infty} \leq \tilde{K}.$$

Proof. Let $\xi \in L^\infty(\mathbb{R}^n)$ and $\tilde{\alpha} \in \mathcal{L}^\infty$ such that

$$\|\xi\|_\infty \leq C_1 \quad \text{and} \quad \left\| \int_0^T \tilde{\alpha}_t dt \right\|_\infty \leq C_2 \quad (4.21)$$

for two positive constants C_1 and C_2 . Denote respectively by n_1 , n_2 and n_3 the number of elements in J_1 , J_2 and J_3 such that $n_1 + n_2 + n_3 = n$. Since the generator g satisfies Assumption (C1b), in view of (i) of Remark 2.13, we need to consider the following three cases.

(1) For $i \in J_1$, g^i satisfies (i) of Assumption (C1b). In this case, for each $t \in [T - h, T]$, we have

$$g^i(\omega, t, Y_t(\omega), Z_t(\omega)) \operatorname{sgn}(Y_t^i(\omega)) \leq \tilde{\alpha}_t(\omega) + \beta|Y_t(\omega)| + \lambda \sum_{j \in J_1} |Z_t^j(\omega)|^{1+\delta} + \frac{\gamma}{2} |Z_t^i(\omega)|^2$$

and

$$g^i(\omega, t, Y_t(\omega), Z_t(\omega)) \geq \frac{\bar{\gamma}}{2} |Z_t^i(\omega)|^2 - \tilde{\alpha}_t(\omega) - \beta|Y_t(\omega)| - \lambda \sum_{j \in J_1} |Z_t^j(\omega)|^{1+\delta}.$$

Note that $(Y^i, Z^i) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{1 \times d})$ is a solution of BSDE (2.2) on $[T - h, T]$. According to Lemma A.2 in Fan et al. [17] and in view of the last two inequalities together with (4.21), we can get that for each $i \in J_1$ and $t \in [T - h, T]$,

$$\begin{aligned} \exp(\gamma|Y_t^i|) &\leq \exp\left(\gamma(C_1 + C_2) + \beta\gamma \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty} ds\right) \\ &\quad \times \mathbb{E}_t\left[\exp\left(\gamma\lambda \sum_{j \in J_1} \int_t^T |Z_s^j|^{1+\delta} ds\right)\right] \end{aligned} \quad (4.22)$$

and

$$\begin{aligned} &\mathbb{E}_t\left[\exp\left(\frac{\bar{\gamma}}{2}\varepsilon_0 \int_t^T |Z_s^i|^2 ds\right)\right] \\ &\leq \mathbb{E}_t\left[\exp\left(6\varepsilon_0\|Y^i\|_{\mathcal{S}_{[t, T]}^\infty} + 3\varepsilon_0\left\|\int_0^T \tilde{\alpha}_t ds\right\|_\infty\right.\right. \\ &\quad \left.\left.+ 3\varepsilon_0\beta \int_t^T |Y_s| ds + 3\varepsilon_0\lambda \sum_{j \in J_1} \int_t^T |Z_s^j|^{1+\delta} ds\right)\right] \\ &\leq \exp\left(6\varepsilon_0 \sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty} + 3\varepsilon_0 C_2 + 3\varepsilon_0\beta \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty} ds\right) \\ &\quad \times \mathbb{E}_t\left[\exp\left(3\varepsilon_0\lambda \sum_{j \in J_1} \int_t^T |Z_s^j|^{1+\delta} ds\right)\right], \end{aligned} \quad (4.23)$$

where

$$\varepsilon_0 := \frac{\bar{\gamma}}{9} \wedge \frac{\gamma}{24} > 0.$$

Furthermore, in view of (4.23), by Hölder's inequality we obtain that for $t \in [T - h, T]$,

$$\begin{aligned} &\mathbb{E}_t\left[\exp\left(\frac{\bar{\gamma}\varepsilon_0}{2n_1} \sum_{j \in J_1} \int_t^T |Z_s^j|^2 ds\right)\right] \\ &\leq \exp\left(6\varepsilon_0 \sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty} + 3\varepsilon_0 C_2 + 3\varepsilon_0\beta \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty} ds\right) \\ &\quad \times \mathbb{E}_t\left[\exp\left(3\varepsilon_0\lambda \sum_{j \in J_1} \int_t^T |Z_s^j|^{1+\delta} ds\right)\right]. \end{aligned} \quad (4.24)$$

By Young's inequality, observe that for each pair of $a, b > 0$,

$$ab^{1+\delta} = \left(\left(\frac{1+\delta}{2}\right)^{\frac{1+\delta}{1-\delta}} a^{\frac{2}{1-\delta}}\right)^{\frac{1-\delta}{2}} \left(\frac{2}{1+\delta} b^2\right)^{\frac{1+\delta}{2}} \leq b^2 + \frac{1-\delta}{2} \left(\frac{1+\delta}{2}\right)^{\frac{1+\delta}{1-\delta}} a^{\frac{2}{1-\delta}}. \quad (4.25)$$

Letting $a = 12n_1\lambda/\bar{\gamma}$ and $b = |Z_s|$ in (4.25), we have for each $j \in J_1$,

$$3\varepsilon_0\lambda|Z_s^j|^{1+\delta} = \frac{\bar{\gamma}\varepsilon_0}{4n_1} \left(\frac{12n_1\lambda}{\bar{\gamma}} |Z_s^j|^{1+\delta}\right) \leq \frac{\bar{\gamma}\varepsilon_0}{4n_1} |Z_s^j|^2 + C_3, \quad s \in [0, T], \quad (4.26)$$

with

$$C_3 := \frac{\bar{\gamma}\varepsilon_0(1-\delta)}{8n_1} \left(\frac{1+\delta}{2}\right)^{\frac{1+\delta}{1-\delta}} \left(\frac{12n_1\lambda}{\bar{\gamma}}\right)^{\frac{2}{1-\delta}}.$$

Coming back to (4.24), by (4.26) and Hölder's inequality we derive that for $t \in [T-h, T]$,

$$\begin{aligned} \mathbb{E}_t \left[\exp \left(\frac{\bar{\gamma}\varepsilon_0}{2n_1} \sum_{j \in J_1} \int_t^T |Z_s^j|^2 ds \right) \right] \\ \leq \exp \left(12\varepsilon_0 \sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t,T]}^\infty} + 6\varepsilon_0 C_2 + 2C_3 T + 6\varepsilon_0 \beta \int_t^T \|Y\|_{\mathcal{S}_{[s,T]}^\infty} ds \right). \end{aligned} \quad (4.27)$$

On the other hand, it follows from (4.22) and Jensen's inequality that

$$\begin{aligned} \exp \left(\gamma \sum_{j \in J_1} |Y_t^j| \right) &\leq \exp \left(n_1 \gamma (C_1 + C_2) + n_1 \gamma \beta \int_t^T \|Y\|_{\mathcal{S}_{[s,T]}^\infty} ds \right) \\ &\times \mathbb{E}_t \left[\exp \left(n_1 \gamma \lambda \sum_{j \in J_1} \int_t^T |Z_s^j|^{1+\delta} ds \right) \right], \quad t \in [T-h, T]. \end{aligned} \quad (4.28)$$

By letting $a = 2n_1^2 \gamma \lambda / \bar{\gamma} \varepsilon_0$ and $b = |Z_s|$ in (4.25), we have for each $j \in J_1$,

$$n_1 \gamma \lambda |Z_s^j|^{1+\delta} = \frac{\bar{\gamma}\varepsilon_0}{2n_1} \left(\frac{2n_1^2 \gamma \lambda}{\bar{\gamma}\varepsilon_0} |Z_s^j|^{1+\delta} \right) \leq \frac{\bar{\gamma}\varepsilon_0}{2n_1} |Z_s^j|^2 + C_4, \quad s \in [0, T], \quad (4.29)$$

where

$$C_4 := \frac{\bar{\gamma}\varepsilon_0(1-\delta)}{4n_1} \left(\frac{1+\delta}{2}\right)^{\frac{1+\delta}{1-\delta}} \left(\frac{2n_1^2 \gamma \lambda}{\bar{\gamma}\varepsilon_0}\right)^{\frac{2}{1-\delta}}.$$

Combining (4.27)-(4.29) yields that for each $t \in [T-h, T]$,

$$\begin{aligned} \sum_{j \in J_1} |Y_t^j| &\leq n_1(C_1 + C_2) + \frac{C_4 T}{\gamma} + \frac{6\varepsilon_0 C_2 + 2C_3 T}{\gamma} + \frac{12\varepsilon_0}{\gamma} \sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t,T]}^\infty} \\ &+ \beta \left(n_1 + \frac{6\varepsilon_0}{\gamma} \right) \int_t^T \|Y\|_{\mathcal{S}_{[s,T]}^\infty} ds. \end{aligned}$$

And, it follows from the definition of ε_0 that

$$\sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t,T]}^\infty} \leq C_5 + 2\beta \left(n_1 + \frac{1}{4} \right) \int_t^T \|Y\|_{\mathcal{S}_{[s,T]}^\infty} ds, \quad t \in [T-h, T], \quad (4.30)$$

where

$$C_5 := 2n_1(C_1 + C_2) + \frac{2C_4 T}{\gamma} + \frac{12\varepsilon_0 C_2 + 4C_3 T}{\gamma}.$$

By taking square in both sides of (4.30) and using Hölder's inequality, we can conclude that for each $t \in [T-h, T]$,

$$\sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t,T]}^\infty}^2 \leq \left(\sum_{j \in J_1} \|Y^j\|_{\mathcal{S}_{[t,T]}^\infty} \right)^2 \leq 2C_5^2 + 8\beta^2 T (n_1 + 1)^2 \int_t^T \|Y\|_{\mathcal{S}_{[s,T]}^\infty}^2 ds. \quad (4.31)$$

(2) For $i \in J_2$, g^i satisfies (ii) of Assumption (C1b). In this case, for each $t \in [T-h, T]$, we have

$$g^i(\omega, t, Y_t(\omega), Z_t(\omega)) \operatorname{sgn}(Y_t^i(\omega)) \leq \tilde{\alpha}_t(\omega) + \beta |Y_t(\omega)| + \bar{v}_t(\omega) |Z_t^i(\omega)| + \frac{\gamma}{2} |Z_t^i(\omega)|^2$$

with

$$\bar{v}_t(\omega) := v_t(\omega) + \phi(|Y_t(\omega)|) + c \sum_{j=1}^{i-1} |Z_t^j(\omega)| + \lambda \sum_{j=i+1}^n |Z_t^j(\omega)|^\delta \in \text{BMO}_{[T-h, T]}(\mathbb{R}).$$

Note that $(Y^i, Z^i) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{1 \times d})$ is a solution of BSDE (2.2) on $[T-h, T]$. In view of the last inequality, by using Itô-Tanaka formula and Girsanov's transform, a similar argument to that from (A.11) to (A.12) yields that for each $j \in J_2$ and $t \in [T-h, h]$,

$$|Y_t^i| \leq \|\xi^i\|_\infty + \|\tilde{\alpha}\|_{\mathcal{L}^\infty} + \beta \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty} ds,$$

and then, in view of (2.27),

$$\sum_{j \in J_2} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty} \leq n_2(C_1 + C_2) + n_2\beta \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty} ds.$$

By taking square in both sides of the last inequality and using Hölder's inequality, we can conclude that for each $t \in [T-h, T]$,

$$\sum_{j \in J_2} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty}^2 \leq \left(\sum_{j \in J_2} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty} \right)^2 \leq 2n_2^2(C_1 + C_2)^2 + 2Tn_2^2\beta^2 \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty}^2 ds. \quad (4.32)$$

(3) For $i \in J_3$, g^i satisfies (iii) of Assumption (C1b). In this case, for each $t \in [T-h, T]$, we have

$$g^i(\omega, t, Y_t(\omega), Z_t(\omega)) \operatorname{sgn}(Y_t^i(\omega)) \leq \tilde{\alpha}_t(\omega) + \beta|Y_t(\omega)| + \lambda \sum_{j \in J_3} |Z_t^j(\omega)|,$$

and then, in view of inequality $2ab \leq 2\varepsilon a^2 + \frac{1}{2\varepsilon} b^2$ for each $a, b \geq 0$ and $\varepsilon > 0$,

$$\begin{aligned} & 2 \sum_{j \in J_3} Y_t^j(\omega) g^j(\omega, t, Y_t(\omega), Z_t(\omega)) \\ & \leq 2 \sum_{j \in J_3} |Y_t^j(\omega)| \left(\tilde{\alpha}_t(\omega) + \beta|Y_t(\omega)| + \lambda \sum_{j \in J_3} |Z_t^j(\omega)| \right) \\ & \leq \sum_{j \in J_3} \left[2\tilde{\alpha}_t(\omega) |Y_t^j(\omega)| + 2\beta|Y_t(\omega)|^2 + \sum_{j \in J_3} \left(2n_3\lambda^2 |Y_t^j(\omega)|^2 + \frac{1}{2n_3} |Z_t^j(\omega)|^2 \right) \right] \\ & \leq 2\tilde{\alpha}_t(\omega) \sum_{j \in J_3} |Y_t^j(\omega)| + 2n_3(\beta + n_3\lambda^2) |Y_t(\omega)|^2 + \frac{1}{2} \sum_{j \in J_3} |Z_t^j(\omega)|^2. \end{aligned}$$

Note that for each $i \in J_3$, $(Y^i, Z^i) \in \mathcal{S}_{[T-h, T]}^\infty(\mathbb{R}) \times \text{BMO}_{[T-h, T]}(\mathbb{R}^{1 \times d})$ is an adapted solution of BSDE (2.2) on $[T-h, T]$. In view of the last inequality together with (4.21), applying Itô's formula to $\sum_{j \in J_3} |Y_s^j|^2$ yields that for each $t \in [T-h, h]$,

$$\begin{aligned} & \sum_{j \in J_3} |Y_t^j|^2 + \frac{1}{2} \mathbb{E}_t \left[\int_t^T \sum_{j \in J_3} |Z_s^j(\omega)|^2 ds \right] \\ & \leq \mathbb{E}_t \left[\sum_{j \in J_3} |\xi^j|^2 \right] + \mathbb{E}_t \left[\int_t^T \left(2\tilde{\alpha}_s \sum_{j \in J_3} |Y_s^j| + 2n_3(\beta + n_3\lambda^2) |Y_s|^2 \right) ds \right] \\ & \leq C_1^2 + 2C_2 \sum_{j \in J_3} \|Y^j\|_{\mathcal{S}_{[t, T]}^\infty} + 2n_3(\beta + n_3\lambda^2) \int_t^T \|Y\|_{\mathcal{S}_{[s, T]}^\infty}^2 ds, \end{aligned}$$

and then, in view of

$$2C_2 \sum_{j \in J_3} \|Y^j\|_{S_{[t,T]}^\infty}^2 \leq 2n_3 C_2^2 + \frac{1}{2n_3} \left(\sum_{j \in J_3} \|Y^j\|_{S_{[t,T]}^\infty} \right)^2 \leq 2n_3 C_2^2 + \frac{1}{2} \sum_{j \in J_3} \|Y^j\|_{S_{[t,T]}^\infty}^2,$$

we have

$$\sum_{j \in J_3} \|Y^j\|_{S_{[t,T]}^\infty}^2 \leq 2C_1^2 + 4n_3 C_2^2 + 4n_3 (\beta + n_3 \lambda^2) \int_t^T \|Y\|_{S_{[s,T]}^\infty}^2 ds. \quad (4.33)$$

Finally, adding (4.31), (4.32) and (4.33) up together yields that

$$\|Y\|_{S_{[t,T]}^\infty}^2 \leq C_6 + C_7 \int_t^T \|Y\|_{S_{[s,T]}^\infty}^2 ds, \quad t \in [T-h, T]$$

with $C_6 := 2C_5^2 + 2n_2^2(C_1 + C_2)^2 + 2C_1^2 + 4n_3 C_2^2$ and

$$C_7 := 8\beta^2 T (n_1 + 1)^2 + 2T n_2^2 \beta^2 + 4n_3 (\beta + n_3 \lambda^2).$$

It then follows from Gronwall's inequality that

$$\|Y\|_{S_{[t,T]}^\infty}^2 \leq C_6 \exp(C_7(T-t)) \leq C_6 \exp(C_7 T) =: \tilde{K}, \quad t \in [T-h, T],$$

which yields the desired conclusion. The proof is complete. $\square$

Proof of Theorem 2.14. With Theorem 2.3 and Proposition 4.2 in hands, we can closely follow the proof of Theorem 4.1 in Cheridito and Nam [9] to prove our Theorem 2.10. All the details are omitted here. $\square$

5 Global unbounded solution: Proof of Theorem 2.27

Identical to the proof of Theorem 2.26, we see that in view of Theorem 2.10, for completing the proof of Theorem 2.27 it suffices to prove that the generator $\bar{g}$ defined in (2.26) also satisfies Assumptions (D1) and (D2) with $\beta = 0$ and some other appropriate parameters when the constant θ is smaller than a given constant $\bar{\theta}_0$ depending only on p , $\bar{p}$ and γ . Clearly, $\bar{g}$ satisfies (D2) since g satisfies it.

In the sequel, we will prove that the generator $\bar{g}$ also satisfies Assumption (D1) with $\beta = 0$ and some other appropriate parameters under the given conditions. Assume that $\alpha \in \mathcal{E}^\infty(p\gamma)$ for some real $p > 1$, and $|H|^2 \in \mathcal{E}^\infty(2\bar{p}(q\gamma)^2)$ for some $\bar{p} > 1$ with $q = p/(p-1)$ such that $1/p + 1/q = 1$. Define the following constants:

$$\varepsilon := \frac{p-1}{p+1}, \quad \hat{p} := \frac{(p+3)\bar{p} + p-1}{2(p+2\bar{p}-1)} > 1, \quad \tilde{p} := \frac{p+1}{2\hat{p}} > 1 \quad \text{and} \quad \tilde{q} := \frac{p+1}{p-2\hat{p}+1} > 1.$$

It is clear that $1/\tilde{p} + 1/\tilde{q} = 1$. Note that g satisfies Assumption (D1) with $\beta = 0$. By a similar argument to (iv) of Remark 2.2, in Assumption (D1) we can without loss of generality assume that $f = g^i$ satisfies any one of the three assertions (D1)(i), (D1)(ii) and (D1)(iii) with $\beta = 0$ for all $i = 1, \dots, n$. Thus, for $i = 1, \dots, n$, we need to consider the following three cases.

(1) g^i satisfies (D1)(i) with $\beta = 0$. For this case, it follows from the definition of $\bar{g}$ that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$\bar{g}^i(\omega, t, y, z) \leq \alpha_t(\omega) + \sum_{j \neq i} \left(\lambda |z^j + H_t^j(\omega)| + \theta |z^j + H_t^j(\omega)|^2 \right) + \frac{\gamma}{2} |z^i + H_t^i(\omega)|^2$$

and

$$\begin{aligned}\bar{g}^i(\omega, t, y, z) &\geq \frac{\bar{\gamma}}{2}|z^i + H_t^i(\omega)|^2 - \bar{\alpha}_t(\omega) - \sum_{j=i+1}^n \left(\bar{\lambda}|z^j + H_t^j(\omega)|^{1+\delta} + \theta|z^j + H_t^j(\omega)|^2 \right) \\ &\quad - \bar{c} \sum_{j=1}^{i-1} |z^j + H_t^j(\omega)|^2.\end{aligned}$$

Note that for each $a, b \geq 0$, it holds that

$$\begin{aligned}(a+b)^2 &\leq 2a^2 + 2b^2, \quad (a+b)^{1+\delta} \leq 2a^{1+\delta} + 2b^{1+\delta}, \\ (a+b)^2 &\leq (1+\varepsilon)a^2 + \left(1 + \frac{1}{\varepsilon}\right)b^2 \quad \text{and} \quad (a+b)^2 \geq \frac{1}{2}a^2 - b^2.\end{aligned}$$

We know that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$,

$$\begin{aligned}\frac{\bar{\gamma}}{4}|z^i|^2 - \hat{\alpha}_t(\omega) - \sum_{j=i+1}^n (2\bar{\lambda}|z^j|^{1+\delta} + 2\theta|z^j|^2) - 2\bar{c} \sum_{j=1}^{i-1} |z^j|^2 &\leq \bar{g}^i(\omega, t, y, z) \\ &\leq \check{\alpha}_t(\omega) + \sum_{j \neq i} (\lambda|z^j| + 2\theta|z^j|^2) + \frac{\gamma(1+\varepsilon)}{2}|z^i|^2\end{aligned}\tag{5.1}$$

with

$$\check{\alpha}_t(\omega) := \alpha_t(\omega) + \sum_{j \neq i} \left(\lambda|H_t^j(\omega)| + 2\theta|H_t^j(\omega)|^2 \right) + \frac{\gamma(1+\varepsilon)}{2\varepsilon}|H_t^i(\omega)|^2$$

and

$$\hat{\alpha}_t(\omega) := \bar{\alpha}_t(\omega) + 2\bar{c} \sum_{j=1}^{i-1} |H_t^j(\omega)|^2 + \frac{\bar{\gamma}}{2}|H_t^i(\omega)|^2 + \sum_{j=i+1}^n \left(2\bar{\lambda}|H_t^j(\omega)|^{1+\delta} + 2\theta|H_t^j(\omega)|^2 \right).$$

By virtue of $\bar{\alpha} \in \mathcal{M}^\infty$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$ together with the fact that

$$\hat{\alpha} \leq 2n\bar{\lambda} + \bar{\alpha} + \left(2\bar{c} + \frac{\bar{\gamma}}{2} + 2\bar{\lambda} + 2\theta \right) |H|^2,$$

we can deduce that

$$\hat{\alpha} \in \mathcal{M}^\infty.\tag{5.2}$$

On the other hand, from the definitions of constants $q, \hat{p}, \tilde{q}, \varepsilon$ and $\check{\alpha}$ it is not difficult to verify that $p\gamma = \tilde{p}\hat{p}\gamma(1+\varepsilon)$ and that there exists two very small positive constants $\bar{\varepsilon}$ and $\bar{\theta}_0$ such that

$$\check{\alpha} \leq \alpha + \frac{n\lambda^2}{4\bar{\varepsilon}} + (\bar{\varepsilon} + 2\theta + q\gamma)|H|^2 \quad \text{and} \quad (\bar{\varepsilon} + 2\bar{\theta}_0 + q\gamma)\tilde{q}\hat{p}\gamma(1+\varepsilon) = 2\bar{p}(q\gamma)^2.$$

Then, it follows from the integrability condition of α and $|H|^2$ that $\alpha \in \mathcal{E}^\infty(\tilde{p}\hat{p}\gamma(1+\varepsilon))$ and $(\bar{\varepsilon} + 2\theta + q\gamma)|H|^2 \in \mathcal{E}^\infty(\tilde{q}\hat{p}\gamma(1+\varepsilon))$ for each $\theta \in [0, \bar{\theta}_0]$. Note that $1/\tilde{p} + 1/\tilde{q} = 1$. By (iii) of Remark 2.1 we know that for each $\theta \in [0, \bar{\theta}_0]$,

$$\check{\alpha} \in \mathcal{E}^\infty(\hat{p}\gamma(1+\varepsilon)).\tag{5.3}$$

Combining (5.1), (5.2) and (5.3) yields that the generator $\bar{g}^i$ satisfies (i) of Assumption (D1) with parameters $(\check{\alpha}, \hat{\alpha}, 0, \gamma(1+\varepsilon), \bar{\gamma}/2, 2\bar{c}, \lambda, 2\bar{\lambda}, 2\theta)$ instead of $(\alpha, \bar{\alpha}, \beta, \gamma, \bar{\gamma}, c, \lambda, \bar{\lambda}, \theta)$.

(2) g^i satisfies (D1)(ii) with $\beta = 0$ and violates (D1) (i) and (iii). Note that we have $H^i \equiv 0$ in this case. It follows from the definition of $\bar{g}$ that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$|\bar{g}^i(\omega, t, y, z)| \leq \tilde{\alpha}_t(\omega) + \frac{\gamma}{2}|z^i|^2 + \left\{ \bar{v}_t(\omega) + c \sum_{j=1}^{i-1} |z^j| + \sum_{j=i+1}^n (\lambda|z^j|^\delta + \theta|z^j|) \right\} |z^i|$$

with

$$\bar{v}_t(\omega) := v_t(\omega) + n\lambda + (c + \lambda + \theta) \sum_{j \neq i} |H_t^j(\omega)| \leq v_t(\omega) + n\lambda + (c + \lambda + \theta) \sqrt{n} |H_t(\omega)|.$$

And, it follows from $v \in \text{BMO}(\mathbb{R})$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$ that $\bar{v} \in \text{BMO}(\mathbb{R})$, which means that $\bar{g}^i$ satisfies (ii) of Assumption (D1) with parameters $(\bar{\alpha}, 0, \bar{v}, c, \gamma, \lambda, \theta)$ instead of $(\bar{\alpha}, \beta, v, c, \gamma, \lambda, \theta)$.

(3) $\bar{g}^i$ satisfies (D1)(iii) with $\beta = 0$. In this case, it follows from the definition of $\bar{g}$ that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, we have

$$|g^i(\omega, t, y, z)| \leq \bar{\alpha}_t(\omega) + \bar{\lambda}|z| + 2\theta \sum_{j \neq i} |z^j|^2$$

with

$$\bar{\alpha}_t(\omega) := \bar{\alpha}_t(\omega) + \bar{\lambda}|H_t(\omega)| + 2\theta \sum_{j \neq i} |H_t^j(\omega)|^2 \leq \bar{\alpha}_t(\omega) + \bar{\lambda}|H_t(\omega)| + 2\theta|H_t(\omega)|^2.$$

And, it follows from $\bar{\alpha} \in \mathcal{M}^\infty$ and $H \in \text{BMO}(\mathbb{R}^{n \times d})$ that $\bar{\alpha} \in \mathcal{M}^\infty$, which means that $\bar{g}^i$ satisfies (iii) of Assumption (D1) with parameters $(\bar{\alpha}, 0, \bar{\lambda}, 2\theta)$ instead of $(\bar{\alpha}, \beta, \bar{\lambda}, \theta)$.

All in all, we have proved the desired conclusion. Theorem 2.27 is then proved.

A Auxiliary results on bounded solutions of scalar-valued BSDEs with unbounded coefficients

We collect here some general results concerning the uniform estimate of (bounded) solution to scalar-valued BSDEs. We give some brief proofs for completeness.

Let the function $f(\cdot, y, z) : \Omega \times [0, T] \rightarrow \mathbb{R}$ be $(\mathcal{F}_t)$ -progressively measurable for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$. Consider the following scalar-valued BSDE:

$$Y_t = \eta + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad t \in [0, T], \quad (\text{A.1})$$

with $\eta \in L^\infty(\mathbb{R})$. The solution (Y, Z) is a pair of $(\mathcal{F}_t)$ -progressively measurable processes, taking values in $\mathbb{R} \times \mathbb{R}^{1 \times d}$ such that $Y \in \mathcal{S}^\infty(\mathbb{R})$.

Assume that $\bar{\beta}, \bar{\delta} \geq 0$ are two constants, $\varphi(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ is a nondecreasing continuous function, and $\bar{\alpha} \in \mathcal{E}^\infty(\gamma \exp(\bar{\beta}T))$, $\dot{\alpha} \in \mathcal{M}^\infty$, $\ddot{\alpha} \in \mathcal{L}^\infty$, $\bar{u} \in \text{BMO}(\mathbb{R})$, $\bar{v} \in \text{BMO}(\mathbb{R})$ are five real-valued non-negative $(\mathcal{F}_t)$ -progressively measurable processes.

We introduce the following assumptions on the generator f :

(A1) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$\frac{\bar{\gamma}}{2}|z|^2 - \dot{\alpha}_t(\omega) - \bar{\beta}|y| \leq f(\omega, t, y, z) \leq \bar{\alpha}_t(\omega) + [\bar{\beta}|y| \mathbf{1}_{y>0} + \varphi(|y|) \mathbf{1}_{y<0}] + \frac{\gamma}{2}|z|^2.$$

(A2) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$\frac{\bar{\gamma}}{2}|z|^2 - \dot{\alpha}_t(\omega) - \bar{\beta}|y| \leq -f(\omega, t, -y, -z) \leq \bar{\alpha}_t(\omega) + [\bar{\beta}|y| \mathbf{1}_{y>0} + \varphi(|y|) \mathbf{1}_{y<0}] + \frac{\gamma}{2}|z|^2.$$

(A3) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$\begin{aligned} -[\bar{\beta}|y| \mathbf{1}_{y<0} + \varphi(|y|) \mathbf{1}_{y>0}] - h(\omega, t, z) &\leq f(\omega, t, y, z) \\ &\leq [\bar{\beta}|y| \mathbf{1}_{y>0} + \varphi(|y|) \mathbf{1}_{y<0}] + h(\omega, t, z) \end{aligned}$$

with

$$h(\omega, t, z) := \ddot{\alpha}_t(\omega) + \bar{u}_t(\omega)|z| + \frac{\gamma}{2}|z|^2.$$

(A4) Almost everywhere in $\Omega \times [0, T]$, for any $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, we have

$$\begin{aligned} -\dot{\alpha}_t(\omega) - [\bar{\beta}|y| \mathbf{1}_{y < 0} + \varphi(|y|) \mathbf{1}_{y > 0}] - \bar{\lambda}|z| &\leq f(\omega, t, y, z) \\ &\leq \dot{\alpha}_t(\omega) + [\bar{\beta}|y| \mathbf{1}_{y > 0} + \varphi(|y|) \mathbf{1}_{y < 0}] + \bar{\lambda}|z|. \end{aligned}$$

(A5) Almost everywhere in $\Omega \times [0, T]$, for any $(y, \bar{y}, z, \bar{z}) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{1 \times d} \times \mathbb{R}^{1 \times d}$, we have for some $k > 0$,

$$|f(\omega, t, y, z) - f(\omega, t, \bar{y}, \bar{z})| \leq \bar{\beta}|y - \bar{y}| + k(\bar{v}_t(\omega) + |z| + |\bar{z}|)|z - \bar{z}|.$$

Proposition A.1. Let the generator f satisfy Assumption (A1) (resp. (A2)).

(i) For a solution (Y, Z) of BSDE (A.1) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$, we have $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$ and for each $t \in [0, T]$,

$$\begin{aligned} \|Y\|_{\mathcal{S}_{[t, T]}^\infty}^2 + \|Z\|_{\text{BMO}_{[t, T]}}^2 \\ \leq \frac{2(1 + \bar{\beta}T) + \bar{\gamma}}{\bar{\gamma}} \exp(2\bar{\beta}T) \left(3\|\eta\|_\infty + \|\dot{\alpha}\|_{\mathcal{E}_{[t, T]}^\infty(\gamma \exp(\bar{\beta}T))} + 2\|\dot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} \right). \end{aligned} \quad (\text{A.2})$$

And, if the generator f only satisfies the second inequality for the case of $y > 0$ and the first inequality in Assumption (A1) (resp. (A2)), the inequality (A.2) is still true.

- (ii) BSDE (A.1) admits a minimal (resp. maximal) solution (Y, Z) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$ in the sense that for any solution $(\bar{Y}, \bar{Z})$ of BSDE (A.1) such that $\bar{Y} \in \mathcal{S}^\infty(\mathbb{R})$, we have for each $t \in [0, T]$, $\mathbb{P} - a.s.$, $Y_t \leq \bar{Y}_t$ (resp. $Y_t \geq \bar{Y}_t$). Moreover, $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.
- (iii) If the generator f further satisfies Assumption (A5), then BSDE (A.1) admits a unique solution (Y, Z) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$. Moreover, $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.

Proof. We only give the proof when the generator f satisfies Assumption (A1). The other case is proved in the same way.

(i) For each integer $m \geq 1$ and each stopping time $\tau \in \mathcal{T}_{[0, T]}$, define the following stopping time

$$\sigma_m^\tau := T \wedge \inf \left\{ s \in [\tau, T] : \int_\tau^s |Z_s|^2 ds \geq m \right\}$$

with convention $\inf \emptyset = \infty$. It follows from the first inequality in Assumption (A1) that for each $m \geq 1$ and each $t \in [0, T]$,

$$\frac{\bar{\gamma}}{2} \mathbb{E}_\tau \left[\int_\tau^{\sigma_m^\tau} |Z_s|^2 ds \right] \leq |Y_\tau| + \mathbb{E}_\tau \left[|Y_{\sigma_m^\tau}| + \int_\tau^{\sigma_m^\tau} (\dot{\alpha}_s + \bar{\beta}|Y_s|) ds \right], \quad \tau \in \mathcal{T}_{[t, T]}.$$

Sending $m \rightarrow +\infty$ in previous inequality and using Fatou's lemma yields that for each $t \in [0, T]$,

$$\frac{\bar{\gamma}}{2} \mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \leq \|\eta\|_\infty + \|\dot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + (1 + \bar{\beta}T) \|Y\|_{\mathcal{S}_{[t, T]}^\infty}, \quad \tau \in \mathcal{T}_{[t, T]},$$

which means that $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$, and for each $t \in [0, T]$,

$$\frac{\bar{\gamma}}{2} \|Z\|_{\text{BMO}_{[t, T]}}^2 \leq \|\eta\|_\infty + \|\dot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty} + (1 + \bar{\beta}T) \|Y\|_{\mathcal{S}_{[t, T]}^\infty}. \quad (\text{A.3})$$

Furthermore, define the function

$$u(t, x) := \exp \left(\gamma \exp(\bar{\beta}t)x + \gamma \int_0^t \exp(\bar{\beta}s) \dot{\alpha}_s ds \right), \quad (t, x) \in [0, T] \times \mathbb{R}.$$

In view of the second inequality for the case of $y > 0$ in Assumption (A1), by applying Itô-Tanaka formula to $u(s, Y_s^+)$ we can deduce that for each $m \geq 1$ and $t \in [0, T]$,

$$\begin{aligned} \exp(\gamma Y_t^+) &\leq \mathbb{E}_t \left[\exp \left(\gamma \exp(\bar{\beta}T) \eta^+ + \gamma \exp(\bar{\beta}T) \int_t^T \bar{\alpha}_s ds \right) \right], \\ &\leq \exp(\gamma \exp(\bar{\beta}T) \|\eta\|_\infty) \exp \left(\gamma \exp(\bar{\beta}T) \|\bar{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(\gamma \exp(\bar{\beta}T))} \right) \end{aligned}$$

and then,

$$Y_t^+ \leq \exp(\bar{\beta}T) \left(\|\eta\|_\infty + \|\bar{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(\gamma \exp(\bar{\beta}T))} \right). \quad (\text{A.4})$$

On the other hand, from the first inequality in Assumption (A1) we can also get that for each $m \geq 1$ and $t \in [0, T]$,

$$Y_t^- = (-Y_t)^+ \leq \mathbb{E}_t \left[\eta^- + \int_t^T (\bar{\alpha}_s + \bar{\beta}|Y_s|) ds \right],$$

which together with (A.4) yields that for each $t \in [0, T]$,

$$|Y_t| \leq \exp(\bar{\beta}T) \left(\|\eta\|_\infty + \|\bar{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(\gamma \exp(\bar{\beta}T))} \right) + \|\eta\|_\infty + \|\bar{\alpha}\|_{\mathcal{M}_{[t,T]}^\infty} + \bar{\beta} \mathbb{E}_t \left[\int_t^T |Y_s| ds \right].$$

And, it follows from Gronwall's inequality that

$$\|Y\|_{\mathcal{S}_{[t,T]}^\infty} \leq \exp(2\bar{\beta}T) \left(2\|\eta\|_\infty + \|\bar{\alpha}\|_{\mathcal{E}_{[t,T]}^\infty(\gamma \exp(\bar{\beta}T))} + \|\bar{\alpha}\|_{\mathcal{M}_{[t,T]}^\infty} \right), \quad t \in [0, T]. \quad (\text{A.5})$$

Finally, the desired conclusion (A.2) follows from (A.5) and (A.3) immediately.

(ii) In view of Assumption (A1), it is easy to verify that for each integer $m \geq 1$, the following function

$$\begin{aligned} f^m(\omega, t, y, z) &:= \inf \left\{ f(\omega, t, \bar{y}, \bar{z}) + (m + \bar{\beta})|y - \bar{y}| + m|z - \bar{z}| : (\bar{y}, \bar{z}) \in \mathbb{R} \times \mathbb{R}^{1 \times d} \right\}, \\ &(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d} \end{aligned}$$

is well defined and an $(\mathcal{F}_t)$ -progressively measurable process for each (y, z) . It is also not difficult to prove that f^m is uniformly Lipschitz continuous in the state variables (y, z) and also satisfies Assumption (A1) with the same parameters for each $m \geq 1$, and that the sequence $\{f^m\}_{m=1}^\infty$ converges increasingly uniformly on compact sets to the generator f as m tends to $+\infty$. Then, $dP \times dt - a.e.$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$ and $m \geq 1$, we have

$$|f^m(\omega, t, y, z)| \leq \bar{\alpha}_t(\omega) + \dot{\alpha}_t(\omega) + \varphi(|y|) + \bar{\beta}|y| + \frac{\gamma}{2}|z|^2, \quad (\text{A.6})$$

and then

$$|f^m(\omega, t, 0, 0)| \leq \bar{\alpha}_t(\omega) + \dot{\alpha}_t(\omega) + \varphi(0), \quad (\text{A.7})$$

which means that

$$\mathbb{E} \left[\left(\int_0^T |f^m(s, 0, 0)| ds \right)^2 \right] < +\infty.$$

Consequently, by the classical results (see for example Theorems 3 and 2 in Fan et al. [18]) we know that for each $m \geq 1$, the following BSDE

$$Y_t^m = \eta + \int_t^T f^m(s, Y_s^m, Z_s^m) ds - \int_t^T Z_s^m dB_s, \quad t \in [0, T]$$

admits a unique solution $(Y^m, Z^m) \in \mathcal{S}^2(\mathbb{R}) \times \mathcal{H}^2(\mathbb{R}^{1 \times d})$, and Y^m converges increasing pointwisely to a process Y . Moreover, by the classical a priori estimate on the L^2 solution (see for example Proposition 3.2 in Briand et al. [5]) we know that there exists a uniform constant $c_0 > 0$ such that for each $m \geq 1$ and $t \in [0, T]$,

$$\mathbb{E}_t \left[\sup_{s \in [t, T]} |Y_s^m|^2 \right] \leq c_0 \exp(2(m + \bar{\beta})T + 2m^2T) \mathbb{E}_t \left[|\eta|^2 + \left(\int_t^T |f^m(s, 0, 0)| ds \right)^2 \right],$$

which together with (A.7) and the facts that $\eta \in L^\infty(\mathbb{R})$, $\check{\alpha} \in \mathcal{E}^\infty(\gamma \exp(\bar{\beta}T))$ and $\dot{\alpha} \in \mathcal{M}^\infty$ yields that $Y^m \in \mathcal{S}^\infty(\mathbb{R})$ for each $m \geq 1$, and then $Z^m \in \text{BMO}(\mathbb{R}^{1 \times d})$ by (i).

In the sequel, it follows from (i) that there exists a uniform constant $A > 0$ which is independent of m such that $d\mathbb{P} \times dt - a.e.$, we have

$$\sup_{m \geq 1} |Y_t^m(\omega)| \leq A$$

and, in view of (A.6),

$$\forall (y, z) \in [-A, A] \times \mathbb{R}^{1 \times d}, \quad |f^m(\omega, t, y, z)| \leq \check{\alpha}_t(\omega) + \dot{\alpha}_t(\omega) + \varphi(A) + \bar{\beta}A + \frac{\gamma}{2}|z|^2.$$

Thus, applying the monotonic stability result Proposition 3.1 in Luo and Fan [36], we have the existence of a process $Z \in \mathcal{H}^2(\mathbb{R}^{1 \times d})$ such that $Y \in \mathcal{S}^\infty(\mathbb{R})$ and (Y, Z) is a solution of BSDE (A.1). In view of (i), we have $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.

It remains to show that (Y, Z) is the minimal solution. For this, let $(\bar{Y}, \bar{Z})$ be any solution of BSDE (A.1) such that $\bar{Y} \in \mathcal{S}^\infty(\mathbb{R})$. By (i) again we know that $\bar{Z} \in \text{BMO}(\mathbb{R}^{1 \times d})$. This means that $(\bar{Y}, \bar{Z}) \in \mathcal{S}^2(\mathbb{R}) \times \mathcal{H}^2(\mathbb{R}^{1 \times d})$. Then, since f^m is uniformly Lipschitz continuous in (y, z) and $f^m \leq f$ for each $m \geq 1$, it follows from the classical comparison theorem on the L^2 -solution that for each $m \geq 1$ and $t \in [0, T]$, $\mathbb{P} - a.s.$, $Y_t^m \leq \bar{Y}_t$, and letting $m \rightarrow \infty$ yields that $Y \leq \bar{Y}$, which is the desired conclusion.

(iii) Let the generator f further satisfy Assumption (A5), and (Y, Z) and $(\bar{Y}, \bar{Z})$ be the solution of BSDE (A.1) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$ and $\bar{Y} \in \mathcal{S}^\infty(\mathbb{R})$. First of all, from (i) we know that $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$ and $\bar{Z} \in \text{BMO}(\mathbb{R}^{1 \times d})$. Furthermore, define $\hat{Y} := Y - \bar{Y}$ and $\hat{Z} := Z - \bar{Z}$. By virtue of Itô-Tanaka formula and (A5) we can deduce that for each $t \in [0, T]$,

$$\begin{aligned} |\hat{Y}_t| &\leq \int_t^T \left(\bar{\beta} |\hat{Y}_s| + k(\bar{v}_s + |Z_s| + |\bar{Z}_s|) |\hat{Z}_s| \right) ds - \int_t^T \text{sgn}(\hat{Y}_s) \hat{Z}_s dB_s \\ &\leq \bar{\beta} \int_t^T |\hat{Y}_s| ds - \int_t^T \text{sgn}(\hat{Y}_s) \hat{Z}_s \left[dB_s - k \text{sgn}(\hat{Y}_s) (\bar{v}_s + |Z_s| + |\bar{Z}_s|) \frac{\hat{Z}_s^\top}{|\hat{Z}_s|} \mathbf{1}_{|\hat{Z}_s| \neq 0} ds \right]. \end{aligned} \quad (\text{A.8})$$

Since all of processes $\bar{v}, Z, \bar{Z}$ belong to $\text{BMO}(\mathbb{R}^{1 \times d})$, it is easy to verify that the process

$$M_t := k \int_0^t \text{sgn}(\hat{Y}_s) (\bar{v}_s + |Z_s| + |\bar{Z}_s|) \frac{\hat{Z}_s^\top}{|\hat{Z}_s|} \mathbf{1}_{|\hat{Z}_s| \neq 0} dB_s, \quad t \in [0, T]$$

is a BMO martingale. Define

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} := \exp \left\{ M_T - \frac{1}{2} \langle M \rangle_T \right\}$$

and

$$\tilde{B}_t := B_t - k \int_0^t \text{sgn}(\hat{Y}_s) (\bar{v}_s + |Z_s| + |\bar{Z}_s|) \frac{\hat{Z}_s^\top}{|\hat{Z}_s|} \mathbf{1}_{|\hat{Z}_s| \neq 0} ds, \quad t \in [0, T].$$

Then, $\tilde{\mathbb{P}}$ is a new probability, and $\tilde{B}$ is a Brownian motion with respect to $\tilde{\mathbb{P}}$. Then, taking the obvious conditional mathematical expectation with respect to $\tilde{\mathbb{P}}$ in (A.8) and utilizing Gronwall's inequality yields that $|\hat{Y}_t| = 0$ for each $t \in [0, T]$, which is the desired conclusion. $\square$

Proposition A.2. Let the generator f satisfies Assumption (A3).

- (i) For a solution (Y, Z) of BSDE (A.1) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$, we have $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$ and for each $0 \leq t \leq r \leq T$,

$$\begin{aligned} \|Y\|_{\mathcal{S}_{[t,r]}^\infty} + \|Z\|_{\text{BMO}_{[t,r]}}^2 &\leq \frac{4(\gamma+1)}{\gamma^2} \exp \left\{ 4\gamma \exp(\bar{\beta}T) \left(\|\eta\|_\infty + \|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} \right) \right\} \\ &\times \left\{ 1 + \bar{\beta}T \exp(\bar{\beta}T) \left(\|\eta\|_\infty + \|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} \right) + \|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} + \|\bar{u}\|_{\text{BMO}_{[t,r]}}^2 \right\}. \end{aligned} \quad (\text{A.9})$$

And, if the generator f only satisfies the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in Assumption (A3), the above inequality (A.9) is still true.

- (ii) BSDE (A.1) has a solution (Y, Z) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$. Moreover, $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.
 (iii) If the generator f further satisfies Assumption (A5), then BSDE (A.1) admits a unique solution (Y, Z) such that $Y \in \mathcal{S}^\infty(\mathbb{R})$. Moreover, $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.

Proof. (i) For each $0 \leq t \leq r \leq T$, each stopping time $\tau \in \mathcal{T}_{[t,r]}$ and each integer $m \geq 1$, define the following stopping time

$$\sigma_m^{\tau,r} := r \wedge \inf \left\{ s \in [\tau, r] : \int_\tau^s |Z_s|^2 ds \geq m \right\}.$$

Using Itô-Tanaka formula to compute $\exp(2\gamma|Y_t|)$ and utilizing the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in Assumption (A3), we have

$$\begin{aligned} \exp(2\gamma|Y_\tau|) + 2\gamma^2 \mathbb{E}_\tau \left[\int_\tau^{\sigma_m^{\tau,r}} \exp(2\gamma|Y_s|) |Z_s|^2 ds \right] \\ \leq \mathbb{E}_\tau [\exp(2\gamma|\eta|)] + 2\gamma \mathbb{E}_\tau \left[\int_\tau^{\sigma_m^{\tau,r}} \exp(2\gamma|Y_s|) \left(\ddot{\alpha}_s + \bar{\beta}|Y_s| + \bar{u}_t|Z_s| + \frac{\gamma}{2}|Z_s|^2 \right) ds \right]. \end{aligned}$$

Therefore, in view of the elementary inequality $2ab \leq 2a^2 + b^2/2$ for $a, b \geq 0$,

$$\begin{aligned} &\gamma^2 \mathbb{E}_\tau \left[\int_\tau^{\sigma_m^{\tau,r}} |Z_s|^2 ds \right] \\ &\leq \exp(2\gamma\|\eta\|_\infty) + 2\gamma \exp \left(2\gamma\|Y\|_{\mathcal{S}_{[t,r]}^\infty} \right) \left(\|\ddot{\alpha}\|_{\mathcal{L}_{[t,r]}^\infty} + \bar{\beta}T\|Y\|_{\mathcal{S}_{[t,r]}^\infty} \right) \\ &\quad + 2\exp \left(4\gamma\|Y\|_{\mathcal{S}_{[t,r]}^\infty} \right) \|\bar{u}\|_{\text{BMO}_{[t,r]}}^2 + \frac{\gamma^2}{2} \mathbb{E}_\tau \left[\int_\tau^{\sigma_m^{\tau,r}} |Z_s|^2 ds \right]. \end{aligned}$$

Sending $m \rightarrow +\infty$ and using Fatou's lemma yields that $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$, and we have

$$\begin{aligned} \|Z\|_{\text{BMO}_{[t,r]}}^2 &\leq \frac{2}{\gamma^2} \exp(2\gamma\|\eta\|_\infty) + \frac{4}{\gamma} \exp \left(2\gamma\|Y\|_{\mathcal{S}_{[t,T]}^\infty} \right) \left(\|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} + \bar{\beta}T\|Y\|_{\mathcal{S}_{[t,T]}^\infty} \right) \\ &\quad + \frac{4}{\gamma^2} \exp \left(4\gamma\|Y\|_{\mathcal{S}_{[t,T]}^\infty} \right) \|\bar{u}\|_{\text{BMO}_{[t,r]}}^2. \end{aligned} \quad (\text{A.10})$$

Furthermore, using Itô-Tanaka formula we also have

$$\begin{aligned} |Y_t| &\leq |\eta| + \int_t^T \left(\ddot{\alpha}_s + \bar{\beta}|Y_s| + \bar{u}_t|Z_s| + \frac{\gamma}{2}|Z_s|^2 \right) ds - \int_t^T \operatorname{sgn}(Y_s)Z_s dB_s \\ &\leq \|\eta\|_\infty + \|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} + \bar{\beta} \int_t^T |Y_s| ds \\ &\quad - \int_t^T \operatorname{sgn}(Y_s)Z_s \left[dB_s - \operatorname{sgn}(Y_s) \left(\bar{u}_s \frac{1}{|Z_s|} \mathbf{1}_{|Z_s| \neq 0} + \frac{\gamma}{2} \right) Z_s^\top ds \right]. \end{aligned} \quad (\text{A.11})$$

Since both processes $\bar{u}$ and Z belong to $\text{BMO}(\mathbb{R}^{1 \times d})$, we see that the process

$$M_{\bar{t}} := \int_0^{\bar{t}} \operatorname{sgn}(Y_s) \left(\bar{u}_s \frac{1}{|Z_s|} \mathbf{1}_{|Z_s| \neq 0} + \frac{\gamma}{2} \right) Z_s dB_s, \quad \bar{t} \in [0, T]$$

is a BMO martingale. Define

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} := \exp \left\{ M_T - \frac{1}{2} \langle M \rangle_T \right\}$$

and

$$\tilde{B}_{\bar{t}} := B_{\bar{t}} - \int_0^{\bar{t}} \operatorname{sgn}(Y_s) \left(\bar{u}_s \frac{1}{|Z_s|} \mathbf{1}_{|Z_s| \neq 0} + \frac{\gamma}{2} \right) Z_s^\top ds, \quad \bar{t} \in [0, T].$$

Then, $\tilde{\mathbb{P}}$ is a new probability, and $\tilde{B}$ is a Brownian motion with respect to $\tilde{\mathbb{P}}$. Then, taking the obvious conditional mathematical expectation with respect to $\tilde{\mathbb{P}}$ in (A.11) and using Gronwall's inequality yields that

$$\|Y\|_{\mathcal{S}_{[t,T]}^\infty} \leq \exp(\bar{\beta}T) \left(\|\eta\|_\infty + \|\ddot{\alpha}\|_{\mathcal{L}_{[t,T]}^\infty} \right). \quad (\text{A.12})$$

Then, the desired inequality (A.9) immediately follows from (A.10) and (A.12).

(ii) It is easy to check that for each pair of integers $m, l \geq 1$, the following function

$$\begin{aligned} f^{m,l}(\omega, t, y, z) &:= \inf \left\{ f^+(\omega, t, \bar{y}, \bar{z}) + m|y - \bar{y}| + m|z - \bar{z}| : (\bar{y}, \bar{z}) \in \mathbb{R} \times \mathbb{R}^{1 \times d} \right\} \\ &\quad - \inf \left\{ f^-(\omega, t, \bar{y}, \bar{z}) + l|y - \bar{y}| + l|z - \bar{z}| : (\bar{y}, \bar{z}) \in \mathbb{R} \times \mathbb{R}^{1 \times d} \right\}, \\ &\quad (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d} \end{aligned}$$

is well defined and $(\mathcal{F}_t)$ -progressively measurable for each (y, z) . In view of Assumption (A3), we see that $f^{m,l}$ is uniformly Lipschitz continuous in the last two arguments and satisfies Assumption (A3) with the same parameters for $m, l \geq 1$, and that for each $l \geq 1$, the sequence $\{f^{m,l}\}_{m=1}^\infty$ converges increasingly and uniformly on compact sets to a function $f^{\infty,l}$ as $m \rightarrow +\infty$, and $\{f^{\infty,l}\}_{l=1}^\infty$ converges decreasingly and uniformly on compact sets to the generator f as $l \rightarrow +\infty$. Then, $d\mathbb{P} \times dt$ -a.e., for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$ and $m, l \geq 1$, we have

$$|f^{m,l}(\omega, t, y, z)| \leq \ddot{\alpha}_t(\omega) + \frac{1}{2} \bar{u}_t^2(\omega) + \varphi(|y|) + \bar{\beta}|y| + \frac{\gamma+1}{2}|z|^2, \quad (\text{A.13})$$

and then

$$|f^{m,l}(\omega, t, 0, 0)| \leq \ddot{\alpha}_t(\omega) + \frac{1}{2} \bar{u}_t^2(\omega) + \varphi(0), \quad (\text{A.14})$$

which means that

$$\mathbb{E} \left[\left(\int_0^T |f^{m,l}(s, 0, 0)| ds \right)^2 \right] < +\infty.$$

Consequently, by the classical results we know that for each $m, l \geq 1$, the following BSDE

$$Y_t^{m,l} = \eta + \int_t^T f^{m,l}(s, Y_s^{m,l}, Z_s^{m,l}) ds - \int_t^T Z_s^{m,l} dB_s, \quad t \in [0, T]$$

admits a unique solution $(Y^{m,l}, Z^{m,l}) \in \mathcal{S}^2(\mathbb{R}) \times \mathcal{H}^2(\mathbb{R}^{1 \times d})$, $Y^{m,l}$ converges increasing pointwisely to a process $Y^{\infty,l}$ as $m \rightarrow +\infty$, and $Y^{\infty,l}$ converges decreasing pointwisely to a process Y as $l \rightarrow +\infty$. Moreover, by the classical a priori estimate on the L^2 solution (see Proposition 3.2 in Briand et al. [5]) we know that there exists a uniform constant $c_0 > 0$ such that for each $m, l \geq 1$ and $t \in [0, T]$,

$$\mathbb{E}_t \left[\sup_{s \in [t, T]} |Y_s^{m,l}|^2 \right] \leq c_0 \exp(2(m \vee l)T + 2(m \vee l)^2 T) \mathbb{E}_t \left[|\eta|^2 + \left(\int_t^T |f^{m,l}(s, 0, 0)| ds \right)^2 \right],$$

which together with (A.14), the energy inequality for BMO martingales (see for example Section 2.1 in Kazamaki [29]) and the facts that $\eta \in L^\infty(\mathbb{R})$, $\bar{\alpha} \in \mathcal{L}^\infty$ and $\bar{u} \in \text{BMO}(\mathbb{R})$ yields that $Y^{m,l} \in \mathcal{S}^\infty(\mathbb{R})$ for each $m, l \geq 1$, and then $Z^{m,l} \in \text{BMO}(\mathbb{R}^{1 \times d})$ by (i).

Finally, it follows from (i) that there exists a uniform constant $A > 0$ which is independent of m and l such that $d\mathbb{P} \times dt - a.e.$, we have

$$\sup_{m, l \geq 1} |Y_t^{m,l}(\omega)| \leq A$$

and, in view of (A.13),

$$\forall (y, z) \in [-A, A] \times \mathbb{R}^{1 \times d}, \quad |f^{m,l}(\omega, t, y, z)| \leq \bar{\alpha}_t(\omega) + \frac{1}{2} \bar{u}_t^2(\omega) + \varphi(A) + \bar{\beta}A + \frac{\gamma+1}{2} |z|^2.$$

Thus, we can apply twice the monotonic stability result Proposition 3.1 in Luo and Fan [36] to obtain the existence of a process $Z \in \mathcal{H}^2(\mathbb{R}^{1 \times d})$ such that $Y \in \mathcal{S}^\infty(\mathbb{R})$ and (Y, Z) is a desired solution of BSDE (A.1). And, it follows from (i) that $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$.

(iii) In view of (i), (iii) is proved in the same way as (iii) of Proposition A.1. $\square$

When the generator f has a linear growth in the last argument z , we have

Proposition A.3. Let the generator f satisfy Assumption (A4).

- (i) For any solution (Y, Z) of BSDE (A.1) such that $Y \in \mathcal{S}^2(\mathbb{R})$, we have $Y \in \mathcal{S}^\infty(\mathbb{R})$ and $Z \in \text{BMO}(\mathbb{R}^{1 \times d})$ and for each $t \in [0, T]$, there exists a uniform constant $c_0 > 1$ such that

$$\|Y\|_{\mathcal{S}_{[t, T]}^\infty} + \|Z\|_{\text{BMO}_{[t, T]}}^2 \leq 1 + c_0 \exp(2\bar{\beta}T + 2\bar{\lambda}^2 T) \left(\|\eta\|_\infty^2 + 2\|\bar{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty}^2 \right). \quad (\text{A.15})$$

And, if the generator f only satisfies the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in Assumption (A4), the above conclusion (A.15) still holds.

- (ii) BSDE (A.1) admits a minimal (resp. maximal) solution (Y, Z) such that $Y \in \mathcal{S}^2(\mathbb{R})$ in the sense that for any solution $(\bar{Y}, \bar{Z})$ of BSDE (A.1) such that $\bar{Y} \in \mathcal{S}^2(\mathbb{R})$, we have for each $t \in [0, T]$, $\mathbb{P} - a.s.$, $Y_t \leq \bar{Y}_t$ (resp. $Y_t \geq \bar{Y}_t$). Moreover, $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}) \times \text{BMO}(\mathbb{R}^{1 \times d})$.
- (iii) If the generator f further satisfies Assumption (A5), then BSDE (A.1) admits a unique solution (Y, Z) such that $Y \in \mathcal{S}^2(\mathbb{R})$. Moreover, $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}) \times \text{BMO}(\mathbb{R}^{1 \times d})$.

Proof. (i) In view of the first inequality for the case of $y < 0$ and the second inequality for the case of $y > 0$ in Assumption (A4) and in virtue of the classical a priori estimate on the L^2 solution (see Proposition 3.2 in [5]), we have the existence of a uniform constant $c_0 > 1$ such that for each $t \in [0, T]$ and each $\tau \in \mathcal{T}_{[t, T]}$,

$$\mathbb{E}_\tau \left[\sup_{s \in [\tau, T]} |Y_s|^2 + \int_\tau^T |Z_s|^2 ds \right] \leq c_0 \exp(2\bar{\beta}T + 2\bar{\lambda}^2T) \mathbb{E}_\tau \left[|\eta|^2 + \left(\int_\tau^T \dot{\alpha}_s ds \right)^2 \right],$$

and then, by the energy inequality for BMO martingales and (i) of Remark 2.1,

$$\|Y\|_{\mathcal{S}_{[t, T]}^\infty}^2 + \|Z\|_{\text{BMO}_{[t, T]}}^2 \leq c_0 \exp(2\bar{\beta}T + 2\bar{\lambda}^2T) \left(\|\eta\|_\infty^2 + 2\|\dot{\alpha}\|_{\mathcal{M}_{[t, T]}^\infty}^2 \right).$$

Then, the desired inequality (A.15) follows from the previous inequality immediately, and then $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}) \times \text{BMO}(\mathbb{R}^{1 \times d})$.

(ii) Define

$$M := 1 + c_0 \exp(2\bar{\beta}T + 2\bar{\lambda}^2T) (\|\eta\|_\infty^2 + 2\|\dot{\alpha}\|_{\mathcal{M}^\infty}^2), \quad \rho^M(x) := \frac{Mx}{M \vee |x|}, \quad x \in \mathbb{R}$$

and

$$f^M(\omega, t, y, z) := f(\omega, t, \rho^M(y), z), \quad (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d}.$$

It is easy to verify that the generator f^M also satisfies Assumption (A4), and that $dP \times dt - a.e.$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$, $|f^M(\omega, t, y, z)| \leq \dot{\alpha}_t(\omega) + \varphi(M) + \bar{\beta}M + \bar{\lambda}|z|$. Then, by Lepeltier and San Martin [35] we know that the following BSDE

$$Y_t = \eta + \int_t^T f^M(s, Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad t \in [0, T] \quad (\text{A.16})$$

admits a maximal solution $(\bar{Y}, \bar{Z})$ and a minimal solution $(\underline{Y}, \underline{Z})$ in the space of processes $\mathcal{S}^2(\mathbb{R}) \times \mathcal{H}^2(\mathbb{R}^{1 \times d})$.

We now show that $(\bar{Y}, \bar{Z})$ and $(\underline{Y}, \underline{Z})$ are also the desired maximal and minimal solution of BSDE (A.1). Indeed, since f^M satisfies Assumption (A4), it follows from (i) and the definition of f^M that both of them belong to the space $\mathcal{S}^\infty(\mathbb{R}) \times \text{BMO}(\mathbb{R}^{1 \times d})$ and are also solutions of BSDE (A.1). Furthermore, let (Y, Z) be any solution of BSDE (A.1) such that $Y \in \mathcal{S}^2(\mathbb{R})$. Then, it follows from (i) and the definition of f^M again that $(Y, Z) \in \mathcal{S}^\infty(\mathbb{R}) \times \text{BMO}(\mathbb{R}^{1 \times d})$ and it is also a solution of BSDE (A.16) in the space $\mathcal{S}^2(\mathbb{R}) \times \mathcal{H}^2(\mathbb{R}^{1 \times d})$. Consequently, for each $t \in [0, T]$, we have $\underline{Y}_t \leq Y_t \leq \bar{Y}_t$, $\mathbb{P} - a.s.$, which is the desired conclusion.

(iii) In view of (i), (iii) is proved in the same way as (iii) of Proposition A.1. $\square$

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